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Master's Degree Thesis

A Study of Algorithms for Monitoring Heart Rate and Agility Index

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Graduate School of Chosun University

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심박수 및 운동량 모니터링 알고리즘 연구

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This is to certify that the Master's degree thesis of Luan Dinh

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Acronyms

ECG	Electrocardiogram
RA	Right Arm
LA	Left Arm
LF	Left Foot
APC	Atria Premature Contraction
VPC	Ventricular Premature Contraction
ADC	Analog-to-Digital Converter
ASP	Acceleration Start Point
APP	Acceleration Peak Point
AEP	Acceleration End Point
-thr	Negative Threshold
HR	Heart Rate
+thr	Positive Threshold
SA	Sinoatrial
AV	Atrioventricular

초록

심박수 및 운동량 모니터링 알고리즘 연구

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유비쿼터스 헬스케어 시스템에서 건강상태 관찰과 측정의 융통성 있는 방법은 필수 기능이다. 신체와 심장의 활동을 모니터링 할 수 있다면 심장 순환계 손상과 신체 활동정도, 건강상태의 관찰신체 활동을 관찰하는 의학 진단가에게 도움을 줄 수 있다.

이 논문에서는 패치형태의 센서모듈에 내장된 심박수와 운동량의 실시간 무선 모니터링을 위한 이론적인 알고리즘을 기술하였다. 심박수 측정을 위해 제안하는 방법은 디지털필터를 통한 잡음제거 효과와 QRS 신호 측정을 위한 임계값, T-파와 QRS 구별하는 기술과 VPC와 QRS 신호를 구별하는 표준정합 방법들의 복합적인 방법으로 개발되었다. 운동의 수행을 평가하기위한 유용한 지표로써 운동량을 제시한다. 운동량 계산을 위한 알고리즘은 다축가속신호(Multi-axis accelerometer signals)의 다양한 디지털처리방법들을 기반으로 한다. 건강측정과 평가를 위한 실시간 무선 시스템(AirBeat^{TM})은 센서모듈을 위한 효과적인 알고리즘과 컴퓨터에서의 분석을 위한 분석 프로그램을 기반으로 개발되었다. 평가를 위해 CASE 시스템은 기본시스템하고 Bruce 프로토콜을 적용하였다. 심박수 측정 오차는 6.2% 이내였다. 평균 운동량과 종래 운동점수의 상관계수는 0.8를 넘었다. 심박수와 운동량 변화는 매우 잘 관찰되었고, 동시에 이 시스템에서 공급하는 건강상태 측정 능력도 측정되었다.

ABSTRACT

A Study of Algorithms for Monitoring Heart Rate and Agility Index

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For ubiquitous healthcare systems, health monitoring and evaluation in a flexible way are essential functions. The ability to monitor physical activities and cardiac activities can aid in determining clinical diagnoses, observing the physical demand of human circulatory system, assessing the intensity of physical activity, and evaluating general health conditions.

In this thesis, algorithms for real-time wireless monitoring of heart rate and agility index, which were implemented in the patched-type sensor module are presented. The proposed method for detecting heart rate was developed by the combination of the following techniques: digital filtering to reduce the effect of noises, threshold QRS detection, T-wave discrimination technique, which provides high ability to distinguish QRS complex from T-wave and template matching method to discriminate QRS complex from VPC. We suggested agility index as a helpful indicator for the evaluation of exercise performance. The algorithm used to calculate agility index was

based on several digital processing techniques of multi-axis accelerometer signals. We developed a real-time wireless system (AirBeat^{TM}) for health monitoring and evaluation based on the effective algorithms executed in sensor module and the analysis software. For testing, the CASE system (from General Electric Medical Co., USA) was used as a reference system, and the Bruce protocol was conducted. For heart beat, the error rate was within 6.2%. The correlation coefficient between average agility index and a conventional agility point was over 0.8. The change of heart rate and agility index can be monitored very effectively and measured at the same time during exercise. Such monitoring and measurement capabilities can provide our AirBeat^{TM} system with the ability to evaluate health condition.

I. Introduction

A. General Description of ECG and Agility Index

1. Electrocardiogram (ECG)

1.1 Introduction

The electrocardiogram (ECG) describes the electrical activity of the heart. It is obtained by placing electrodes on the chest, arms, and legs. With every heartbeat, an impulse through the heart, which determines its rhythm and rate, causes the heart muscle contract to pump blood. The voltage variations measured by the electrodes are caused by the action potentials of the excitable cardiac cells, as they make the cells contract. An ECG is characterized by a series of waves, the morphology and timing provide information used for diagnosing diseases that can be identified by disturbances of the electrical activity of the heart. The pattern that characterizes the occurrence of successive heartbeats is also very important.

The Dutch physiologist, Willem Einthoven, developed using a string galvanometer that was sensitive enough to record electrical potentials on the body surface. He also defined sites for electrode placement on the arms and legs which remain in use today. ECG recording has developed incredibly and become an indispensable tool in many different contexts. The ECG record is used today in a wide variety of clinical applications.

1.2 The Integrated ECG

The heart comprises several different types of tissues such as: sinoatrial (SA) and atrioventricular (AV) nodal tissue, atrial, purkinje, and ventricular tissue. Each type of cells exhibits its own characteristic of action potential. During pumping action, the ECG corresponds to the sum of the electrical activities come from different tissues, as shown in Fig. 1.

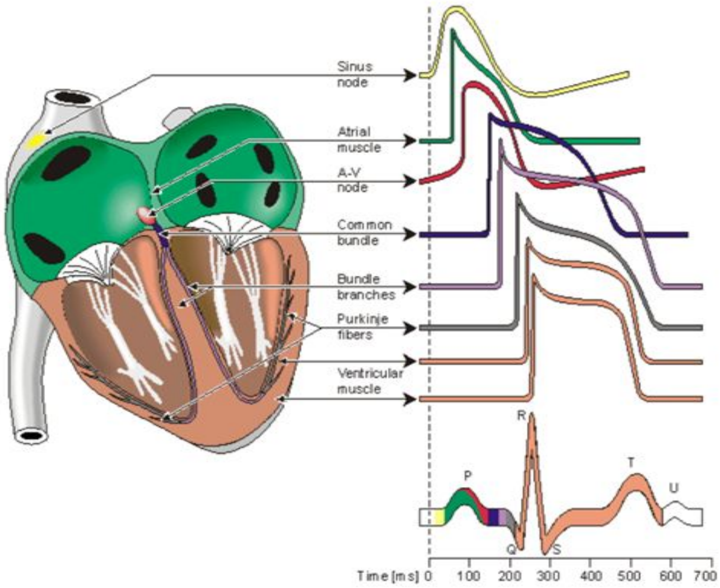


Fig. 1. The integrated ECG.

For monitoring of health condition, heartbeat is a very useful parameter, because it provides the vital signs of heart functions. The morphologies and inter-beat intervals of heartbeats can reveal the condition of heart contraction.

The ECG is used to analyze heart condition for many years. The use of ECG signal is the most accurate method for detecting heart rate. The QRS complex is the most outstanding waveform in the ECG, because it is related to the electrical activities in the heart during the ventricular contraction. Much useful information about current state of the heart is provided by the shape of the QRS complex and the time at which it occurs.

The electrodes used for ECG recording are positioned so that the spatiotemporal variations of the cardiac electrical field are sufficiently well-reflected. The difference in voltage between a pair of electrodes is referred as a lead. The ECG is typically recorded with a multiple-lead configuration.

1.3 The Standard 12-Lead ECG

The heart's electrical activity can be recorded from electrodes on the surface of the body. The lead system, which is used to record the ECG, is formed by 12 positions of a pair-electrode. The 12-lead ECG provides spatial information about the heart's electrical activity in three, approximately-orthogonal directions ie., right and left; superior and inferior; anterior and posterior. Each of the 12 leads represents a particular orientation in space, as shown in Fig. 2.

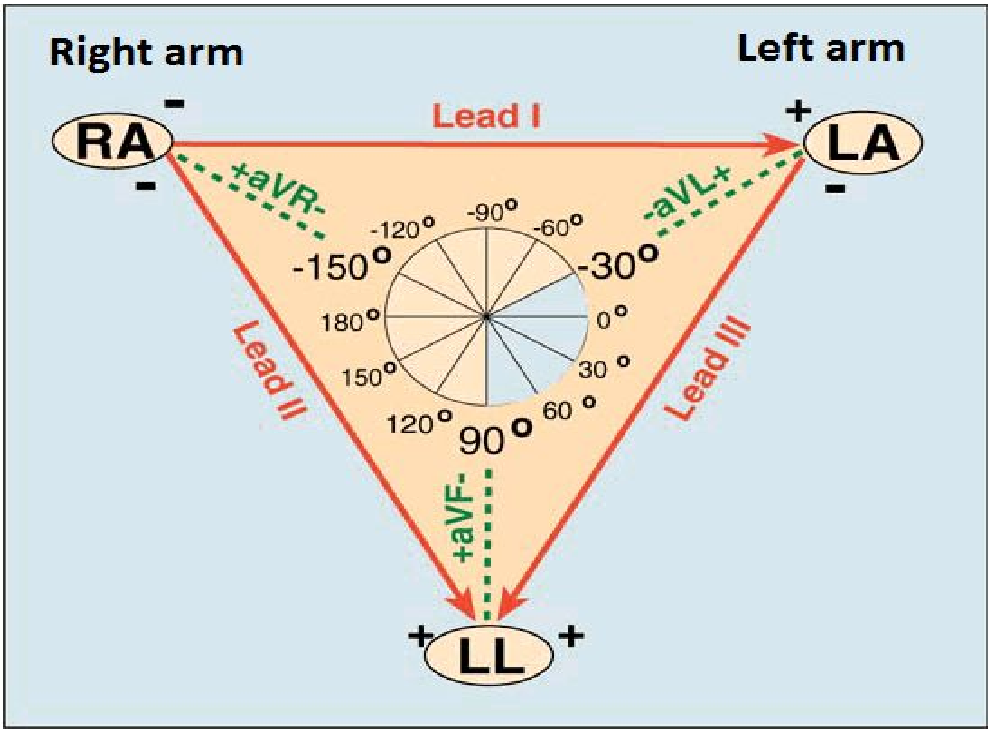


Fig. 2. Lead placement diagram.

Right Arm : RA, Left Arm: LA, Left Foot: LF.

Bipolar limb leads (frontal plane):

+ Lead I: RA (−) to LA (+)
+ Lead II: RA (−) to LF (+)
+ Lead III: LA (−) to LF (+)

Augmented unipolar limb leads (frontal plane):

+ Lead aVR: RA (+) to [LA&LF] (−)
+ Lead aVL: LA (+) to [RA&LF] (−)
+ Lead aVF: LF (+) to [RA&LA] (−)

And others unipolar (+) chest leads in horizontal plane include leads V1, V2, and V3 (posterior anterior), and leads V4, V5, and V6 (right left).

2. Agility Index

The Agility index is the ability to change the position of body in an efficient and effective way, which requires the combination of balance, coordination, speed, reflexes and strength. Balance is the ability to maintain the equilibrium during physical activity, which includes several coordinated actions controlled by body's sensory functions. For an athlete, the ability to maintain balance under changing condition of body movement (dynamic balance) is more important than ability to keep the center of mass above the base of support in stable conditions. Coordination is the ability to control the movement of body in cooperation with the body's sensory functions. Speed is the ability to move all or part of the body quickly. Strength is the ability of muscles to overcome a resistance.

In sports, agility can indicate the potential of athletes, in order to improve agility many training exercises have been proposed. Agility index is manually calculated by a supervisor. The measurement of agility is based on the time it takes an athlete to complete the prescribed exercise or how many steps he or she can complete during a given period of time.

B. Research Necessity

The increasing of social interest in healthcare has resulted in enhanced public demand for programs and activities that promote health and physical development. Currently, health-related devices are becoming the commercial products. Various types of devices have been developed for use in physical fitness training for assessing the physical state of the body. However, most of them are only used in the limited space (indoors) or cause some disadvantages in the exercise. In our modern society, people are confined in a stuffy, high press work environment and they have to deal with the lifestyle imposed by crowded cities. So, outdoor spaces, such as parks or nature-friendly areas, are preferred. The demand for the development of devices that can be used outdoors to measure and manage physical fitness and exercise training is increasing rapidly. For a healthcare system, especially in sports activities, a wireless patched-type sensor module that can provide the desired information and analyses is essential, because it provides subjects a flexible and freely-moving condition. The information provided by the combination of simultaneous monitoring of heart rate and body movement intensity comprises a valuable set of physiological and behavioral data related to the physical activity and cardiac activity. It is widely known that the intensity of body movement has a close relationship with the physical demand of the human circulatory system. The added knowledge about the subject's activity intensity provides valuable information for monitoring, diagnostics, or health alerts.

The manual measurement, calculation, and analysis of the agility index is inconvenient. The accuracy of heart rate monitoring during intensive physical activity is not high, which means that it is difficult to determine with appropriate accuracy when an emergency situation is occurring by just analyzing heat stress. Every year, many athletes and students have strokes and die after intensive exercises or athletic competition due to our inability to forecast such problems or manage them properly when they occur.

In this thesis, algorithms for flexible, high-accuracy monitoring of heart rate and agility index are proposed. A wireless system for evaluating, training and monitoring of athletes, as well as providing emergency forecast of potential problems by assessing the combination of heart rate and agility index information, was developed.

C. Research Goals

As mention above, the purposes of our study were to provide a suitable and high accuracy method for heart rate monitoring in harsh conditions of athletes encounter; develop a new indicator that correlates well with conventional agility index; focus on the development of a small, light-weight, patched-type sensor module to collect vital information about health conditions of subject. The aims of this study are:

(1) To increase the accuracy of Heart Rate detection of athletes by using multiple techniques.

(2) To provide a new method to calculate agility index in flexible, highly-sensitive way.

(3) To develop a real-time, wireless system for health monitoring and evaluation base on effective algorithms executed by a sensor module and by the analysis software in a computer.

D. Thesis Organization

In Chapter I, background knowledge and the principle of ECG signal and agility index are introduced. Also, the reasons for conducting the research and its final target are presented. In Chapter Ⅱ, we present the common methods for detecting heart rate, and calculating agility index. The problems and limitation of each methods are discussed, and our proposed methods are described. The system we developed and the results of its performance evaluation are described in Chapter III. Our conclusion are provided in Chapter IV.

II. Algorithms for Monitoring Heart Rate and Agility Index

A. Heart Rate Detection Algorithm

Normally, an ECG signal contains several types of waves with differing periods, as shown in Fig. 3. The P wave represents the initiation of the heartbeat in the upper chambers of the heart (atria); the duration of the P wave is usually less than 0.12 second. The QRS complex corresponds to the lower chambers (ventricular) depolarization. Its duration is normally 0.04 to 0.12 second. The T-wave represents the recovery phase.

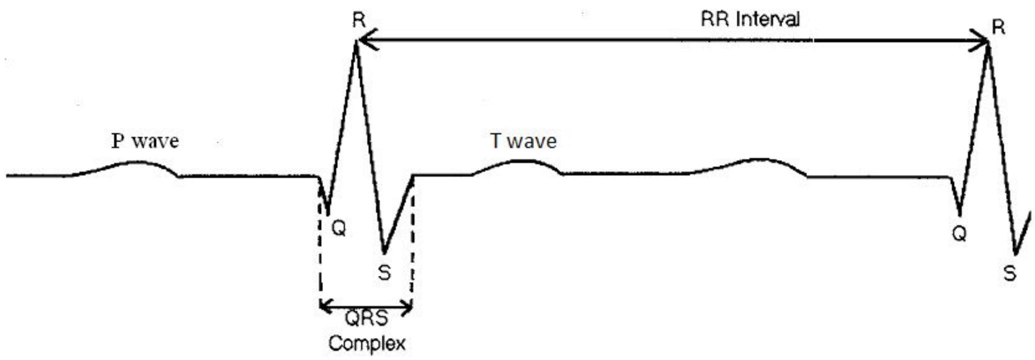


Fig. 3. The components of ECG signal.

In order to calculate heart rate, we must specify the position of the QRS complex and the RR interval precisely. (RR interval is the time that elapses between two consecutive R waves).

1. Problems in QRS Detection

Highly-accuracy QRS detection is difficult because various types of noises occur in the ECG signal. Muscle action, electrode motion, baseline instability can be the sources of noises. The occurrence of T waves with high frequency characteristics similar to those QRS complex can be obstacles for QRS detection. This is especially an issue for athletes engaged in high intensity activities, because the ECG may have a lot of noise. Unexpected noise with random amplitude and frequency easily can cause false detection. In addition, the physiological variability of the QRS complex requires more effort to detect heart rate. A detailed description of each noise signal and its characteristic is given bellow:

Motion artifacts: During the movement of subject, the motion of electrodes changes the electrode-skin impedance lead to the transient (not step) baseline changes. As electrode-skin impedance changes, the ECG amplifier recognizes a different source impedance, which form a voltage divider with the amplifier input impedance. Therefore, the amplifier input voltage depends on the source impedance. Normally, we consider motion artifacts as vibrations or movement of the subject. The shape of the baseline disturbance caused by motion artifacts can be assumed as a biphasic signal resembling on cycle of a sine wave. The peak amplitude varies up to 500 percent of peak-to-peak ECG amplitude, and the duration of the artifact ranges from 100 to 500 ms.

Electrode contact noise: Electrode contact noise is transient interference caused by loss of contact between the electrode and skin, which effectively disconnects the measurement system from the subject. Movements and vibration of subject bring the loose electrode in and out of contact with the skin, this loss of contact can be permanent or intermittent. The switching action at the input of measurement system can result in large artifacts since the ECG signal is usually capacitively coupled to the system. The 60-Hz interference may be significant as the amplifier input is disconnected. Electrode contact noise can be modeled as a randomly-occurring, rapid baseline transition (step), which decays exponentially to the baseline value and has a superimposed 60-Hz component. The transition may occur only one, or rapidly occur several times. The characteristics of this noise signal include the amplitude of the initial transition, the amplitude of the 60-Hz component, and the time constant of the decay. The duration can be 1 s and the amplitude is up to the maximum output of the recorder.

Electrical activity of muscles: Muscle contractions cause artifactual milivolt-level potentials to be generated. The baseline of electromyogram is in the microvolt range, so it is usually insignificant. The signals that result from muscle contraction can be considered as transient bursts of zero-mean, band-limited Gaussian noise.

Baseline drift and ECG amplitude modulation with respiration: The drift of the baseline with respiration can be represented as a sinusoidal component at the frequency of respiration added to the ECG signal. The frequency and amplitude of this component should be variable. The amplitude of the ECG signal also varies about 15 percent with respiration. The variation could be reproduced by amplitude modulation of the ECG and the sinusoidal component that was added to the baseline.

Sometimes atria premature contraction (APC) and ventricular premature contraction (VPC) occur in detected signal. The APC and QRS complex have similar wave form to a normal contraction while the VPC wave is difference from the shape of QRS. Using threshold method, one can recognize QRS and VPC waves, but cannot distinguish them because of the lack of information about wave form.

2. Related Works

Software QRS detection has been a research topic for long time, and, with the great advances in computer technology, the computational load is no longer a concern. Many new approaches of QRS detection have been proposed and implemented using computer such as algorithms from the field of artificial neutral networks, genetic algorithms and wavelet transforms. However, in this study, the algorithm for the detection of QRS was implemented in a battery-driven device. The computational load and power consumption need to be considered, and, for this reason we conducted studies to determine the strong points of the following approaches, and we incorporated those points in our algorithm, called the **First Derivative based Algorithm**: Holsinger *et al.* [5] proposed a QRS detection scheme base on first derivative, the equation they used is shown below :

$$Y_{(n)} = X_{(n+1)} - X_{(n-1)} \tag{1}$$

This array is searched until a point is found that exceeds the slope threshold $Y_{(i)} > 0.45$. QRS candidate occurs if another point in the next three sample points also exceeds the threshold:

$$Y_{(i+1)} \text{ and } Y_{(i+2)} \text{ and } Y_{(i+3)} > 0.45. \tag{2}$$

Amplitude and First Derivative based Algorithm: This algorithm was proposed by Gustafson [6]. The first derivative is calculated at each point of the ECG:

$$Y_{(n)} = X_{(n+1)} - X_{(n-1)} \tag{3}$$

The first derivative array is then searched for points which exceed a constant threshold: $Y_{(i)} > 0.15$, also the next three derivative value

$$Y_{(i+1)} \text{ and } Y_{(i+2)} \text{ and } Y_{(i+3)} > 0.15. \tag{4}$$

If all above conditions are met, and if the next two sample points have positive slope amplitude product

$$Y_{(i+1)}X_{(i+1)} \text{ and } Y_{(i+2)}X_{(i+2)} > 0 \tag{5}$$

point i is considered as a QRS complex.

Digital Filters based Algorithm: Okada *et al.* [7] proposed a QRS detection algorithm, in which the first stage smooths the ECG using three-point moving average filter:

$$Y0_{(n)} = [X_{(n-1)} + 2X_{(n)} + X_{(n+1)}]/4 \tag{6}$$

then the output is passed through a low-pass filter:

$$Y1_{(n)} = [1/(2m+1)]\sum_{k=n-m}^{n+m}(Y0_{(k)}) \tag{7}$$

the difference between input and output of the low-pass filter is squared:

$$Y2_{(n)} = (Y0_{(n)} - Y1_{(n)})^2, \tag{8}$$

then the squared difference is filtered:

$$Y3_{(n)} = Y_{(2)}[\sum_{k=n-m}^{n+m} Y2_{(k)}]^2, \tag{9}$$

finally, the fourth array is formed by:

$$Y4_{(n)} = Y3_{(n)} \text{ if } [Y0_{(n)} - Y0_{(n-m)}][Y0_{(n)} - Y0_{(n+m)}] > 0 \tag{10}$$

$$otherwise\ Y4_{(n)} = 0$$

the threshold is determined by:

$$Threshold = 0.125\max[Y4_{(n)}]. \tag{11}$$

The QRS complex occur when a point in $Y4_{(n)}$ exceeds the threshold.

3. Proposed Algorithm

As mention above, several methods have been suggested for eliminating noise in the ECG signal, with positive results. However, in our study we aim to find an effective method, that takes advantage of the strong points of these methods and that is suitable for low processing capacity of a patched-type sensor. The algorithm for detecting heart rate was focused on analyzing the athlete's heart rhythm during exercise.

We have redesigned a well-known, band pass filter-based R-wave detection algorithm to make it more robust during motion. This algorithm processes acquired Lead II signal according to the flowchart shown in Fig. 4.

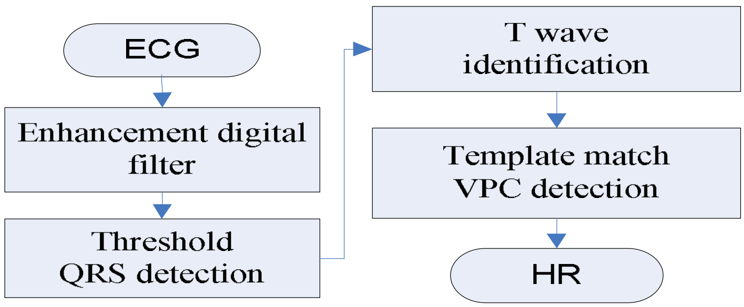


Fig. 4. The flowchart of heart rate detection.

The ECG signal is converted into digital signal by an analog-to-digital converter (ADC) with a sampling rate of 200 Hz, and this 12-bit A/D stores data for a period of 3 seconds before converting them to 8-bit data and sending them to processing components. The digitalized signal is passed through the enhancement digital filter stage to reduce the effect of noise and obtain other major information about QRS complex. The filter stage includes the following processing steps (Fig. 5).

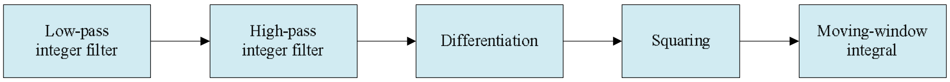


Fig. 5. Enhancement digital filter.

The combination of low and high pass creates an effective band-pass filter. The passband is up to the maximum energy of QRS complex, which is approximately 5 to 15 Hz. Because we only use the integer coefficient, which reduces the operation of the micro-processor by a considerable amount, the real-time filtering, and other calculations to identify the QRS can be implemented by micro-controller (MSP430F149). **Low-pass filter:** The transfer function of the low-pass filter is described by:

$$H_{(z)} = \frac{(1-z^{-6})^2}{(1-z^{-1})^2}. \tag{12}$$

The amplitude response is:

$$|H_{(wT)}| = \frac{\sin^2(3wT)}{\sin^2(3wT/2)}. \tag{13}$$

Where T is the sampling period (units). The difference equation of the filter is given by:

$$y_{(nT)} = 2y_{(nT-T)} - y_{(nT-2T)} + x_{(nT)} - 2x_{(nT-6T)} + x_{(nT-12T)}. \tag{14}$$

High-pass filter: The high-pass filter will subtract the low frequency components from the output signal of previous low-pass filter, the transfer function is:

$$H_{(z)} = \frac{(-1+32z^{-16}+z^{-32})}{1+z^{-1}}. \tag{15}$$

The amplitude response is:

$$|H_{(wT)}| = \frac{[256+\sin^2(16wT)]^{1/2}}{\cos(wT/2)}. \tag{16}$$

The difference equation is:

$$y_{(nT)} = 32x_{(nT-16T)} - [y_{(nT-T)} + x_{(nT)} - x_{(nT-32T)}] \tag{17}$$

The differentiation step will provide the QRS complex slope information, the transfer function is:

$$H_{(z)} = (1/8T)(-z^{-2} - 2z^{-1} + 2z + z^2.) \tag{18}$$

The amplitude response:

$$|H_{(wT)}| = (1/4T)[\sin(2wT) + 2\sin(wT)] \tag{19}$$

The difference equation is given by:

$$y_{(nT)} = (1/8T)[-x_{(nT-2T)} - 2x_{(nT-T)} + 2x_{(nT+T)} + x_{(nT+2T)}] \tag{20}$$

After differentiation, the signal is passed through squaring process in order to enhance the slope of the frequency-response curve of the derivative and reduce false positives caused by highly abnormal energy T wave. In this stage, the signal is squared point by point with the equation:

$$y_{(nT)} = [x_{(nT)}]^2. \tag{21}$$

Finally, the moving-window integral process was used to obtain information about slope and width of QRS complex. The equation of this process is

$$y_{(nT)} = (1/N)[x_{(nT-(N-1)T)} + x_{(nT-(N-2)T)} + \ldots + x_{(nT)}] \tag{22}$$

The width (number of samples) of moving-window is very important; it should be the same as the widest possible QRS complex in ECG signal. If the window is too narrow, the integrated waveform will produce several peaks. On the other hand, if the window is too wide, the QRS and T wave will be merge to form the integrated signal. These problems can cause difficulty for detecting the QRS complex in the next step.

The filtered signal is passed through an adaptive-threshold, QRS detection procedure to detect the QRS candidate (QRS^*). Through the QRS^* detection stage, thresholds are chosen and adjusted to float over the noise. Taking the advantages of high signal-to-noise provided by digital filter in previous stage, both low and high threshold are used. First, the higher threshold is used to detect QRS*, and, if there is no found signal within pre-defined period, the back-searching process conducted using the lower threshold. The details of this process are shown in Fig. 6.

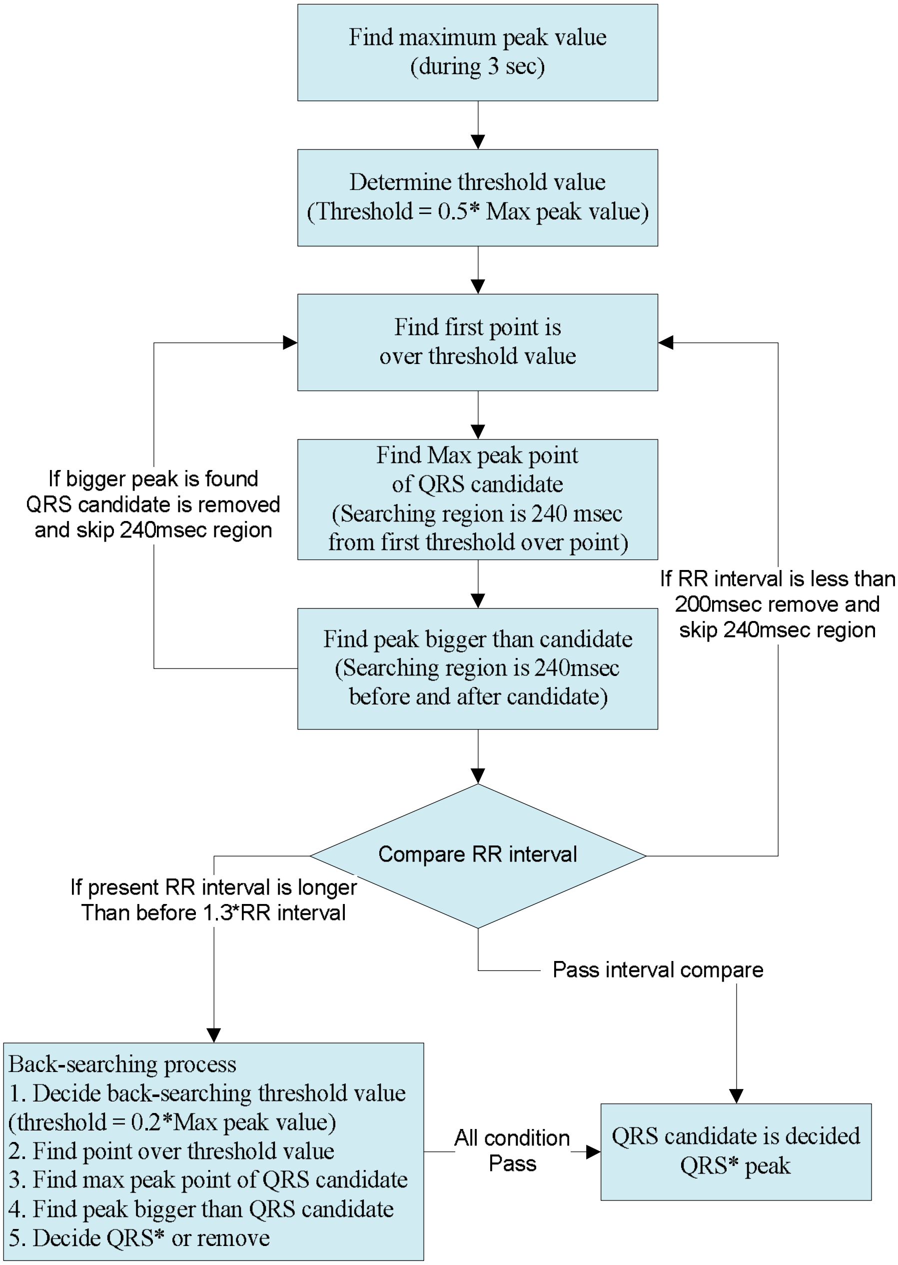


Fig. 6. Adaptive threshold QRS detection.

Find the maximum peak value: The input data are stored in three seconds, and, then max-value is selected by measuring the peak to peak. The high threshold is a half of max peak value. **Find the QRS candidate**: Search the max-value in 240-msec interval region from the first point over the threshold value. **Peak value comparison**: The comparison to other peak value is conducted, and the QRS candidate will be considered as a QRS complex if it is the only maximum value in the region of 240 msec before and after it. If there is a peak value greater than the QRS candidate, 240-msec interval region is skipped, and the candidate is removed. **QRS interval comparison**: The RR interval between the decided QRS candidate and previous QRS will be compared to the previous RR interval. If current RR interval is too short, it means that the T wave or other abnormal signal is detected. If current RR interval is longer than 1.3 times of the previous RR interval, there is a possibility of missing the QRS complex. So, the back-searching process is performed. **Back-searching process**: As back-searching process is conducted, the low threshold is decided by 0.2 times of the maximum peak value. After that, the decision of QRS complex is performed as above process. After going through the steps above, the QRS^* is decided and will be inspected again before being identified as a QRS complex by the template match stage. We calculate the cross correlation coefficients between QRS* and QRS template, defined as

$$Coe = \frac{\sum_{i=1}^{N}(x_i - X)(y_i - Y)}{\sqrt{\sum_{i=1}^{N}(x_i - X)^2}\sqrt{\sum_{i=1}^{N}(y_i - Y)^2}} \tag{23}$$

Where x_i and y_i are QRS^* and QRS template, respectively. The coefficient has the values between −1 and +1. The values that are approximately equal to 1 show a high similarity to the QRS template. On the other hand, the lower numbers refer to VPC, which has a different wave shape. To minimize the calculation, the window size N of QRS template only spans the duration of detected QRS* and it is updated after each QRS is decided. In the above steps, the QRS peaks are dropped if they are lower than 50% of the previous value. The QRS complex which does not show heart rate within 30 to 250 bpm is rejected. Finally, the algorithm calculates the heart rate.

B. Agility Index Algorithm

1. Physical Activities Monitoring

Human motion analysis has been primarily dominated by video camera. C. R. Taylor *et al.* [8] digitized the captured video of motion states, and the movements, can be enhanced further for analysis through motion analysis software. In a study conducted by K. Aninian *et al.* [9], physical activities were monitored based on accelerometers and compared with video observation. T. Ryan Burchfield *et al.* [10] developed a wireless sensor application that detects abnormal human movements, such as seizure. In another study [11], a kinematic sensor that was composed of a multi-axial accelerometer and gyroscope was used to monitor the daily physical activities of elderly people. J. Ng *et al.* [12] developed a device that can record information about body position along with ECG signal in order to aid clinical diagnosis. However, motion detection with multiple cameras requires complicated setup in a limited space, so it is unsuitable for monitoring people in outdoor or mobile environments. The above studies, which were related to body posture, were limited by the accuracy of the system. In order to acquire more accurate depiction of body posture and position, it would be necessary to mount more sensors at various places on the body.

2. Proposed Algorithm

We developed an agility index for evaluating the exercise performance and potential of athletes. This new indicator has high correlation coefficient with conventional agility index. In this study, the change of acceleration in exercise was detected by using a 3-axis accelerometer (MMA7260Q). With the fine-grained sensitivity of this sensor, we were able to adjust the sensitivity range at low gravity levels or to acquire more coarse-grained data acquisition at high gravity levels. Because our study aim was to measure the motion signal of athletes, the accelerometer was configured at -6G to 6G range. The proposed algorithm that was used to calculate the agility index was based on several digital signal processing techniques, as shown in Fig. 7.

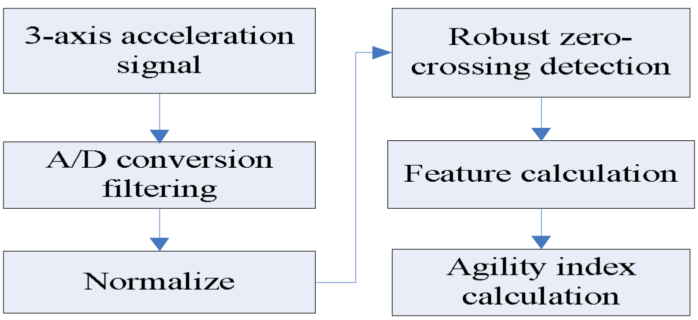


Fig. 7. The flowchart of the agility index calculation.

First, the acceleration signal is digitalized by using a 12-bit, analog-to-digital converter with a 100-Hz sampling rate. Because the signal is not noise free, a pre-processing stage was required to reduce the effect of noise. For digital filtering, a band-pass filter and the moving average were used.

For each axis, the accelerometer generates an analog voltage that is relative to the accelerometer force parallel to that axis. The g-value of each axis was computed independently and then combined into a single value that required less memory and computational resources of micro-controller. The root-mean squared formula is as follow:

$$g = \sqrt{g_x^2 + g_y^2 + g_z^2} \tag{24}$$

Base on the output characteristics of the MMA7260Q sensor, the detected signal was always positive. However, the real acceleration can be positive or negative, depending how the velocity changed. Therefore, the filtered signal was passed through a normalize stage, as shown in Fig. 8.

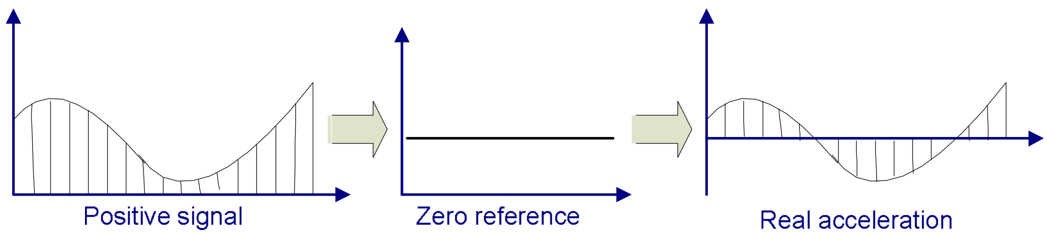


Fig. 8. Normalize stage.

In oder to reduce the effects of noises, the zero reference point is the output of accelerometer at non-movement condition. The accelerometer output varies from the minimum (V_{ddmin}) to the maximum (V_{ddmax}) value. The zero value is near to V_{ddmin} + (V_{ddmax} - V_{ddmin})/2. Values that are greater than the reference point represent positive values of acceleration, and the lower values represent deceleration.

In the above steps, zero-crossing detection is the most important step. Actually, "zero-crossing" is a common term in electronics, mathematics, and image processing. In math, this usually means the change of sign, e.g., from positive to negative, that occurs when the graph of a particular function crosses the axis (zero-value). In actuality, many noise factors affect the acceleration signal. So, we have to design the robust zero-crossing detector.

As shown in Fig. 9, our zero-crossing detection algorithm uses the following steps.

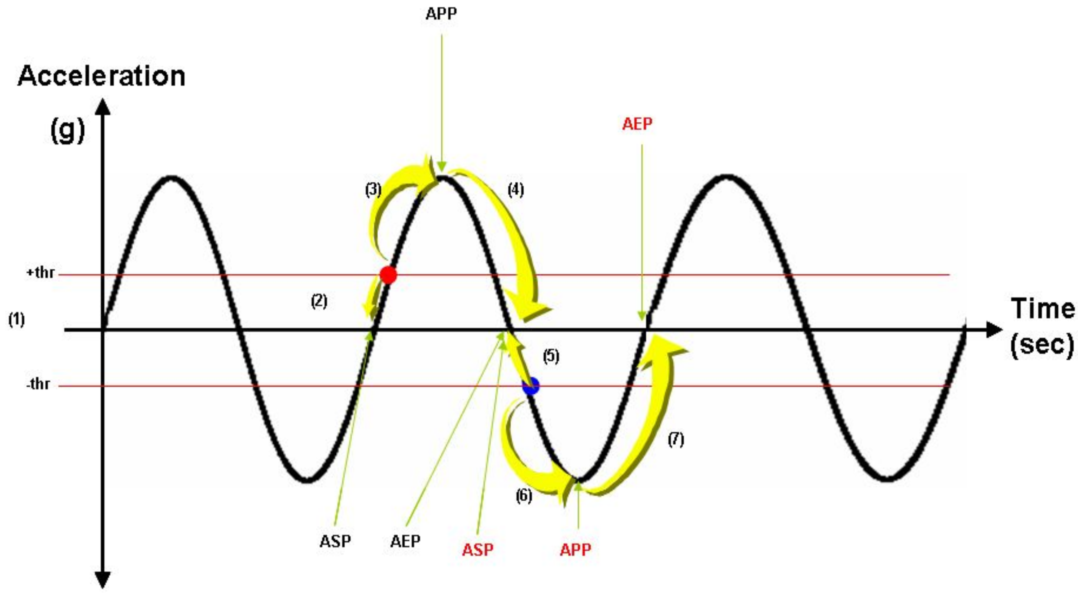


Fig. 9. The procedural drawing of the robust zero-crossing detection algorithm.

(1) Determine the positive and negative threshold (+thr and -thr) considering the noise margins.
(2) When signal strength goes above +thr, we can find zero-crossing point by decreasing sample index from the intersecting point. Then, the zero-crossing point is named as the Acceleration Start Point (ASP).
(3) After finding ASP, the Acceleration Peak Point (APP) can be detected as a peak value by increasing the sample index from +thr intersecting point.
(4) Then, we consequently find zero-crossing point as Acceleration End Point (AEP) by increasing the sample index.
(5), (6), (7) We reiterate these steps in the negative threshold, -thr.

With this procedure we can reduce the possibility of the extraction of the false value due to motion artifact noise. The results of algorithm's performance tests show that it achieved correct detections on the simulated acceleration signal 99% of the time. We can calculate feature values by using the extracted feature points that were provided by robust zero-crossing detection. The feature value is experimentally selected as tri-angular area of ASP, APP and AEP, since this area value shows the best correlation with agility performance test in zigzag run, side step test, burpee test and shuttle run test. Finally, agility index was defined as the average triangular areas of the graph of accelerations per second, as shown in Fig. 10.

We evaluated the new indicator by calculating the correlation coefficient between the agility index in this study and the conventional agility index used for exercise.

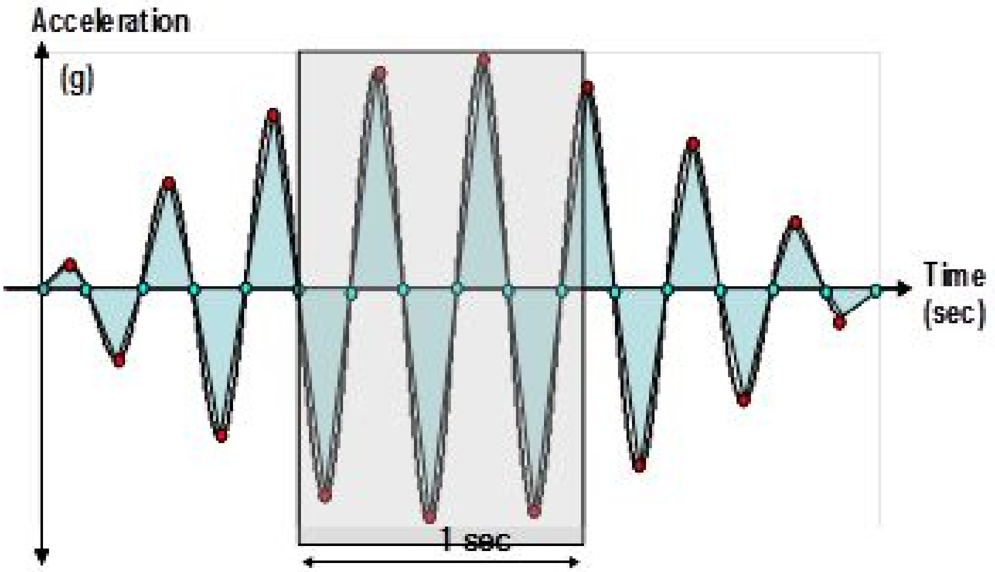


Fig. 10. The conceptual acceleration waveform with feature points, feature value (triangular area) and agility index (the sum of area per sec).

III. System and Performance Evaluation

A. AirBeat^{TM} System

Our health evaluating system includes a wireless sensor module with two proposed algorithms implemented and a computer with analysis software. The sensor is attached to the chest of the subject to measure heart rate and agility index. The measured information is sent to host computer where analysis software performs processing, analyzing, and displaying functions. The system is shown in Fig. 11. From left to right are the receiver, the sensor module, and the analysis software.

Fig. 11. AirBeat^{TM} system.

The patched-type sensor module for real-time monitoring of heart rate and agility index is shown in detail in Fig. 12.

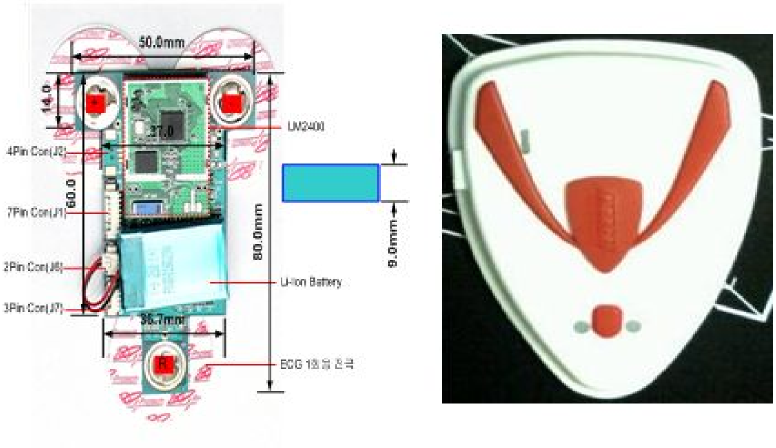


Fig. 12. The patched-type sensor module.

Table I shows the specification of sensor module

Table I. Specifications of the sensor module.	
Item	Description
System clock	4MHz
Program memory	64Kbyte flash memory
Data memory	2Kbyte RAM
AD converter	12-Bit internal reference
Sampling rate	200 sample/s
Program language	C
Compiler	IAR C/C++ compiler for MSP430
Power supply	DC 3.3V
Comm. module	Zigbee
Comm. distance	400m
Channel	8 ch

With the support of analysis software, a technician can observe both heart rate and agility index at the same time in real time or save subject's information for further offline analysis and evaluation. Detailed software specification are provided in table II.

Table II. Specifications of AirBeat™ software.

Compiler	MS Visual studio .NET
Program language	C#
PC OS	Microsoft windows XP Professional V.2002 Service Pack 2
PC	Intel pentium® 4 CPU 2.60 GHz
RAM	1.0 GB
Port	USB 2.0

Fig. 13 is an example of heart rate (line graph) and agility index (bar graph) displayed by AirBeat™ system.

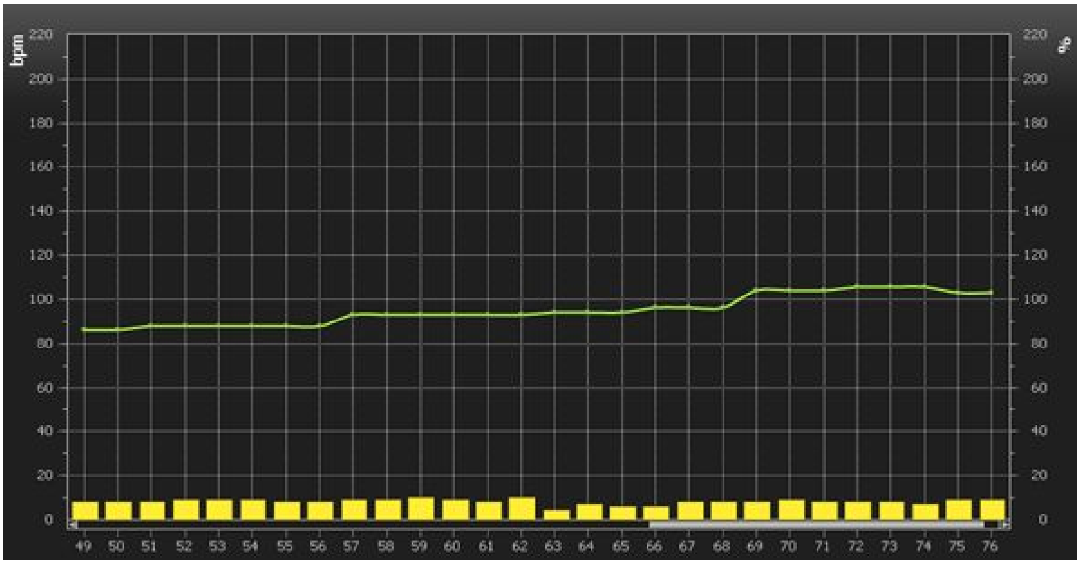


Fig. 13. Displayed information (heart rate and agility index).

B. Case System

CASE system, a commercial product of General Electric Medical Co., USA. is a well known system for heart rate monitoring and diagnosis.
The CASE system enables clinical excellence with outstanding data quality and accuracy in an easy-to-use system. Diagnostic support provided by the CASE system includes 1) full-disclosure data that enable the review and re-analysis of every beat and arrhythmia for enhanced clinical confidence and 2) risk-predicting algorithms that assist technician in predicting patients are at risk for sudden cardiac death.
The system is shown in Fig. 14.

Fig. 14. CASE system.

C. Evaluation of The Heart Rate Detection Algorithm

In order to evaluate the performance of proposed algorithm, we performed a test in which a runner's speed was change, and we monitored the test simultaneously with the reference system and with our system (AirBeat™), then compared the results.
We use the CASE system (GE medical, USA) as a reference system. Heart rate information was real time monitored and gathered simultaneously from both the reference system and our system. Fig. 15 shows the equipment setup.

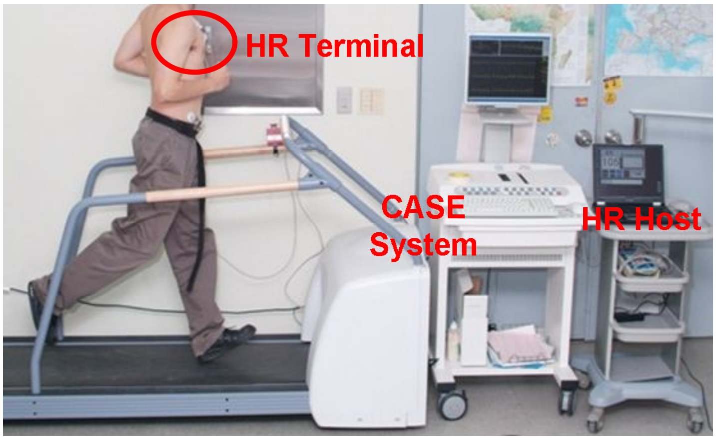
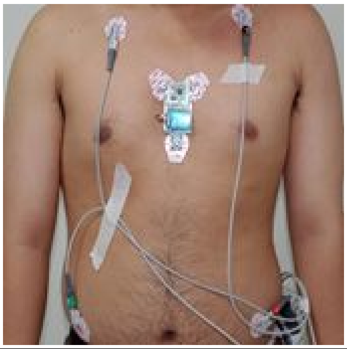


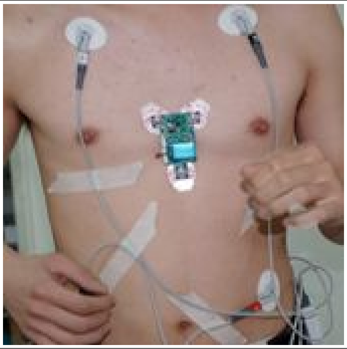
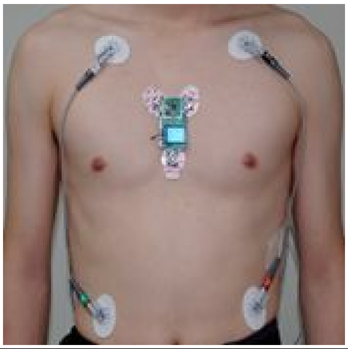
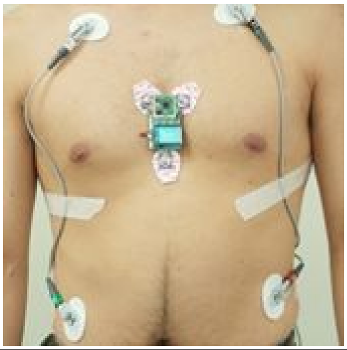
Fig. 15. Equipment setup for the comparative test.

During the test, five male subjects (Table III) at the ages 27 and 28, ran on the treadmill machine at speeds that ranged from 0.2km/h to 13km/h, with the speed increased incrementally as shown in Table IV.

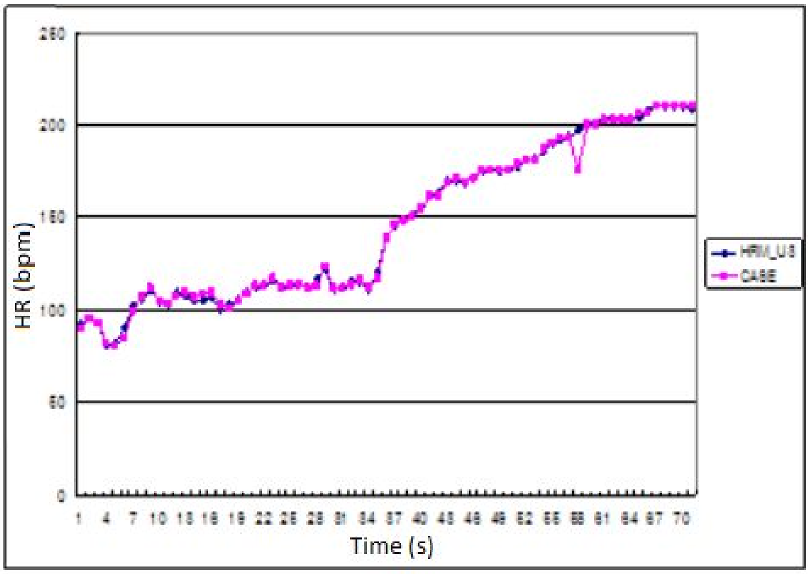
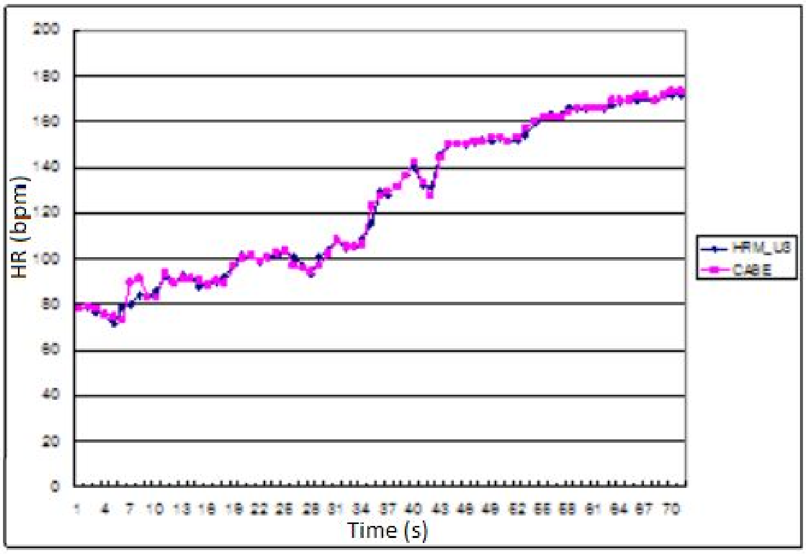
Table III. Subjects information.					
Subject	Gender	Age	Height	Weight	Heart disease
S1	Male	28	168cm	76kg	no
S2	Male	28	176cm	71kg	no
S3	Male	27	177cm	71kg	no
S4	Male	28	168cm	64kg	no
S5	Male	27	170cm	68kg	no

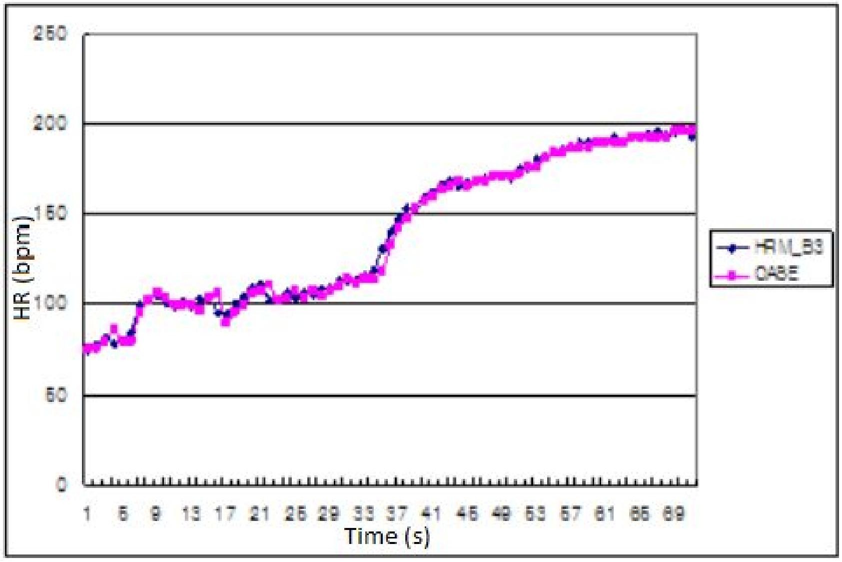
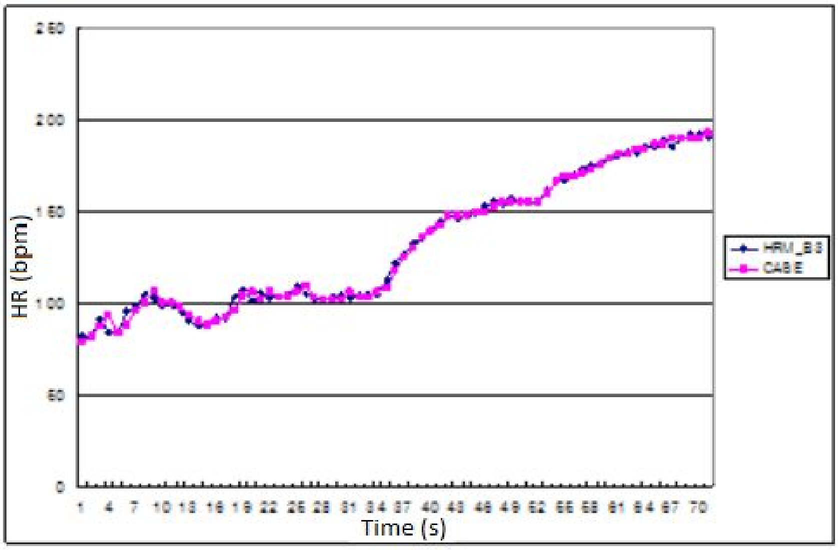
Table IV. Speed profile.		
Step	Duration (second)	Speed (km/h)
1	0 – 60	0.2
2	60 – 180	3
3	180 – 300	5
4	300 – 600	10
5	600 – 780	13

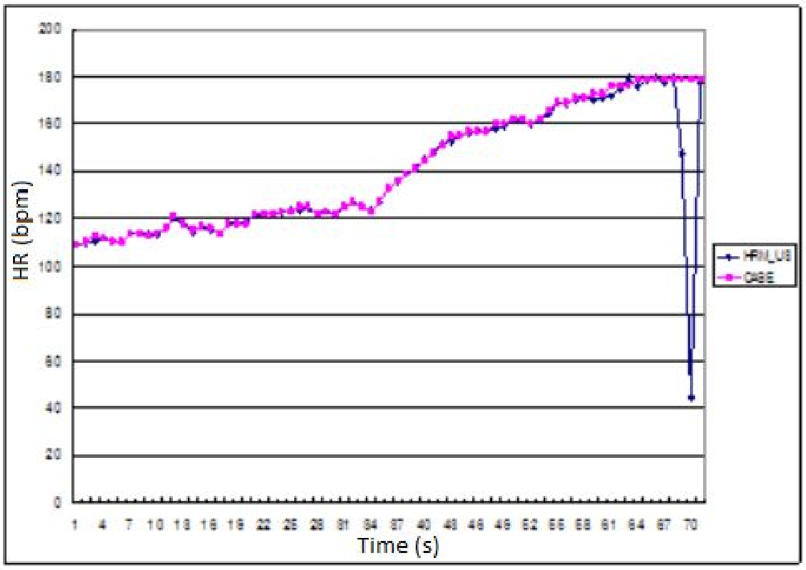
Table V. Electrode position on subjects.	
Subject	
S1	

S2	
S3	
S4	
S5	

The results in comparison with CASE system are shown as follow:

Table VI. HR comparison results.	
Subject	
S1	
S2	

S3	250 200 150 100 50 0 HR (bpm) 1 5 9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 Time (s) HRM_B3 OABE
S4	250 200 150 100 50 0 HR (bpm) 1 4 7 10 13 16 19 22 25 28 31 34 37 40 43 46 49 52 55 58 61 64 67 70 Time (s) HRM_B3 OABE

S5	

These results show that during wide range of exercise speed, AirBeat™ system can provide highly accuracy of heart rate detection that was within 6.2% difference (compared to the CASE system). The average errors are shown in Table VII.

Table VII. Average errors.	
Subject	Error
S1	1.2%
S2	1.4%
S3	5.2%
S4	1.5%
S5	6.2%

As mentioned above, CASE system is well-known for heart rate monitoring and diagnosis which provides highly accurate information about heart rate. However, CASE system uses wired electrodes which attached to body surface of subject to measure the heart rate. The wired electrode system only works in limited space (indoor) and causes inconvenient for subject. According to the results of the comparison our wireless system and CASE system which shown in table VII, one can see the error range is from 1.2% to 6.2%. It means the difference between our AirBeat^{TM} system and CASE system in measured heart rate is in acceptable range (less than 6.2%). In addition, our wireless system can provide a flexible way to measure heart rate. Without the constraint of wired electrode, our system can perform heart rate monitoring with high accuracy in outdoor condition.

D. Evaluation of Agility Index Algorithm

As mention above, in our approach to calculate agility index, detecting the zero-crossing is a very important process that extracts feature points of the acceleration signal. This function was evaluated by using simulated signal in an environment that included noise. We use a generator to simulate the signals with noises level control shown in Fig. 16. Our robust zero-crossing detector showed good extraction of feature points with 1% error.

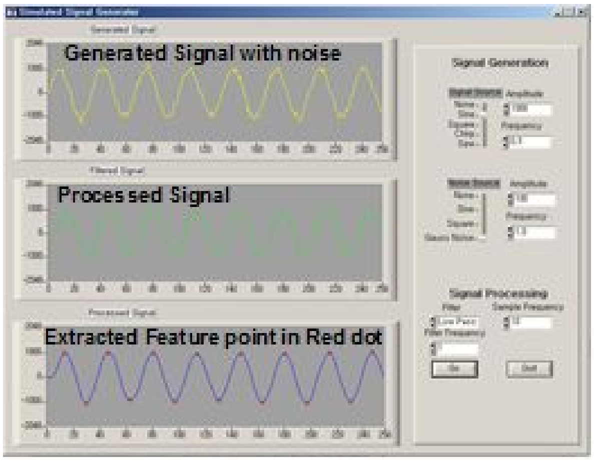


Fig. 16. Screen shot of the simulated signal generator program for evaluating the feature point extraction.

After zero-crossing detection, we conducted real-time monitoring of agility index. We evaluated the accuracy of our agility index that were determined based on the athlete's exercise states. In this experiment, 10 athletes performed several kinds of tasks, such as a zigzag run, the burpee test and side step, and a 20-m shuttle run. The zigzag run was a run in a zigzag pattern, using long strides with each step, swinging both elbows from side-to-side and leaning the body toward the direction of motion. The burpee test is a full body exercise that is performed in five steps, ie., (1) standing position, (2) drop into a squat position with hands on the ground, (3) kick foot back while lowering yourself without a pushup, (4) return foot to the squat position while straightening arms, (5) leap up as high as possible from the squat position with arms overhead. The Side-step test is performed by standing at a center of line, then jumping to the side and touching the line with the closest feet, jumping back to the center then jumping to the other side. The shuttle run is performed by sprinting as fast as possible from the starting line to the finishing line, picking up a block, and returning the block to the ground behind the starting line. In each task the conventional agility point and our agility index were determined by a supervisor (base on time or counts) and the AirBeat^{TM} system. We calculated the correlation coefficient between them, and the results are shown in table VIII.

Table VIII. Correlation coefficient between average agility index measured by AirBeat^{TM} and conventional agility point.

Test	Correlation coefficient
Zigzag run	0.90
Burpee test	0.89
Side step test	0.91
Shuttle run	0.80

Originally, the correlation coefficient was a statistical measure of the strength of the linear relationship between two variables. Positive values indicate a relationship between x and y variables, such that as values for x increase, values for y also increase. If x and y have a strong positive linear correlation, the correlation coefficient is close to +1.

With the correlation coefficients from 0.8 to 0.9, as shown in table VIII, it is apparent that the proposed agility index shows good correlation with conventional agility test. Therefore, it can be applied in sport and exercise programs for evaluating the athlete's exercise states.

E. Evaluation of Heart Rate and Agility Index Relationship Using the Bruce Protocol

In this step, we show the relation between heart rate and agility index. In keeping with the Bruce protocol, an exercise with multiple steps was performed. After each step (each of which had a duration of three minutes) the exercise intensity was increased.

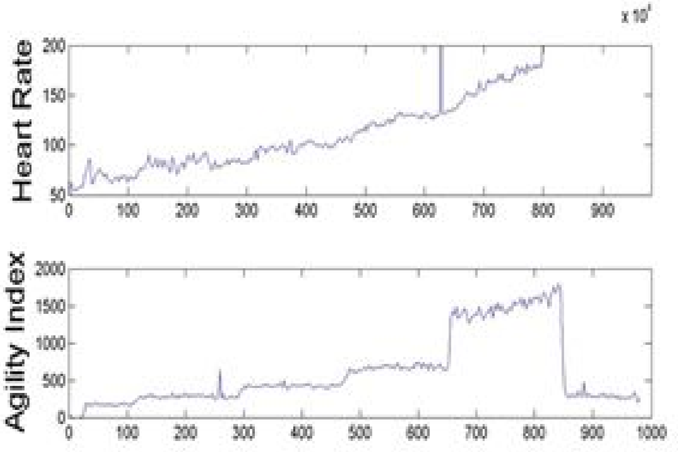


Fig. 17. The result field test using the bruce protocol.

Fig. 17 shows the result of the field test using the Bruce protocol. According to the test results, both the agility index and heart rate increase after each steps. The change of agility index can cause a corresponding change in heart rate. Heart rate and agility index can be monitored and measured accurately during the exercise.

IV. Conclusion

We confirmed that the minute measurement of heart rate and agility index was possible by using the patched-type sensor module that we designed. The proposed agility index showed good correlation with conventional agility test and it can be used to evaluate the athlete's exercise state. The change of the heart rate and the agility index according to the change of the exercise intensity was confirmed through the analysis of heart rate variability and 3-axis acceleration. The ability of the proposed module to detect heart rate and agility accurately using a wireless patched-type sensor provides a reliable and flexible way to monitor and evaluate the health status of athletes and others engaged in strenuous physical activity.

Our experiment showed that agility index is a vitally important parameters for determining the heart rate during exercises. Heart rate has been shown to be higher as agility index increases. Because of the ability to provide useful information about the status of the athletes during periods of exercise, we aim to apply our wireless system training, monitoring, and evaluating athletes in various sports, as well as an emergency forecast method for monitoring in-situ heart rate.

However, in the conditions when subjects were engaging in highly intense activities, the error rate of heart rate detection was still high, about 6.2% compared to the CASE system. Also, the current size of the sensor module (100*50*5 mm) is quite large, and it must be reduced in size for future applications. Therefore, in future work, we plan to increase the accuracy of heart rate detection and reduce the size of the sensor module.

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List of Publications

Luan Dinh, Li Meina, Ji Hwan Lee, and Youn Tae Kim, "Algorithm for real-Time wireless monitoring of heart rate and agility index," Proceedings of AMA-IEEE Medical Technology International Conference on Individualized Healthcare, Washington, DC. USA, Mar. 2010.

Li Meina, **Luan Dinh**, Ji Hwan Lee, and Youn Tae Kim, "In-situ monitoring of energy expenditure by the application of wireless patch type sensor module," Proceedings of AMA-IEEE Medical Technology International Conference on Individualized Healthcare, Washington, DC. USA, Mar. 2010.

Luan Dinh, Youn Tae Kim, "Algorithm for monitoring of heart rate and agility index apply in real-time wireless health evaluating system," Submitted to Computer Methods and Programs in Biomedicine.

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Although I tried my best to make this thesis as good and as meaningful as possible, I know there will inevitably be some mistakes and omissions. In that case, as always, it will be my pleasure to receive informative advise and thoughtful guidance from my professors, teachers, and friends.

Gwangju, May 2011

Luan Dinh

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논문제목	한글 : 심박수 및 운동량 모니터링 알고리즘 연구 영어 : A Study of Algorithms for Monitoring Heart Rate and Agility Index				

본인이 저작한 위의 저작물에 대하여 다음과 같은 조건아래 조선대학교가 저작물을 이용할 수 있도록 허락하고 동의합니다.

\- 다 음 -

1. 저작물의 DB구축 및 인터넷을 포함한 정보통신망에의 공개를 위한 저작물의 복제, 기억장치에의 저장, 전송 등을 허락함
2. 위의 목적을 위하여 필요한 범위 내에서의 편집ㆍ형식상의 변경을 허락함. 다만, 저작물의 내용변경은 금지함.
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4. 저작물에 대한 이용기간은 5년으로 하고, 기간종료 3개월 이내에 별도의 의사 표시가 없을 경우에는 저작물의 이용기간을 계속 연장함.
5. 해당 저작물의 저작권을 타인에게 양도하거나 또는 출판을 허락을 하였을 경우에는 1개월 이내에 대학에 이를 통보함.
6. 조선대학교는 저작물의 이용허락 이후 해당 저작물로 인하여 발생하는 타인에 의한 권리 침해에 대하여 일체의 법적 책임을 지지 않음
7. 소속대학의 협정기관에 저작물의 제공 및 인터넷 등 정보통신망을 이용한 저작물의 전송ㆍ출력을 허락함.

동의여부 : 동의(○) 반대( )

2011년 8월

저작자: 딘 루안 (서명 또는 인)

조선대학교 총장 귀하

AUSTRALIAN COMMISSION ON SAFETY AND QUALITY IN HEALTH CARE

PATIENT-CENTRED CARE:

IMPROVING QUALITY AND SAFETY BY FOCUSING CARE ON PATIENTS AND CONSUMERS

Discussion paper

Draft for public consultation

September 2010

This discussion paper is available on the website of the Australian Commission on Safety and Quality in Health Care (ACSQHC), www.safetyandquality.gov.au

The ACSQHC will be accepting written submissions up to 17 December 2010. Submissions marked ‘Patient- and Consumer-Centred Care’ should be forwarded to:

GPO Box 5480

SYDNEY NSW 2001

Or emailed to:

mail@safetyandquality.gov.au

Please be aware that in order to ensure transparency and promote a robust discussion, all submissions will be published on the ACSQHC website, including the names of individuals or organisations making the submission. The ACSQHC will consider requests to withhold the contents of any submissions made in whole or part.

Acknowledgments

This discussion paper is based on preliminary work undertaken on consumer engagement funded by the Australian Commission on Safety and Quality in Health Care (ACSQHC) and should be considered in the context of the ACSQHC’s development of a National Safety and Quality Framework and the National Safety and Quality Healthcare Service Standards.

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Contents

Acronyms and abbreviations

ABS	Australian Bureau of Statistics
ACSQHC	Australian Commission on Safety and Quality in Health Care
AHRQ	Agency for Healthcare Research and Quality
CAHPS	Consumer Assessment of Healthcare Providers and Systems
CMS	Centers for Medicare and Medicaid Services (United States)
CQUIN	Commissioning for Quality and Innovation payment framework
EBD	experience-based design
GP	general practitioner
H-CAHPS	Hospital Consumer Assessment of Healthcare Providers and Systems
IHI	Institute for Healthcare Improvement
IOM	Institute of Medicine (United States)
IPFCC	Institute for Patient- and Family-Centered Care (United States)
MCG	Medical College of Georgia
NHS	National Health Service (United Kingdom)
OECD	Organisation for Economic Co-operation and Development
P4P	pay for performance
WHO	World Health Organization

Executive summary

Patient-centred care is health care that is respectful of, and responsive to, the preferences, needs and values of patients and consumers. The widely accepted dimensions of patient-centred care are respect, emotional support, physical comfort, information and communication, continuity and transition, care coordination, involvement of family and carers, and access to care. Surveys measuring patients' experience of health care are typically based on these domains.

Research demonstrates that patient-centred care improves patient care experience and creates public value for services. When healthcare administrators, providers, patients and families work in partnership, the quality and safety of health care rise, costs decrease, and provider satisfaction increases and patient care experience improves. Patient-centred care can also positively affect business metrics such as finances, quality, safety, satisfaction and market share.

Patient-centred care is recognised as a dimension of high-quality health care in its own right and is identified in the seminal Institute of Medicine report, *Crossing the Quality Chasm,*[1] as one of the six quality aims for improving care. In recent years, strategies used in the US and UK to improve overall healthcare quality, such as public reporting and financial incentives, have emerged as policy-level drivers for improving patient-centred care.

In Australia, a patient-centred approach is supported by the Australian Charter of Healthcare Rights, the National Safety and Quality Framework, other national service standards, reports of state-based inquiries, and a range of jurisdictional and private sector initiatives.

Recent national health reform arrangements (such as the Performance and Accountability Framework of the 2010 National Health and Hospitals Network Agreement) provide further incentives to improve patient-centred care by linking it to performance and funding. Another driver for improving patient-centred care is the establishment of a National Performance Authority to report transparently on a range of performance indicators, including 'patient satisfaction' for every Local Hospital Network, public hospital, private hospital and primary healthcare organisation.

Against this background, Australian healthcare organisations are becoming increasingly interested in patient-centred care. Most organisations can readily put patient charters and informed consent policies in place, but many find it hard to actively change the way care is delivered, and struggle to involve patients and learn from their experience.[2] Key strategies from leading patient-centred care organisations include demonstrating committed senior leadership; regular monitoring and reporting of patient feedback data; engaging patients, families and carers as partners; resourcing improvements in care delivery and environment; building staff capacity and a supportive work environment; establishing performance accountability; and supporting a learning organisation culture.

Internationally, healthcare services use a range of strategies to promote patient-centred care, including staff development, leadership, collecting and reporting patient feedback, redesigning and co-designing service delivery, implementing patient rights charters, and engaging patients and carers as partners in improving care. A range of international organisations provide frameworks and tools to help organisations implement these strategies, such as the US-based Institute for Patient- and Family-Centered Care, and Planetree.

Based on these strategies and frameworks (and taking into account Australia's healthcare system, with its mix of public and private sectors), practical policy and service-level recommendations to foster patient-centred care are outlined in this discussion paper.

Summary of recommendations

Policy level

Recommendation 1	Policy makers and regulators should include patient-centred care as a dimension of quality in its own right in strategic and other policy documentation.
Recommendation 2	Patient survey tools should include a core set of items standardised at a national level to enable the collation and comparison of patient care experience data in key healthcare settings.
Recommendation 3	Patient surveys used to assess patient care experience need to include questions specifically addressing recognised patient-centred care domains and assess more than patient 'satisfaction'.
Recommendation 4	Implementation of healthcare funding models incorporating performance-based payments should include 'improving patient care experience' as an integral indicator of health service quality improvement.
Recommendation 5	To improve transparency, Australian policy makers and regulators should make data regarding patient care experience in health services publicly available via websites.

Organisational level

Healthcare service executives and managers should:

Recommendation 6	Ensure that systems are in place for the regular collection and reporting of patient care experience data through quantitative patient surveys and qualitative, narrative-based sources.
Recommendation 7	Ensure that organisational approaches to quality improvement include feedback about patient care experience — alongside clinical and operational data — when determining health service action plans.
Recommendation 8	Contribute to the evidence base for patient-centred care by recording and publishing changes in key organisational and patient outcome metrics over time.
Recommendation 9	Develop a shared patient-centred mission that senior leaders continually articulate to staff to promote the implementation of patient-centred care.
Recommendation 10	Develop and implement policies and procedures for involving patients, families and carers in their own care and, at a service level, in policy and program development, quality improvement, patient safety initiatives and healthcare design.
Recommendation 11	Ensure that the service meets the ACSQHC National Safety and Quality Healthcare Service Standard for 'Partnering with Consumers'.
Recommendation 12	Resource patient-centred changes to care delivery based on patient feedback and consumer input.
Recommendation 13	Implement training strategies tailored to building the capacity of all staff to support patient-centred care.
Recommendation 14	Focus on work environment, work culture and satisfaction of staff as an integral strategy for improving patient-centred care. Workforce surveys and review of staff recruitment and retention rates should be undertaken at regular intervals to monitor work environment.
Recommendation 15	Integrate accountability for the care experience of patients into staff performance review processes.
Recommendation 16	Foster a culture of learning within the organisation, equally learning from successes and failures, including tragic events, to promote patient-centred care.

Introduction

The health and health experiences of Australians compare well with those of other countries. Australia's life expectancy at birth remains among the highest in the world. Death rates are falling for many of our major health problems such as cancer, cardiovascular disease, chronic obstructive pulmonary disease, asthma and injuries, and survival from these conditions continues to improve.[3]

In a recent Commonwealth Fund comparison[4] of seven international health systems, Australia ranked highly demonstrating health care professionals' commitment to high quality care. However, this achievement should be tempered with the awareness that there is still work to be done. In 2008-09 a large proportion of the complaints made to Australian healthcare commissioners were about health care professionals' attitude and manner.[5] It has been suggested that these types of complaints may represent a failure to appreciate that in some circumstances the emotional need of the patient may be as important as their physical state.[6] If health care is to become truly responsive to the needs and desires of the patient then it will be necessary to refine the skills and capacity of health professionals as evidence for, and our understanding of, how to implement patient-centred care becomes clearer.

In 2009, the Australian Commission on Safety and Quality in Health Care (ACSQHC) released a proposed National Safety and Quality Framework[7] that identified 'patient-focused care' as the first of three dimensions required for a safe and high-quality health system in Australia. Including this dimension in the framework reflected a growing recognition of the importance of placing the patient at the centre of the healthcare system. When the ACSQHC talked to patients, carers, consumers, clinicians, managers and policy makers about the proposed framework, there was strong support for the inclusion of this dimension and an overall framework that will set the strategic direction for safety and quality in Australia over the next decade.

The proposed National Safety and Quality Framework contains a number of strategies for providing care that is respectful of, and responsive to, the preferences, needs and values of patients and consumers. The ACSQHC is currently developing a series of guides that will support use of the framework in practice.

In Australia there is wide and strong commitment to a healthcare system that focuses on the needs of patients and consumers. Many health services and health-service providers have taken steps to embed patient-centred care principles into practice, and provide care that addresses for the needs and desires of the patient, as well as their immediate treatment requirements. However, for a range of reasons others can struggle to carry out patient-centred care in practice and more specific and focused guidance in this area may be useful.

Internationally, there has been a focus on patient-centred care for some time, particularly in the US and UK. However, the health systems in these countries are different from the Australian system and questions arise as to whether the lessons and approaches that are used internationally can be applied here.

To address these needs, the ACSQHC has prepared this discussion paper as the first comprehensive review of patient-centred care for Australia. The purpose of the paper is to inform practitioners, managers and policy makers who want to re-orient their systems and practices to be more patient centred, and to work more closely with patients, consumers and carers. The paper recommends strategies for how to do this, and provides background information and helpful resources.

The paper provides an in-depth discussion of the concepts and evidence regarding patient-centred care; a comprehensive review of international approaches and activities; and an examination of some of the national, jurisdictional and other policies and activities in Australia to support patient-centred care. It focuses on strategies that healthcare organisations can use to support patient-centred care.

This is the consultation edition of this discussion paper. The ACSQHC wants to know how the approaches discussed in this paper may apply in Australia, whether the recommendations are useful, and how to move forward in supporting patient-centred care. Details about how to make a submission to this consultation process are provided in Section 6.

Within this paper, the term ‘patient-centred care’ is used, as this is the most commonly used term in research and literature in this area. The proposed National Safety and Quality Framework used the term ‘patient-focused care’ when it was released in 2009 and currently uses ‘consumer-centred care’. All these terms are conceptually similar, and emphasise the central roles of the patient, family, carer and consumer in individual aspects of care, and also in the broader approach to improving health service planning and delivery.

Much of the evidence and many of the examples provided in the discussion paper come from tertiary healthcare settings, as this has been the main focus and application of patient-centred care at a service level. However, many of the lessons and all the recommendations are equally applicable across the range of healthcare settings.

Section 1 of this discussion paper defines the concepts and origins of patient-centred care. Section 2 examines international approaches and initiatives to patient-centred care, and Section 3 examines its relevance to the Australian healthcare system. Section 4 highlights current patient-centred activity in Australia. In light of the international and current Australian approaches, Section 5 discusses the way forward for patient-centred care in Australia, and provides recommendations. Section 6 sets out the consultation process. The appendices include checklists to help organisations assess their readiness for improving patient care experience, and a list of patient-centred care resources.

1 What is patient-centred care?

This section provides an overview of the concept of patient-centred care as a dimension of high-quality health care, including commonly used terms and their definitions, and a summary of the evidence base for this approach.

> Patient-centred care is:
>
> 'is an innovative approach to the planning, delivery, and evaluation of health care that is grounded in mutually beneficial partnerships among health care providers, patients, and families. Patient- and family-centered care applies to patients of all ages, and it may be practiced in any health care setting.'[8]

1.1 Concepts of patient-centred care

There are numerous proposed definitions of patient-centred care that encompass many of the same core concepts, but there is no globally accepted definition.[9] Modern concepts of patient-centred care are based largely on research conducted in 1993 by the Picker Institute, in conjunction with the Harvard School of Medicine. This research identified eight dimensions of patient-centred care that were originally documented in the book, *Through the Patient's Eyes: Understanding and Promoting Patient-Centered Care*.[10] These dimensions are:

- respect for patients' preferences and values
- emotional support
- physical comfort
- information, communication and education
- continuity and transition
- coordination of care
- the involvement of family and friends
- access to care.

This framework clearly defined the patient's perspective for the first time and served as the foundation for the NRC Picker surveys measuring patient experience of health care.[11]

Patient-centred care also has a focus on staff. To succeed, a patient-centred approach should also address the staff experience, because the staff's ability and inclination to care effectively for patients is compromised if they do not feel cared for themselves.[12]

Organisation-specific concepts of 'patient-centred care' have also emerged. Some organisations identify individual elements of patient-centred care as part of an overall patient-centred care framework. An overview of leading organisations promoting strategies for patient-centred care is presented in Section 2.

The World Health Organization (WHO) uses the term 'responsiveness' in preference to 'patient-centred care'. Responsiveness describes how a healthcare system meets people's expectations regarding respect for people and their wishes, communication between health workers and patients, and waiting times.[13] WHO states that recognising responsiveness as an intrinsic goal of health systems reinforces the fact that health systems are there to serve people.

Several studies reviewing patient-centred care in the US[14] (cited in Goodrich and Cornwell 2008[15]) have identified the core elements in these frameworks as:

- education and shared knowledge
- involvement of family and friends
- collaboration and team management
- sensitivity to nonmedical and spiritual dimensions of care
- respect for patient needs and preferences
- the free flow and accessibility of information.

In reviewing definitions of patient-centred care more generally, Cronin[16] (cited in International Alliance of Patients' Organizations 2007[9]) identified the same core elements.

Similarly, Robb and Seddon[17] identified the following common concepts in definitions of patient-centred care:

- informing and involving patients
- eliciting and respecting patient preferences
- engaging patients in the care process
- treating patients with dignity
- designing care processes to suit patient needs, not providers
- ready access to health information
- continuity of care.

According to the International Alliance of Patients' Organizations, the most common element in definitions of patient-centred care is respect for the needs, wants, preferences and values of patients.[9] While some definitions acknowledge patients' rights and responsibilities (such as the New Zealand Ministry for Health's definition referred to later in this section), most do not, and safety is rarely included in these definitions or frameworks.[9]

In 2000, a five-day seminar was held in Salzburg, Austria, where 64 people from 29 countries examined what health care could become in a utopian land called PeoplePower. They envisaged informed and shared decision making, mutual commitments to quality and health outcomes, and patient partnership in governance.[18] The phrase 'nothing about me without me' was their guiding principle; this phrase has since been popularised by authors and regulators and is considered synonymous with efforts to advance a vision for patient-centred care.[19]

1.2 Some definitions of associated terms

Several terms are used interchangeably or associated with 'patient-centred care'; these are described below.

Consumer-centred care

The term 'consumer-centred care' is sometimes preferred to 'patient-centred care' to acknowledge that care should focus on people who are actual or potential users of healthcare services. For some, the term 'patient' has passive overtones. In contrast, the term 'consumer' is seen as a more active term, encompassing the need to engage people as partners in health service delivery.[20] The term 'consumer' also aligns with 'client' and 'user' in business and management models of service delivery.[21]

Person-centred care

The term 'patient-centred care' is often used interchangeably in primary care settings with terms such as 'person-centred care', 'person-centredness', 'relationship-centred care' and 'personalised care'. This term appears more frequently in literature on the care of older people,[22] and focuses on developing relationships and plans of care collaboratively between staff and patients.[22-23] This term values the needs of patients, carers and staff, with emphasis on the reciprocal nature of all relationships.[24-25]

Personalised care

'Personalised care' is the integrated practice of medicine and patient care based on one's unique biology, behaviour and environment. Personalised care uses genomics and other molecular-level techniques in clinical care; as well as health information technology, to integrate clinical care with the individualised treatment of patients.[26]

Family-centred care

This term emerged in the US in the 1980s in response to the needs of families with children who could not leave hospital. These families sought to work more collaboratively with healthcare professionals and successfully advocated for changes to enable them to care for their children in home and community settings.[27] More generally, children's hospitals in the US adopted the concept of family-centred care in recognition of input from parents and family members to improve the care of patients who were too young to tell physicians and nurses how they felt.[28-29] In the UK, family-centred care also relates to children's health care, and encompasses the concepts of parental participation; partnership and collaboration between the healthcare team and parents in decision making; family-friendly environments that normalise family functioning within the healthcare setting as much as possible; and care of other family members.[29]

1.3 The evidence for patient-centred care

Recent research has shown that there are many benefits to patient-centred care, broadly categorised as care experience, clinical and operational benefits. Studies show that when healthcare administrators, providers, patients and families work in partnership, the quality and safety of health care rises, costs decrease, and provider and patient satisfaction increase.[9 28 30-32]

A study by Stone[33] (cited in Charmel and Frampton 2008[34]) examined data for inpatient units at two similar hospitals in the US over five years. One hospital introduced an extensive program of patient-centred practices, and the other continued their usual practices. The study found that the patient-centred inpatient unit consistently demonstrated a shorter average length of stay; a statistically significantly lower cost per case; a shift in emphasis from the use of higher cost staff to lower cost staff; and higher than average overall patient satisfaction scores. Similarly, Iacono[35] found that patient satisfaction rates in 12 hospitals using the Planetree method (see Section 2.5) increased after implementing the patient-centred approach.

Other benefits associated with patient-centred care include decreased mortality,[36] decreased emergency department return visits, fewer medication errors, lower infection rates,[37] higher functional status,[38] improved clinical care,[39] and improved liability claims experience.[34]

In the care of patients with chronic conditions, studies indicate that patient-centred approaches can improve disease management; increase both patient and doctor satisfaction; increase patient engagement and task orientation; reduce anxiety; and improve quality of life.[32 40-41] Patient-centred care can also increase efficiency through fewer diagnostic tests and unnecessary referrals, and reduce hospital attendance rates.[32 40] A patient-centred care approach has been linked to improvements in long-term outcomes in cardiac patients.[42] Patient-centred care is therefore regarded as an integral component of preventative care.[40 43]

Increasing patient satisfaction through patient-centred approaches also increases employee satisfaction, and this, in turn, improves employee retention rates and the ability to continue practising patient-centred care.[34] According to Charmel and Frampton[34] the link between patient satisfaction and employee satisfaction is reflected in the fundamental philosophy of patient-centred care: the importance of staff feeling cared for themselves, so they can best care for their patients.

The Institute for Patient- and Family-Centered Care (IPFCC)30 states that patient-centred care has become the business model for the Medical College of Georgia (MCG) Health System in Augusta, Georgia, because it positively affects each of the MCG's business metrics (finances, quality, safety, satisfaction and market share). The following represents three years of quality improvement data:

- patient satisfaction increased from the 10th to 95th percentile
- volume of discharges increased by 15.5 per cent
- length of stay in neurosurgery decreased by 50 per cent
- medical error decreased by 62 per cent
- staff vacancy rate decreased from 7.5 per cent to 0 per cent
- perception of the unit by doctors and staff underwent a positive change.

Such findings led Charmel and Frampton[34] to conclude that patient-centred care is not merely philosophical, it is sound business practice.

1.4 Patient-centred care as a dimension of high-quality health care

Over the past two decades, patient-centred care has become internationally recognised as a dimension of the broader concept of high-quality health care. In 2001, the US Institute of Medicine's (IOM) Crossing the Quality Chasm: A New Health System for the 21st Century defined good-quality care as:[1]

- safe
- effective
- patient-centred
- timely
- efficient
- equitable.

It defined patient-centred care as 'care that is respectful of and responsive to individual patient preferences, needs and values, and ensuring that patient's values guide all clinical decisions'. The report set out several rules to redesign and improve patient-centred care, including ensuring that care is based in continuous, healing relationships; customising care based on patients' needs and values; ensuring the patient is the source of control; sharing knowledge and information freely; and maintaining transparency.

The IOM report outlined four levels that further define quality care and the role of patient-centred care in each level:

1. The **experience level** refers to an individual patient's experience of their care. At this level, care should be provided in a manner that is respectful, that assures the candid sharing of useful information in an ongoing manner, and that supports and encourages the participation of patients and families.
2. The **clinical micro-system level** refers to the service, department or program level of care. At this level, patients and family advisers should participate in the overall design of the service, department or program; for example, as full members of quality improvement and redesign teams and participating in planning, implementing, and evaluating change.
3. The **organisational level** refers to the organisation as a whole. The organisational level overlaps with the clinical micro-system level, in that the organisation is made up of various services, departments and programs. At this level, patients and families should participate as full members of key organisational committees for subjects such as patient safety, facility design, quality improvement, patient or family education, ethics and research.
4. The **environment level** refers to the regulatory level of the health system. Here, the perspectives of patients and families can inform local, state, federal and international agency policy and program development. These agencies, along with accrediting and licensing bodies, are in a position to set expectations and develop reimbursement incentives to encourage and support the engagement of patients and families in healthcare decision making at all levels.

According to Charmel and Frampton,[34] the IOM report reinforces patient-centred care not only as a way of creating a more appealing patient experience, but also as a fundamental practice for providing high-quality care in the US.

Like its evolution in the US, the concept of patient-centred care in the UK became a dimension of high-quality health care over time. From 1997, when the UK government began a 10-year program to reform the National Health Service (NHS), strategy documents and policies increasingly referred to the ambition to create a ‘patient-centred NHS’.[23]

By 2004, the NHS had set out its priorities for the next four years in its *NHS Improvement Plan: Putting People at the Heart of Public Services*.[23] The aim was ‘to ensure that a drive for responsive, convenient and personalised services takes root across the whole of the NHS and for all patients’ (cited in Goodrich 2009[23]). *Creating a Patient-led NHS: Delivering the NHS Improvement Plan*[44] outlined how the quality agenda could be delivered: patient-led services would ensure that patients were treated with respect, dignity and compassion.[23]

The 2008 *NHS Next Stage Review*[45] aimed to improve patients’ overall experience by putting ‘quality at the heart of what we do’. The review committed to providing ‘safe, personalised, clinically effective care’ and ‘locally-led, patient-centred and clinically-driven change’ (cited in Goodrich 2009[23]). In his introduction to the Next Stage Review, Lord Darzi stated: ‘High quality care should be as safe and effective as possible, with patients treated with compassion, dignity and respect. As well as clinical quality and safety, quality means care that is personal to each individual’.[45]

Recently, The King’s Fund[46] assessed the effect of NHS reforms on various factors within the service (accessibility, safety, promoting health and managing long-term conditions, clinical effectiveness, equity, efficiency, accountability and patient experience). The study was based on the best available evidence and measured improvements from patients’ perspectives. The authors found that, overall, the NHS had improved the standards for high-quality, safe health care. Waiting times for hospital care had been reduced, and access to primary care had improved. There had also been progress in making the NHS more accountable and transparent to government and taxpayers. However, there was still some way to go to firmly embed a strong safety culture in some parts of the service.

The effectiveness of efforts to improve patients’ experiences of the NHS presented a mixed picture in this study. Overall, patient experience scores were high and public satisfaction, as a whole, had been improving steadily since 2002. Patient experience had been built into regulation and many providers were monitoring survey results routinely. However, the surveys revealed several weaknesses, including limited progress in delivering greater choice of treatments, information about care, involvement with decisions, and problems with aspects of privacy for inpatients, especially mental health inpatients. To address this issue, The King’s Fund recommended that organisations find more effective ways of encouraging and acting on feedback from patients, carers and staff, ensuring that patients’ experiences have a real impact on the quality of care.[46]

In New Zealand, the Code of Health and Disability Services Consumers’ Rights Regulation 1996[47] (the Health and Disability Code) and the New Zealand Ministry of Health’s *Improving Quality* document[48] established patient-centred care as a priority.[17] *Improving Quality* talks about ‘people-centred’ rather than ‘patient-centred’ care and defines this concept as: ‘…the extent to which a service involves people, including consumers, their families and is receptive to their needs and values’. It includes participation, appropriateness and adherence to the Health and Disability Code, and adherence to other consumer protections, such as the Heath Information Privacy Code 1994.[17 48] According to Robb and Seddon,[17] by referencing the Health and Disability Code, this document acknowledges patients’ rights as integral to patient-centred care, a common omission in other definitions. In addition, the New Zealand Ministry of Health definition, like its IOM counterpart, implies extending patient-centred care beyond the patient–practitioner interaction. This reflects the importance of collaboration between patients, families, healthcare practitioners and hospital leaders in all aspects of health care and at all levels of the healthcare system.[17]

At the federal level in Canada, the term 'collaborative patient-centred care' is used to describe Health Canada's current initiatives in the primary healthcare sector to increase patients' self-care[49] and educate professionals about working together to achieve patient-centred care.[50] At the state and provincial level, many jurisdictions are adopting a patient-centred approach as part of their aim to increase healthcare quality.[51-52]

Overall, many regulators and organisations recognise patient-centred care as a key dimension of quality. Of the Organisation for Economic Co-operation and Development (OECD) member states, the US, UK, Canada and Australia include patient-centred care, patient focus or responsiveness as a dimension of healthcare quality in national documents and frameworks. The OECD and WHO included these concepts as a dimension of healthcare performance.[53] One study found that, in addition to the US, UK, Canada and Australia, several European countries — Germany, France, Denmark, the Netherlands and Switzerland — had all implemented various performance measures that include patient-centred health care.[54]

2 International approaches and initiatives

There is wide agreement about the need to place patients at the centre of their own care, and at the centre of the health system more generally. However, there is a need to highlight appropriate strategies for building and maintaining patient-centred care, as healthcare organisations often have difficulty implementing the type of change necessary to sustain patient-centred care.

This section presents an overview of the policies that drive patient-centred care, and the approaches and initiatives used successfully by international healthcare regulators and organisations to put patient-centred care into operation at different levels and in different sectors of the healthcare system. It includes a summary of the leading organisations that provide tools and resources, and a case study of the patient-centred approach adopted by the Medical College of Georgia (MCG).

A list of the leading organisational websites, together with other websites and resources mentioned in this section, are included in Appendix B.

2.1 Policy-level drivers for patient-centred care

Internationally, there has been a focus on patient-centred care for some time, particularly in the US and UK. These countries have three main policy-level drivers for patient-centred care:

- mandatory government requirements for service providers to collect and publish patient experience data (other jurisdictions, such as Canada and New Zealand, also collect patient experience data, although there is no mandatory requirement to do so)
- publicly available information that enables consumers to choose between service providers
- financial incentives for providers who achieve high measures of patient-centredness.

The different approaches to these policies are discussed below.

Collecting and publishing patient experience data

The UK, the US and some European countries have implemented patient survey programs to systematically collect patient and carer experience feedback at a national level.

United Kingdom

In the UK, the government's vision is for patients and the public to drive the design and delivery of high-quality services. In relation to health care, this will be achieved through 'putting patient experience centre stage[45 55] and a number of regulatory requirements are currently in place to meet this aim. Measuring patient experience is a requirement under the Public Service Agreement Target on patient experience and the Care Quality Commission assessment process.[55-57] It has also been made an equal partner in the *National Health Service (NHS) Quality Framework*, alongside effectiveness; and is a 'vital sign' in the *NHS Operating Framework* for 2010–11. Patient experience is a standard measurement in the NHS Indicators for Quality Improvement, Measuring for Quality Improvement, Quality Accounts and local Commissioning for Quality and Innovation (CQUIN) schemes.

In the UK and Europe, patient experience surveys are conducted using standardised questionnaires and methods based on Picker Institute Europe's methods. Picker Institute Europe develops, coordinates and implements healthcare surveys based on the eight Picker dimensions of patient-centred care (see Section 1.1). Information and some findings from a range of NHS surveys can be found on the Care Quality Commission's website and the Survey Coordination Centre website (see Appendix B).

The UK Department of Health also conducts the General Practice Patient Survey. This survey includes questions on many aspects of primary care patients' experiences, although it focuses strongly on access and has limited questions on experience of the consultation. Various disease-specific patient experience surveys have also been conducted in the UK, including the National Cancer Patient Survey Programme.

United States

In the US, the Centers for Medicare and Medicaid Services (CMS) and the Agency for Healthcare Research and Quality (AHRQ) routinely collect data on patient experience from service providers. This has been expanded under the recently enacted *Patient Protection and Affordable Care Act 2010*.[58]

The US has implemented the Consumer Assessment of Healthcare Providers and Systems survey (CAHPS), developed at Harvard University and based on Picker principles of patient-centred care. The CAHPS program develops and supports a comprehensive and evolving family of standardised surveys that ask patients to report on and evaluate their healthcare experiences. Hospitals use the hospital-CAHPS (H-CAHPS) survey to assess seven domains: communication with doctors; communication with nurses; responsiveness of hospital staff; communication about medicines; pain control; cleanliness and quiet of physical environment; and discharge information. The survey provides an overall rating and assesses the likelihood or willingness of patients to recommend the service to others. Other versions of the CAHPS survey are used by clinician group practices and health insurance plans, and a shorter, real-time version has also been developed.

The National CAHPS Benchmarking Database (also referred to as the CAHPS Database) is the national repository for data from CAHPS surveys. It holds 11 years of data and is one of the key resources for the AHRQ's *National Healthcare Quality Report* and *National Healthcare Disparities Report*. These annual documents help policy makers monitor the nation's progress toward improved healthcare quality.

Various patient experience surveys that focus on specific diseases are also emerging in the US; for example, the National Cancer Institute and the AHRQ are developing a patient care experience survey for cancer patients.

Canada

Some provinces and healthcare organisations in Canada also conduct CAHPS surveys. Statistics Canada measures aspects of patient satisfaction with health services,[59] but there is no standardised system to compare quality of care between health services in Canada.

Performance reporting and choice of providers

Both the UK and the US have public web-based systems that provide information on health service providers' performance, to allow consumers to choose which provider they wish to access. A brief overview of these systems is presented below.

United Kingdom

Since April 2008 in the UK, all patients who need a specialist referral have had the right to choose to go to any hospital, including many private and independent sector hospitals. The

NHS Choices website[a] publishes information on the specialist options available and performance indicators from the national survey data for each hospital. It also allows patients to rate and comment on their healthcare providers online. Regular surveys of patients' experience of choice provide information about the availability and uptake of provider choice.

NHS organisations are now required to obtain real-time patient feedback to improve care and service quality. Several websites are exploring web technology to achieve this. The websites help patients and carers find out what other people think of local hospitals, hospices, mental health services, general practitioners and other individual clinicians. People can submit their stories about their experience when they were ill and their comments on the services, for publication on the website. Providers can automatically direct these to the email of a responsible staff member.

United States

The US CMS has implemented a web-based tool called 'Hospital Compare'[b]. This tool sets out information about how well hospitals care for patients with certain medical conditions or surgical procedures, using data from the Hospital Outcomes of Care Measures, CAHPS and H-CAHPS surveys. CMS is also planning to measure individual physicians on quality and patient experience metrics and publish the results on a new Physician Compare site, under the newly enacted *Patient Protection and Affordable Care Act 2010.*[58]

Financial incentives

Public and private healthcare sectors around the world are now linking service quality with provider payment. Patient care experience is a key outcome metric for quality reporting, and financial incentives can now help drive services towards patient-centred care. Both the UK and the US provide financial incentives to some healthcare providers for adopting improved quality practices, including clinical outcomes and some patient-centred care principles.

This section focuses on the role of 'pay for performance' (P4P) in driving organisations to focus on patient experience. P4P is defined as 'financial incentives that reward providers for the achievement of a range of payer objectives, including delivery efficiencies, submission of data and measures to payer, and improved quality and patient safety'.[60]

United Kingdom

In 2004, the NHS began a major P4P initiative — the Quality Outcomes Framework (QOF). In this framework, general practitioners (GPs) receive income increases based on their performance, which is measured using 146 quality indicators that cover organisation of care, clinical care for ten chronic diseases, and patient experience.

Early analysis of the QOF[61] showed that, overall, family practices achieved high scores in their performance indicators in the first year. Practitioners included in the study earned an average of AU$40 000 more by collecting nearly 97 per cent of the points available. However, a small number of practices achieved high scores by excluding large numbers of patients through exception reporting. More research is needed to determine whether practices like these exclude patients for sound clinical reasons, or to increase their income.[61]

It is unclear whether the high performance scores reflect improvements that were already underway, or whether improvement was accelerated by the introduction of the scheme.[62]

[a] www.nhs.uk
[b] www.hospitalcompare.hhs.gov

Overall, the QOF has seen modest improvements in measured quality of care,[63] but has had little effect on inequalities in chronic disease management.[64]

The NHS introduced a new CQUIN framework to improve the quality of care in hospitals and other healthcare organisations. The CQUIN makes a proportion of providers' income conditional on quality and innovation. To earn from the CQUIN, providers of acute, ambulance, community, mental health and learning disability services that use national contracts must agree on a CQUIN scheme with their funding body. The CQUIN schemes must include goals in the three NHS domains of quality (safety, effectiveness and patient experience) and must reflect innovation. A CQUIN scheme is the agreed package of goals and indicators that, when achieved, earn the provider its full CQUIN payment (1.5 per cent of contract value in 2010–11). From 2011–12, providers that fail to meet agreed patient satisfaction goals may have a proportion of contract payment withheld (up to 10 per cent over time).

United States

The US provides financial incentives to physicians and hospitals that provide data on quality measures, including patient experience, to the CMS.

In 2008, the CMS introduced the Reporting Hospital Quality Data for Annual Payment Update Initiative to strengthen the relationship between payment and quality of service. Under this scheme, hospitals publicly report a range of inpatient data, including quality, mortality and H-CAHPS patient care experience items, via Hospital Compare. In April 2010, the CMS announced that acute care hospitals that choose not to participate in the voluntary reporting program or do not meet the established reporting deadlines, will have a 2 per cent reduction in reimbursement for their Medicare patients (those who are 65 years and over or who meet other special criteria).

For individual physicians, the *Tax Relief and Health Care Act 2006*[65] established a physician quality reporting system, including an incentive payment for eligible physicians who report on quality measures for services covered by Medicare. Payments are also available for better care coordination between home, hospital and offices for patients with chronic illnesses. Ambulatory surgical centres and other health organisations must also comply with Medicare P4P reporting and performance targets under the *Patient Protection and Affordable Care Act 2010.*[58]

Many private providers and purchasers have embraced P4P approaches as an essential way to meet quality improvement goals. For example, in 2009, the large private insurer, Blue Cross Blue Shield Massachusetts, introduced an 'alternative quality contract' with healthcare providers. Participating providers receive performance-based incentives for a broad set of quality and outcome measures for outpatient and inpatient care. Organisations that meet defined targets can earn up to an additional 10 per cent of their total budget. The same performance incentives and measures are used for every provider, including nationally accepted measures of clinical processes, clinical outcomes and patient experiences.

2.2 Emerging views on the appropriate outcomes of services

Patient-centred health care is part of the emerging views on the appropriate outcomes of services. These views include the need to create public value, the need to focus on the experience of care, and the need to improve efficiency.

Creating public value

One of the common themes resulting from high-profile public inquiries into the health systems in Australia (see Section 4) and the UK was a reported loss of trust in administrators and clinical colleagues from patients and the community.[66] To regain public trust and confidence, public sector organisations are adopting a 'public value' approach to service delivery. Public value is the return created by public services for the taxes that people pay. The expected return is more than simply 'value for money'. Consumers expect to be able to trust those delivering the services, the actual service delivered, the way they are included in the process of creating and delivering the service, and the measurement of any process or product of delivery from the service.[67] The power of a public value approach lies in its advocacy of a greater role for consumers in decision making. It also drives public managers to seek out what consumers want and need, creating a quality service.[68]

Research conducted by the Work Foundation in the UK and sponsored by the NHS Institute for Innovation and Improvement and the Quality Improvement Agency cites a range of methods and activities that enable public services to involve consumers and create public value.[69] These are in line with the patient-centred activities outlined in Section 2.3 and include formal mechanisms such as facilitating patient engagement on committees and boards, client surveys, and providing information and improving communication and interactions with consumers.

In the UK, the NHS uses collaboration with people who use their services to create public value. NHS organisations are required to regularly and systematically collect and analyse feedback from service users to inform decisions about service commission and delivery.[55] In addition, public services are empowering individual citizens to shape their services by establishing local networks that support the involvement of consumers and patients in commissioning, providing and scrutinising local care services. The *Local Government and Public Involvement in Health Act 2007*[70] empowers local networks to canvas the views of patients and citizens on their need for and experience of care services, and to make recommendations to health services based on those experiences. Furthermore, overview and scrutiny committees question whether services meet the needs of local people and whether the experience of patients is leading to improvements in service delivery

Valuing the experience

Another approach to seeking out what consumers want and need comes from industry models of business economy theories. In 1999, Harvard Business School published Pine and Gilmore's book, *The Experience Economy*.[71] This work proposed that society was moving away from the industrial economy view to a modern view, where the 'product' that a customer receives in a service economy is 'the experience'. In this new economy, organisations create engaging, personal and memorable experiences for customers. The application of this approach to the healthcare sector leads to a greater focus on patient care experience as an outcome of the 'service' received. Such approaches in health care are typified by work such as *If Disney Ran Your Hospital*,[72] which emphasises that 'hospitals would do well to emulate the most vital things that earn Disney the love of their guests and employees', as Disney does not provide a service — it provides an 'experience'.

Driving efficiency gains

The need for improved efficiency in service delivery is also driving the engagement of patients and consumers more generally. Limited resources in the form of underfunding, low staffing levels and low morale in already overstretched systems are a perceived barrier to the practice of patient-centred care.[9] However, recent research indicates that a patient-centred approach can make health service delivery more efficient.[34] By harnessing consumers' contributions, insights and collaboration, patient-centred care is becoming a critical strategy for public services to deal with rising service costs.[73-74]

2.3 Approaches and strategies to promote patient-centred care

Internationally, a variety of approaches are used to promote patient-centred care. A review by the Picker Institute Europe identifies the most effective strategies ('best-buys') for facilitating patient-centred care, and numerous other approaches are also available.

'Best buys' for improving patient experience

The Picker Institute Europe reviewed the body of evidence for strategies to engage patients and consumers in health care ('the Picker review').[75] They evaluated 31 systematic and high-quality narrative reviews on various initiatives to improve patients' experience, including studies of direct and indirect feedback from patients (including patient experience and satisfaction surveys); service user involvement in evaluations; consultation styles; and communication skills training. Each study was graded according to four outcomes: impact on patients' knowledge, impact on patients' experience, impact on service use and costs, and impact on health behaviour and health status.

According to the Picker review, the most effective ways (best buys) to improve patient experience are patient-centred consultation styles and communication training for health professionals; and patient feedback (surveys, focus groups, complaints) with public reporting of performance data. The body of evidence for these strategies includes:

- Impact on patients' knowledge: training health professionals to communicate information about medicines improves patients' knowledge and understanding. Longer consultations in primary care can increase patients' confidence to take action in relation to their health. Educational material can be helpful for carers.
- Impact on patients' experience: patient surveys can stimulate quality improvements, but provider organisations need additional help to implement changes. Patient feedback surveys need to be well planned and carefully implemented. Patient-centred communication and longer consultations in primary care increase patient satisfaction. Communication skills training for clinicians can lead to improved communication, reduced anxiety and greater patient satisfaction.
- Impact on service use and costs: public reporting of hospital performance data can stimulate providers to improve quality. If it is well disseminated and published in a format that patients can understand, this type of information influences public perceptions of a hospital's reputation, making it more likely that patients will want to go there. One review suggested that improved continuity of care reduces costs.
- Impact on health behaviour and health status: communication skills development for clinicians may improve health outcomes, but some reviews have reported conflicting findings. Reviews of patient-centred consultations found mixed results in relation to impact on health status.

Consultation styles and communication training

Regulators and professionals in the UK, US and Canada are investigating the effects of patient-centred consultation styles on continuity of care, length of consultation and increased support for patients on nonclinical issues and concerns.[75] Recent research in Canada found that the degree to which emergency department staff are courteous, particularly to patients in pain, is the key driver of patient ratings of overall quality of care.[76] Research in the UK and US also demonstrates that overall patient satisfaction strongly correlates with patients' assessment of clinicians' interpersonal skills.[77-78]

To improve these outcomes, initiatives to develop communication skills, such as training courses and coaching strategies that teach staff about the need to establish a connection with patients, have been introduced successfully in a number of settings.[12] Strategies include developing and using verbal communication guidelines for staff, scripting tools, and cues for effectively communicating with patients. *The Patient-Centered Care Improvement Guide*[12] provides examples of scripting tools and cues. Nonverbal tools include patient and family communication boards with information such as the care team member names and the date and time scheduled for specific procedures.

A successful communication strategy that has facilitated patient-centred care is the 'just ask' campaign conducted at Northern Westchester Hospital in the US. Hospital staff posted a sign in every patient room encouraging patients and families with the message, 'if you're thinking it ask it'. Displaying photographs of care givers, distinguishing clothing for staff, and follow-up discharge phone calls are further examples of communication strategies to support patient-centred care.[12]

Patient feedback reporting

As described in Section 2.1, regulators collect patient experience survey and complaints data to improve patient experience at all levels of health care. One-off patient focus groups can also be used as a source of in-depth information.[12] In hospitals, staff can gather 'real-time' information from patients by engaging them in a conversation about their stay during patient safety rounds. The Dana-Farber Cancer Institute has developed an online toolkit to help organisations initiate patient safety rounds. The UK Department of Health publication, *Understanding What Matters — A Guide to Using Patient Feedback to Transform Services,*[79] sets out best practice in terms of collecting, analysing and using patient feedback. Links to these resources are provided in Appendix B.

Patient and carer engagement in personal care

Health service providers are adopting strategies that facilitate family and carer engagement to increase their patient-centred practices.[40] The family or carer knows the patient best and their presence can help reassure patients in times of uncertainty, anxiety or vulnerability. Family members and carers can also provide information about the patient's history, routines or symptoms that may assist in their treatment. Strategies include patient-directed visitation, where restrictions on visiting times are removed and the patient decides the visiting times that best suit them. This strategy benefits overall patient experience and decreases anxiety by 65 per cent.[12] Related strategies include patient or family-initiated rapid response teams, where patients and their families are encouraged to alert staff of changes in a patient's condition; and care partner programs that aim to meet the personal, emotional, spiritual, physical and psychosocial needs of the patient by encouraging members of their support system to be involved in their care.[12]

Patient and carer access to information and education

Patient-centred approaches to empowering patients and families through information can include sharing medical records. Traditionally, medical records have remained the property of the health service providers, but some providers now use medical records as a teaching tool and a tool to encourage patient and carer engagement. The medical record is brought to the bedside during treatment and the results of clinical tests and procedures are shared with the patient. Future treatment plans then involve patients, carers and providers deciding the best course of treatment together.[12] Some providers have also introduced patient progress notes, which give the patient a formal place in the medical record to note any comments or concerns, further encouraging a partnership approach to care.[12]

The Picker review found that printed and electronic information, and educational programs, can benefit patients' knowledge and understanding of their condition. Among cancer patients, question prompt sheets and audiotapes of consultations improve the recall of medical information given in clinical consultations. Providing written and electronic information can increase a patient's sense of empowerment. It can also improve their ability to cope, increase satisfaction and may help to reduce anxiety. Among stroke patients, studies show that information strategies that actively involve patients and their care givers are more effective in reducing anxiety and depression. Telephone help lines, telecounselling and telemonitoring can also reduce social isolation, increase decision-making confidence and self-efficacy, and improve satisfaction.[75]

Pre-operative and pre-discharge information may help to reduce consultations, length of stay in hospital and follow-up visits, with a positive effect on service use and costs. Studies on the effect of targeted information and teleconsultations on diagnostic accuracy, consultation rates, waiting times and out-of-pocket costs to patients show conflicting results. Telephone reminders can help to increase attendance rates and improve medication adherence, and reminder packaging may improve adherence to self-administered long-term medication.[75]

Personalised patient information packets for patients to take with them on discharge are also useful.[12] These provide customised health information to meet the specific needs of the patient and family. Packets may include fact sheets, recent articles, and information on local support groups and relevant complementary modalities.[12]

Implementing rights-based patient constitutions, charters or codes

The UK and US have rights-based constitutions, charters or codes that are provided to patients. These documents facilitate patient-centred health care by informing patients about the level of care they are entitled to expect. The NHS Constitution in the UK[80] includes the following statements that have a bearing on patients' experience:

- You have the right to be treated with a professional standard of care, by appropriately qualified and experienced staff, in a properly approved or registered organisation.
- You have the right to expect NHS organisations to monitor, and make efforts to improve the quality of health care they provide, taking account of the applicable standards.
- You have the right to be treated with dignity and respect.
- You have the right to privacy and confidentiality.
- The NHS will strive to ensure that services are provided in a clean and safe environment that is fit for purpose, based on national best practice.

- The NHS will strive for continuous improvement in the quality of services you receive, identifying and sharing best practice in quality of care and treatments.

New Zealand has a similar code in place (Code of Health and Disability Services Consumer Rights Regulation 1996).[47] In the US, the American Hospital Association has developed a *Patient Care Partnership Brochure.*[81] Medicare patients are given a copy of the *Important Message From Medicare Notice*,[82] outlining patients' rights. There are also many organisation and institution-specific patient rights constitutions or charters operating in the US and Canada.

User-centred design and redesign

In both the US and UK, approaches based on design of the physical care environment are emerging as successful tools to improve patients' experience of care. The approach in the US focuses on user-centred design. User-centred design is based on environmental psychology principles, where spatial design is seen as an important part of the patient experience and wellbeing.[83] Typically, a team of architects, psychologists and sociologists are employed to collect data from staff and patients via observations and interviews. The team maps the entire patient journey from the first point of contact to the last point of contact. The team then identifies core patterns and areas for improving the environment to better meet patients' emotional experience of care.[83]

The leading case-study for user-centred design is Kaiser Permanente's work in the US. By implementing user-centred design, Kaiser Permanente improved interaction and information flow between staff, and between staff and patients; improved staff efficiency; provided examination rooms that could accommodate carers and families; increased attention to privacy and comfort in patient rooms; and provided more comfortable waiting areas and easier check-in procedures and improved signage to help patients orientate themselves more intuitively.[83]

In the UK, service redesign has been implemented in several NHS services. Service redesign focuses on technical aspects of the patients' journey, such as efficiency. Like user-based design, service redesign is based on insights from observing service users or mapping all the steps of a patient's journey. Service redesign brings together all key staff involved in the patient's journey with the patient themselves, to identify the problems throughout the patient journey.[84] Solutions are typically generated by staff. This is in contrast to user-centred design, which employs professionals outside the health service to map the patient journey and generate solutions.

Experience-based co-design

Experience-based design (EBD) is a methodology developed by Bate and Robert[85] for working with groups of patients and staff to improve services. It draws on knowledge and ideas from design sciences and design professions, where the aim of making products or buildings better for the user is achieved by making the user integral to the design process itself. In a healthcare setting, EBD focuses on how patients and staff move (or are moved) through the service and how they interact with its various parts. Unlike user-centred design and service redesign, the focus of co-design is on patients working with staff to 'co-design' improvements in the experience of using the service.

Involving patients and staff on an equal footing in EBD is more integral than in 'patient involvement' projects, where patients are often treated as objects for study, rather than partners. How the service 'feels' or is experienced is seen as equally important as how functional and safe it is. Bate and Robert provide a step-by-step guide to the method and illustrate it with a 12-

month pilot program in head and neck cancer services at Luton and Dunstable NHS Foundation Trust, funded by the NHS Institute for Innovation and Improvement. This method has since been used in other clinical specialties and hospitals in the UK, as well as in Australia.[74]

Patient and carer engagement at the governance level

Patients and carers can be engaged at the governance level by being appointed to advisory councils and committees. Patient and family advisory councils can serve as the 'patient voice' and the institutional infrastructure for including the patient and family member perspective in hospital organisational decision making. Councils typically include patients, family members, executive leaders and staff, working in partnership to assure delivery of the highest standard of comprehensive and compassionate care. Patients and carers may be members of existing advisory and governance committees,[12] or new committees being established. Activities may include:

- providing information to hospital leaders and staff about patients' needs and concerns
- helping plan patient care areas and new programs
- making changes that affect patients and family members
- encouraging patients and families to be involved and to speak up
- strengthening communication among patients, family members, care givers and staff.

The *Massachusetts Health Care Quality Act 2010*[86] requires all acute, chronic and long-term care hospitals in the US state of Massachusetts to establish a patient and family advisory council. The purpose of these councils is for patient and family members to advise the hospital on matters including, but not limited to, patient and provider relationships, institutional review boards, quality improvement initiatives, and patient education on safety and quality matters, to the extent allowed by state and federal law. Membership of a council is determined by each hospital; and orientation, training, and continuing education must be provided to council members. The legislation also allows physicians and healthcare providers to acknowledge a medical error by offering an apology to a patient without fear of that apology being used in a lawsuit.

Leadership and change management strategies

Enabling patient-centred care may require a change to the way organisations have traditionally provided care. Leadership and change-management strategies, including patient and staff rounding, fireside or chair-side chats, and clinician, employee and board engagement, are successful ways to strengthen the foundation for patient-centred care.[12] Staff rounding involves service-provider leaders moving through their organisation and hearing first-hand from patients and staff about what is happening in that unit. The idea is that patients and staff can express their views in a personalised, dialogue-based way that enables managers to understand the hospital experience.[12] Similarly, staff may be invited to meet with managers to share ideas or express concerns in fireside chats or chair-side chats. The strength of these communication methods is the direct interaction they afford between management, frontline staff and patients.[12]

Engaging staff and clinicians, such as involving them in patient-centred advisory councils, and identifying and recognising role models and champions of other patient-centred practices, are further strategies to establish and sustain patient-centred care.[12]

Board or governance-level strategies to facilitate patient-centred care include starting each board meeting with a patient story, shared by the patient or a staff member; including a board member on a patient advisory council; and including patient-centredness as a defined and measurable organisational aim.[12]

Staff and practice development

In patient-centred care, the model of care delivery is not limited to a focus on the patient — considerable attention should also be paid to the experience of care providers.[12] In a patient-centred environment, all employees are care givers, each affecting the patient's experience. Accordingly, there is a need to engage employees in fostering an atmosphere of patient-centredness. Successful strategies to recognise and reward employees include public acknowledgment of a staff member in a newsletter for their impact on a patient, family member or another employee; or an employee-of-the-month program.[12] This can also be an opportunity for staff to share their story with the management team. More informal approaches to staff recognition include a simple acknowledgment or thank you during manager rounding.[12] Some providers have successfully adopted a strategy whereby a senior manager writes an inspirational weekly message that is recorded on a dedicated phone line and can be accessed by any employee. Scrolling screen saver messages and staff rounding are further tools to help employees engage in creating a patient-centred culture.[12]

Values training

Another factor associated with success in patient-centred environments is that employees' behaviour consistently reflects the organisation's values. Only when employees' personal values simulate the core values of the organisation can the culture transform to a patient-centred model.[12] At Sharp Coronado Hospital outside San Diego, California, every employee has completed the sentence, 'I come to work to make a difference by…' and the laminated statement is adhered to their name badge as a constant reminder that what they do is meaningful and is making a difference in people's lives.[12] Directly involving staff in determining the organisational values and defining the behaviours that embody those values fosters a culture of patient-centred care.[12] Making the organisation's values visible can remind staff of the patient-centred behaviours expected of them.[12]

Staff satisfaction strategies

According to Shaller,[14] for healthcare organisations to be patient-centred, they need to create and nurture an environment in which their most important asset — their workforce — is valued and treated with the same level of dignity and respect that the organisation expects them to provide to patients and families. An important way to achieve this commitment and engagement is to conduct staff satisfaction surveys and monitor staff experience through staff rounding and experience-based co-design. Directly involving employees in the design and operation of patient-centred processes may also improve staff satisfaction.

Accountability strategies

To encourage and promote staff ownership of the cultural change required in many organisations to become patient-centred, the organisation should create opportunities for staff to be involved in envisaging and engaging in patient-centred care. Approaches that encourage engagement, communication and accountability include a shared governance structure or employee action committees comprised of both management and non-supervisory staff.[12] To define the way patient-centred care is provided, specific commitments and expectations can be outlined in professional codes of conduct. Some patient-centred organisations in the US have incorporated performance measures to support the behaviours defined in codes of conduct,[12] and performance incentives based on improving quality metrics, including patient experience.[87]

Improved complaints processes

In the UK, complaints about the NHS are a rich source of feedback. The Healthcare Commission and the Parliamentary and Health Services Ombudsman have both commented that the NHS should do more to use complaints feedback to improve services. This is at the heart of the reforms to the complaints process set out in *Listening, Responding, Improving: A Guide to Better Customer Care.*[88] The aim of the reforms is to enable healthcare organisations to improve the way they listen to, respond to and learn from people's experiences. To ensure that complaints are resolved with a more personal and responsive approach, a single, comprehensive complaints system is in place across health and social care. The UK Department of Health has published a guide and advice sheets on their new approach to complaints;[79] links to these resources are provided in Appendix B.

In the US, providers have complaints processes in place at the institutional and organisational levels. In addition, the Joint Commission's Office of Quality Monitoring evaluates complaints made by patients, carers and staff about its accredited providers that relate to safety and quality of care issues, such as patient rights, care of patients and safety.

2.4 Case study: Medical College of Georgia Health Inc, Georgia, US

Located in Augusta, Georgia, MCG Health Inc is a non-profit health service affiliated with the Medical College of Georgia. The medical district, including the university hospital, is the largest employer in Augusta, with more than 25 000 staff. MCG Health is a 632-bed tertiary medical centre with more than 22 000 admissions per year and 455 000 outpatients.

MCG Health's focus on patient-centred care started when they built a new paediatric medical centre in 1998. Patients and families were encouraged to participate in their care and were recognised as care team members and partners in improving care delivery. Because of this, MCG Health is considered a pioneer in the field of patient and family-centred care. Once the benefits of a patient-centred focus were recognised in the paediatric area, strategies were broadened to include adult services.

Strategies used by MCG Health include:

- establishing patient advisory councils; MCG Health has the largest patient adviser program in the US, with more than 130 trained patient and family advisers involved in safety and quality, planning, and design throughout the organisation
- removing all visiting hours; family and carers can visit at any time as preferred by the patient (including in the intensive care unit), and they are not viewed as 'visitors', but as 'partners'
- giving patients and family the right to call out rapid response teams
- ensuring that patient advisers sign off on plans for facility redesign and new fitouts
- introducing executives rounding and patient advisers rounding on wards
- providing patients' feedback to staff on a regular basis
- conducting training across the whole organisation in patient-centred customer service values
- redesigning patient accounts based on patient advice.

More recent undertakings include the redesigning of the MCG Medical Center Breast Cancer unit by patients — within three years, patient care experience ratings for the unit moved from the 40th to 74th percentile, rated using surveys administered by Press Ganey (an independent vendor). The neurology intensive care unit was renovated with patients' input and introduced a 24/7 family access policy — within five years, the patient satisfaction ratings increased from the 10th to 95th percentile and the length of stay for patients decreased by 50 per cent. At this point, staff acknowledged that the chief executive officer 'saw the business case' and became a patient-centred care 'convert'. During this time, MCG Health witnessed a number of clinical and operational benefits, including the overall staff vacancy rate falling from 8 per cent to 0 per cent, with a current waiting list of several hundred people. With a continued focus on patients in 2010, planning has begun for a new cancer centre with patient input into the design (Pat Sodomka, Sebrio Vice President, Patient and Family Centered Care, MCG Health Inc and Director, Center for Patient and Family Centered Care, Medical College of Georgia, pers comm, 2009).

2.5 An overview of leading organisations

Several private or not-for-profit organisations advocate specific patient-centred frameworks, or approaches to health service improvement that are built around a combination of the strategies outlined in Section 2.3. Both public and private health service providers have used these frameworks. The leading organisations advocating patient-centred frameworks to health care are described below. Websites and links to resources on strategies to promote patient-centred care are provided in Appendix B.

World Health Organization

The World Health Organization (WHO) is the United Nation's leading authority on global health matters, and their role includes setting standards and developing evidence-based policy.

As stated in Section 1.1, WHO advocates for a 'responsive' healthcare system that meets people's expectations.[13] In addition, WHO advocates for involving patients and carers as partners in initiatives to improve the safety and quality of care,[89] particularly through its 'Patients for Patient Safety' (PFPS) program. PFPS works with a global network of patients, consumers, care givers and consumer organisations to support patient involvement in patient safety programs at local, regional and international levels. WHO has established a global network of PFPS 'champions', including 13 Australian champions, who work in partnership with health professionals and policy makers across the world to identify problems, design solutions and implement change.

PFPS is designed to ensure that the perspective of patients and families are a central reference point in shaping the safe and high-quality delivery of health services, including health service decision making. PFPS aims to develop opportunities for patient voices to be heard in creating public awareness about inherent healthcare risks; educate the public about systems approaches to risk management; report errors or healthcare failures in ways that contribute to systemic learning; and disseminate research and share solutions that can prevent patient harm.[89]

Picker Institute

The Picker Institute was established in 1994 as an independent non-profit organisation and is based in Boston, Massachusetts, US. The institute's goal is to foster a broader understanding of the practical and theoretical implications of patient-centred care by focusing on the concerns of patients and other healthcare consumers. The Picker Institute pioneered the use of scientifically valid, nationwide surveys and databanks on patient-centred care to educate doctors and hospital staff on improving patient services from a patient's perspective. The surveys cover the eight Picker dimensions of patient care (set out in Section 1.1), and patients report on what happened to them rather than rate how satisfied they were, as Picker research has shown that simple patient satisfaction questionnaires do not produce useful results. Picker Institute surveys are used by regulators in the US, UK, Canada and Australia to measure patient-centred care.

In addition to their surveys, the Picker Institute provides education programs, research grants and publications on patient-centred care topics, including *The Patient-Centered Care Improvement Guide*[12] and a self-assessment tool, written in conjunction with Planetree (see below). The Picker Institute has a sister organisation in Europe — the Picker Institute Europe. A variety of resources on patient-centred care is provided on the Picker Institute Europe website[c].

Institute for Patient- and Family-Centered Care

The Institute for Patient- and Family-Centered Care is a non-profit organisation founded in the US in 1992. The institute's mission is to advance the understanding and practice of patient and family-centred care. In partnership with patients, families, and healthcare professionals, the organisation integrates these concepts into all aspects of health care. It provides education, consultation, technical assistance, materials development, information dissemination and research on patient-centred care.

The institute website[d] provides numerous practical resources, including assessment tools, guidance publications and multimedia resources produced by the institute and other leading patient-centred care organisations. Notable publications include:

- *Advancing the Practice of Patient- and Family-Centered Care: How to Get Started*[30]
- *Partnering with Patients and Families to Design a Patient and Family-Centered Health Care System — Recommendations and Promising Practices*[90] (this publication sets out practical examples of patient-centred care in the US)
- *Partnering with Patients and Families To Design a Patient- and Family-Centered Health Care System: A Roadmap for the Future — A Work in Progress.*[27]

Studer Group

The Studer Group is a US-based organisation that partners with healthcare providers to increase performance by improving employee and clinician satisfaction and experience. Improved performance is achieved through 'the nine principles': commit to excellence; measure the important things; build a culture around service; create and develop leaders; focus on employee satisfaction to improve patient satisfaction; build individual accountability; align behaviours with goals and values; communicate at all levels; and recognise and reward success.

The Studer Group provides on-site resources to demonstrate their approach. Resources include tools for improving communication interactions between staff and patients, such as 'key words at key times', 'discharge phone calls' and 'patient rounding' tools. Others concentrate on improving patient experience by first improving staff satisfaction and experience, such as the

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[c] www.pickereurope.org
[d] www.ipfcc.org

‘staff rounding for outcomes’ and ‘thank you notes’ tools. There are also several supplemental resources available to nonpartners, including leadership development resources and evidence-based tools.

Planetree

Planetree was founded in 1978 by a patient, Angelica Thierot, who was disheartened by her own experience of hospital care, which she felt was de-personalising. Her vision of a different type of hospital was one in which patients would become active participants in their own care and wellbeing. Planetree began by developing information libraries for patients and has grown into a non-profit organisation providing education and information on patient-centred care. It facilitates efforts to create patient-centred care in healthcare organisations in the US, Canada and Europe.

Based on the feedback of thousands of patients and hospital staff members, Planetree has identified core principles that are essential to practising patient-centred care, published in *Putting Patients First: Designing and Practicing Patient-Centered Care*.[91]

The Planetree method for fostering the core principles includes:

- working with senior leaders to develop and share ownership of an organisational strategy to drive improvement in patient-centred care
- multidisciplinary and multiprofessional training for all staff in the core principles, mainly through one and two-day retreats
- expressing the desired culture and values in an accessible language
- positively and continuously reinforcing Planetree values and principles with awards and recognition for individuals
- emphasising the built environment.

Planetree recently partnered with the Picker Institute to produce *The Patient-Centered Care Improvement Guide.*[12] This and other Planetree publications are available on the Planetree website[e].

Institute for Healthcare Improvement

The Institute for Healthcare Improvement (IHI) is an independent not-for-profit organisation, founded in 1991 and based in Cambridge, Massachusetts. The IHI aims to identify best practices and bring about system changes that enable healthcare providers to become and remain patient centred. The IHI has developed tools to assist healthcare providers, including an ‘improvement map’ that sets out the processes for improving patient-centred care in a variety of areas (advanced care planning, communication with patients and families after an adverse event, daily goal setting and planning for treatment and discharge, multidisciplinary patient rounding for coordinated care; shared decision making, patient experience programs, and leadership programs to enable patient-centred care). The IHI provides a variety of publications, resources, tools and improvement stories on its website to help organisations achieve patient-centred care[f].

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[e] www.planetree.org
[f] www.ihi.org

Kenneth B Schwartz Center

The Kenneth B Schwartz Center is a non-profit organisation based at Massachusetts General Hospital. It was founded by the family of a patient (Schwartz) in 1995, with a mission to 'provide support and advance compassionate health care, in which care-givers, patients and their families relate to one another in a way that provides hope to the patient'. The centre runs programs to educate, train and support care givers in 'the art of compassionate health care', and to strengthen the relationship between patients and care givers, including Schwartz Center Rounds, the CarePages website and the Patient Voice for Compassionate Care Program.

Schwartz Center Rounds are a multidisciplinary forum where care givers discuss difficult emotional and social issues that arise in caring for patients. An evaluation of Schwartz Center Rounds[92] found many benefits, including improving communication and teamwork among care givers, and communication between care givers and patients. This study also found that participating in rounds increased the likelihood of staff attending to the psychosocial and emotional aspects of care, and improved their beliefs about the importance of empathy.

The CarePages website[g] contains patient blogs that connect friends and family during a health challenge, helping them support each other. The Patient Voice for Compassionate Care: Schwartz Center Dialogues is a pilot program that brings the experiences and perspectives of patients and their families directly to care givers in a series of facilitated discussions. The program helps patients and their families become stronger advocates and partners in their own care; at the same time, it improves the communication skills of care givers and the quality of compassionate care they deliver. The panel of patients and care givers identify specific changes necessary to improve communication and reconvene over time to ensure that the changes are made and the patient focus is sustained. Information on each of these programs is available on the Schwartz Center website[h].

The King's Fund

The King's Fund is a charity that seeks to understand how the health system in the UK can be improved. The King's Fund Point of Care program aims to transform patients' experience of care in hospital by enabling healthcare staff to deliver the quality of care they would want for themselves and their own families. The King's Fund works with patients and their families, staff and hospital boards to research, test and share new approaches to improving patients' experience. *Seeing the Person in the Patient: Point of Care Review Paper*[15] considered current debates and dilemmas in relation to patients' experience of care in acute hospitals in the UK. Based on this research, The King's Fund developed a framework to analyse the factors that affect patients' experience at four levels: individual interactions with staff members; the 'clinical micro-system' of the team, unit or department; the hospital; and the wider healthcare system, including NHS priorities.[15] These levels are similar to the IOM levels of patient-centred care described in Section 1.4.[1]

[g] www.carepages.com
[h] www.theschwartzcenter.org

3 The relevance of patient-centred care to the Australian health system

Australia has a mixed healthcare system, where funding for, and delivery of, health care is divided between the public (represented by the Australian Government, six states and two territories) and private sectors.[93] Despite the complexity of the Australian healthcare system, many of the strategies and approaches to patient-centred care used internationally are highly applicable in Australia. Current policy and reform initiatives address the need for heath service organisations to adopt a patient-centred approach to health care. This section describes the applicability of patient-centred care approaches and strategies to the Australian health system; provides an overview of the current policy reform context of patient-centred care in Australia; and highlights national-level initiatives, strategies and policies that promote patient-centred care.

3.1 The relevance of patient-centred care to a mixed healthcare system

Australia's health system is primarily funded through general taxation and a small compulsory tax-based health insurance levy. Medicare, the tax-funded national health insurance scheme, offers patients free public hospital care and subsidised access to their doctor of choice for out-of-hospital care.[93] The Australian Government's Pharmaceutical Benefits Scheme provides subsidised access to a wide range of medicines for all Australians. The public system administers public hospitals and a range of other primary and community health services.

The private sector includes the majority of doctors (GPs and specialists), numerous private hospitals and day hospitals, a large diagnostic services industry, pharmacists and a small number of major private health insurance funds.[93] Ownership of private hospitals is centralised, with more than two-thirds of all private beds owned by large for-profit chains or the Catholic Church.[93] GPs and pharmacists are largely self-employed and funded by Medicare through a fee-for-service and the Practice Incentive Program (PIP). The PIP offers financial incentives for accredited GPs to improve the quality and accountability of their medical services.

Both the public and private sectors are seeking to improve the overall quality of health care through activities such as patient experience and patient satisfaction surveys. These initiatives are supported and driven at the national policy level via recent national reforms that seek to embed patient experience as a measurable and reportable component of quality. National level service strategies and plans provide a further lever for the government to promote a patient-centred approach to health care in Australia.

Accreditation of health service facilities is another aspect of Australia's healthcare system that could further facilitate patient-centred care by incorporating patient-centred principles into accreditation requirements. The Australian Council on Healthcare Standards is the main accrediting body; others include the Quality Improvement Council, Aged Care Standards and Accreditation Agency, and Australian General Practice Accreditation Limited. To date, accreditation has been largely voluntary; however, most public sector hospitals are required to seek accreditation and most large hospitals seek accreditation.[93] Meeting accreditation standards provides financial incentives, because private health insurers pay higher reimbursement rates to accredited facilities.[93]

3.2 National health reform

Recent national policy reforms set out in the National Health and Hospitals Network Agreement[94] were focused on the need to improve access to services, quality of service delivery, financial responsibility, patient outcomes and patient experience. The reforms include a new Performance and Accountability Framework that includes national performance indicators, national clinical quality and safety standards developed by the ACSQHC, and new Hospital Performance Reports and Healthy Communities Reports.

The Performance and Accountability Framework builds on the requirement for public reporting on objectives and outcomes of the National Healthcare Agreement (NHA).[95] The NHA identifies a number of long-term goals for Australian governments to improve health outcomes, including that 'Australians have positive health and aged care experiences which take account of individual circumstances and care needs'.[95] Progress will be measured by nationally comparative information that indicates patient satisfaction levels on key aspects of the care they received. In the first NHA report in 2010, agreement had not yet been reached on how to measure this progress and no data were included.[95] In the future, progress reports will be drawn from patients' experience surveys in each jurisdiction (see Section 4.1) and a national population-based survey conducted by the Australian Bureau of Statistics (ABS; see Section 4.1) on the healthcare experiences of individuals in private households.[95]

The national performance indicators set out in the Performance and Accountability Framework provide potential levers for improvement in patient-centred care, and quality more generally. The indicators acknowledge the need to measure traditional operational and clinical performance indicators (such as hospital waiting times and hospital-acquired infections) alongside patient-reported quality based on the patient experience.[94] They also facilitate transparent monitoring across services and jurisdictions, setting a minimum standard of healthcare quality that each provider should adhere to.

A new, independent National Performance Authority will be established in early 2011 with responsibility for local-level performance reporting. The hospital performance reports prepared by the authority will show how Local Hospital Networks, individual public hospitals, and private hospitals perform against the new national standards. Performance indicators will include public hospital emergency department and elective surgery waiting times, adverse events in hospitals, patient satisfaction and financial management.

Performance will be tied to funding in the form of a 'nationally efficient price'. This is a nationally consistent patient level costing and pricing regime, determined by the Independent Hospital Pricing Authority. The authority will establish nationally consistent funding to providers based on the national efficient price of each public hospital. This is calculated according to reasonable access to public hospital services, clinical safety and quality, efficiency and effectiveness, and financial sustainability of the public hospital system.[94] By linking this funding to performance, the nationally efficient price will consider clinical indicators and outcomes, as well as patient experience.[94] Improving the patient experience through patient-centred care therefore has important implications for both patients and providers.

The ACSQHC is exploring options for the national clinical quality and safety standards. These standards will complement the National Safety and Quality Healthcare Service Standards currently being developed by the ACSQHC in their work on accreditation reform. The National Safety and Quality Healthcare Service Standards include a consumer engagement component that encompasses patient-centred care and will become compulsory for high-risk health services, including hospitals, day procedure centres and procedure rooms.

3.3 National strategies and initiatives promoting patient-centred care

A variety of national service-level initiatives, strategies and policies set out a patient-centred approach to health care. National initiatives include the Australian Charter of Healthcare Rights and the National Safety and Quality Framework. National strategies such as the National Primary Care Strategy, the National Chronic Disease Management Strategy, the Fourth National Mental Health Plan and the Fifth Community Pharmacy Agreement, all state that a patient-centred approach to health care is needed to improve the quality of health care in Australia. Current Aboriginal and Torres Strait Islander policies also reflect patient-centred principles and focus on family and community.

Australian Charter of Healthcare Rights

The Australian Charter of Healthcare Rights underpins the provision of safe and high-quality care, and supports a shared understanding of the rights of patients and consumers between those seeking health care and those providing health care.[96] Since 1993, each Australian state was required to develop public patient hospital charters to inform patients of their rights under the Australian Health Care Agreements.[97] In July 2008, Australian health ministers endorsed a single national charter as a clear statement of a minimum set of standards, rights, expectations and entitlements that is uniformly applicable across all states and territories and in all settings of care.[96] The charter and associated information is available on the ACSQHC website[i].

National Safety and Quality Framework

In 2009, the ACSQHC proposed a National Safety and Quality Framework for health care. The overall aim of the proposed framework is to provide direction for improving the healthcare system. In developing the framework, the ACSQHC recognised that consumer engagement in the health system through patient-centred care is vital to safety and quality improvements and should underpin the strategic plan for the system.[7] The actions proposed for health systems and providers to achieve patient-centred care are shown in Table 1.

[i] www.safetyandquality.gov.au

Table 1 Actions proposed to achieve patient-centred care under the ACSQHC Proposed National Safety and Quality Framework

Safe, high-quality health care is always:	What it means for me as a patient or consumer:	Strategies for action by health systems and providers:
1. Patient focused This means providing care that is respectful of and responsive to individual preferences, needs and values. It means a partnership between consumers, family, carers and their healthcare providers. Processes of care are designed to optimise the patient experience.	I can access high quality care when I need it.	• Develop service models that improve access to health care for patients.
	I can obtain and understand health information, so that I can make decisions about my own care and participate in ensuring my safety.	• Increase health literacy. • Involve patients so that they can make decisions about their care and plan their lives. • Provide care that is culturally safe.
	My health care is co-ordinated because people and systems work in partnership with me.	• Enhance continuity of care. • Minimise risks at handover. • Provide case management for complex care. • Facilitate patient-centred service models.
	I know my healthcare rights	• Promote healthcare rights.
	If I am harmed during health care, it is dealt with fairly. I will get an apology and a full explanation of what happened.	• Inform and support patients who are harmed during health care

Source: ACSQHC 2009[7]

National Primary Health Care Strategy

The National Primary Health Care Strategy is a high-level action plan consisting of 10 elements to improve patient-centred care in Australia. It sets the policy direction to better connect hospitals, primary and community care to meet patient needs, improve continuity of care and reduce demand on hospitals. Key priority areas identified in the National Primary Health Care Strategy include:

- improving access and reducing inequity
- better management of chronic conditions
- increasing the focus on prevention
- improving quality, safety, performance and accountability.[98]

Although the Australian primary healthcare system serves many patients well, it has not been specifically designed to cater for the particular health needs and cultural requirements of some groups, such as people from culturally and linguistically diverse backgrounds, Aboriginal and Torres Strait Islander people, or disadvantaged and marginalised populations. These groups may find it difficult to access appropriate services within the system, or to know which services to access and when.[98] The National Primary Health Care Strategy recognises that an inclusive and patient-centred focus is a key element to all future reforms in primary health care.

National Chronic Disease Strategy and National Service Improvement Frameworks

Promoting patient-centred care is a central aim for improving health service delivery, outlined in the National Chronic Disease Strategy (NCDS) and the National Service Improvement Frameworks (NSIF) for asthma, cancer, diabetes, heart disease, stroke and vascular disease.[99] NCDS and NSIF place the patient, their family and carers at the centre of a broader health promotion and disease prevention framework. This includes the core principle of achieving patient-centred care and optimising self-management. Self-management helps people to take responsibility for their own health, to make informed decisions, and to maximise their wellbeing and quality of life.[99]

National Mental Health Plan

The Fourth National Mental Health Plan[100] outlines reporting requirements for service providers against agreed standards of care, including consumer and carer experiences and perceptions. A priority in this plan is ensuring that consumers and carers can access information about service provider performance across the range of health quality domains and compare these to national benchmarks. The principles behind this priority include the following patient-centred concepts:

- Consumers, their carers and families should be actively engaged at all levels of policy and service development. They should be fully informed of service options, anticipated risks and benefits.
- Consumers and carers should be able to access information in a language they understand, or have access to interpreters.
- Families and carers should be informed to the greatest extent consistent with the requirements of privacy and confidentiality about the treatment and care provided to the consumer, the services available and how to access those services. They need to know how to get relevant information and necessary support.

The Fourth National Mental Health Plan also calls for the recognition of social, cultural and geographic diversity and experience of consumers and carers.[100]

Community Pharmacy Agreement

The Fifth Community Pharmacy Agreement[101] commenced on 1 July 2010. It encourages pharmacists to adopt a more patient-centred approach to care by providing support for pharmacists to identify, resolve and document medicine-related issues experienced by patients. The agreement also sets out a new patient service charter that outlines the role and responsibilities of the pharmacist and the pharmacy.

Aboriginal and Torres Strait Islander primary healthcare policy

In addition to the National Primary Health Care Strategy, Aboriginal and Torres Strait Islander policies reflect patient-centred principles and aim to be community and family-centred.

The Office for Aboriginal and Torres Strait Islander Health provides direct grants to approximately 280 Aboriginal community controlled health services and Aboriginal medical services. These organisations are primary healthcare services initiated and operated by the local Aboriginal community. They deliver holistic, comprehensive and culturally appropriate health care to the community that controls it, through a locally elected board of management.

The National Aboriginal Community Controlled Health Organisation (NACCHO) is the national peak Aboriginal health body representing Aboriginal community controlled health services throughout Australia. According to NACCHO, this model of primary health care is in keeping with the philosophy of Aboriginal community control and the holistic view of health that this entails:

> Aboriginal health is not just the physical well being of an individual but is the social, emotional and cultural well being of the whole community in which each individual is able to achieve their full potential thereby bringing about the total well being of their community.[102]

In a review on the link between primary health care and health outcomes for Aboriginal and Torres Strait Islander people, Griew[103] identifies the following patient-centred factors in association with successful local-level primary healthcare interventions:

- genuine local Aboriginal and Torres Strait Islander community engagement to maximise participation, up to and including full community control
- a multidisciplinary team approach employing local community members
- service delivery that harmonises with local Aboriginal and Torres Strait Islander ways of life.

Current reform proposals in relation to Aboriginal and Torres Strait Islander primary health care emphasise the importance of a family-centred approach. Family in this context has a broad community focus and recognises, for example, that Aboriginal and Torres Strait Islander children often have other significant carers in addition to their biological mother and father.[104] Family-centred primary health care takes a life course approach, which, without neglecting adult health, focuses on establishing early life resilience and advantages in child development.[104] Family-centred primary health care is an attempt to draw education and family welfare, usually considered to be part of the social determinants of health, into the foreground of primary healthcare practice.[104]

4 Current jurisdictional and other activity in Australia

This section highlights some of the main patient-centred care initiatives in Australia. Many of these evolved from the need to address the safety and quality of health service delivery, revealed by several recent high-profile state-based inquiries. These inquiries involved patient care and adverse clinical incidents at the King Edward Memorial Hospital in Perth,[105] Canberra Hospital,[106] Royal Melbourne Hospital,[107] Campbelltown and Camden Hospitals,[108] Bundaberg Hospital and Queensland Health,[109-110] Royal North Shore Hospital,[111-112] and, more generally, acute care services provided by NSW Health.[113-114]

Hindle et al[115] reviewed the common features of the King Edward, Royal Melbourne and Campbelltown and Camden inquiries. They found that quality monitoring processes were deficient, and patients and families were not informed members of the team. Patients were not adequately involved in care planning and did not always have an adequate basis for informed consent; they were unsure of their rights and frequently afraid of exercising them; they were sometimes treated in inconsiderate ways (mainly by doctors); and they seldom received sympathetic and helpful support when they made complaints.

All the inquiries concerned allegations of poor clinical practice and poor communication between healthcare organisations and patients, their families or other carers. This led to a loss of trust in the health system from patients and the community.[66]

The response to each of these inquiries was an increased focus on patient-centred care. Several strategies to achieve this are in place at the jurisdictional level, including improving communication with patients, improving complaints processes, promoting healthcare rights, increasing reporting of hospital activity and performance in relation to patient safety and quality issues, and increasing consumer involvement in health service planning and design.[116-117]

The need to improve the safety and quality of health care has led to public and private providers adopting a variety of approaches (typically involving patient feedback surveys) to measure the quality of the patient journey and the experience of interactions across different settings. These and other approaches are discussed in this section.

4.1 Listening to patient feedback

Most Australian jurisdictions conduct surveys to measure patients' experience across different settings. In New South Wales, Patient and Care Experience surveys are conducted annually. Condition-specific surveys are also emerging, with a care experience survey for cancer patients conducted through the Cancer Institute NSW.[118] Other states' surveys include the Victorian Patient Satisfaction Monitor survey, the Queensland Health Patient Satisfaction survey, the Healthcare Survey (Australian Capital Territory), and the Patient Evaluation of Health Services survey (South Australia and Western Australia). Tasmania has a Patient Satisfaction System in place.[119] Links to some of the state and territory survey websites, which contain information about the surveys and some results, are provided in Appendix B.

Survey results are used to identify trends, monitor performance, benchmark results against similar service providers, and inform health service planning and patient safety and quality initiatives.[120] For example, NSW Health requires each of its area health services to produce action plans based on their survey results.

The Australian Bureau of Statistics (ABS) also collects population-based information about healthcare experiences from private households. Part of the annual Multipurpose Household Survey identifies healthcare issues at a national level across a range of key delivery areas, including hospitals (inpatient and emergency), general practice, specialists, pathology and imaging tests, and dentistry. Unlike the state and territory surveys, the ABS asks household members to recall a range of experiences over the preceding 12-month period, rather than a specific service experience. This survey of consumer experience, first conducted in July–December 2009 on a sample of around 6500 people, has the advantage of capturing information about people who could not access services and exploring underlying reasons. Future ABS surveys will be expanded to capture the experiences of more than 20 000 people from 2010–11.

The Council of Ambulance Authorities has conducted annual national patient satisfaction surveys since 2002,[121] which measure the quality of the ambulance service as perceived by its customers. The survey results are compared across states and territories and are used to set performance benchmarks.

The private sector, including GPs, insurers and hospitals, also conduct surveys to capture patients' experiences and feedback. There is little publicly available information on the content or scope of private sector surveys. The draft fourth edition of the Royal Australian College of General Practitioners Standards (due for release in late 2010) requires GPs to obtain feedback from patients by both establishing and maintaining an ongoing feedback mechanism (such as a suggestion box or complaints management system); and collecting patient feedback information every three years (such as through written patient questionnaires, focus group discussion or patient interviews).

4.2 The Australian Institute for Patient- and Family-Centred Care

The Australian Institute for Patient- and Family-Centred Care was recently formed by a group of interested individuals and healthcare professionals. The institute is based on the American Institute for Patient- and Family-Centered Care (see Section 2.5). The purpose of the institute is to:

- advance the understanding and practice of patient and family-centred care
- improve the healthcare experience and outcomes for patients, families and healthcare professionals
- provide onsite training, technical assistance and education packages to all healthcare professionals and students
- undertake research and evaluation
- develop policy statements
- advise healthcare program planners and decision makers.

4.3 Examples of state and territory initiatives

State and territory health departments have made a range of advances in driving and supporting patient-centred care. These are summarised below.

ACT Health

In 2003, ACT Health established the Consumer Feedback Project, which involved comprehensive consultation with the community and resulted in the report *Listening and Learning, ACT Health's Consumer Feedback Standards.*[122] The standards encourage consumers to provide feedback, both positive and negative, on their health experience. The importance of the standards was reinforced with the release of ACT Health's *Consumer Feedback Management Policy*[123] in 2008. The policy outlines the general principles and approach of all ACT Health staff to manage consumer feedback.

ACT Health provides consumer information on its website[j]. This includes the ACT Healthcare Survey, details of how to access medical records, and information on consumer involvement, community consultation, surgery, interpreter services, freedom of information, interstate patient travel, and maps of the campuses.

A patient experience program and a redesign program are currently underway in the ACT. More than two hundred patients and carers across a wide range of clinical settings have told their stories through these programs. This information is used for planning and designing new facilities, redesigning processes, and identifying gaps and barriers to healthcare delivery. ACT Health is developing a framework for partnering with patients in all aspects of the redesign program.

Information from the patient experience and redesign programs is a valuable educational tool. Staff are directly involved in documenting the information received from interviews, and the patient experience reports are provided to senior staff and executives. Furthermore, ACT Health partners with the University of Canberra to educate nursing students about patient experiences and the importance of a patient-centred approach to care delivery.

Since 2006, ACT Health has implemented the Respecting Patient Choices Program, a national advance care planning program developed by Austin Health in Victoria. The Respecting Patient Choices Program enables individuals to discuss and document decisions about their future health care, to prepare for a time when they are unable to participate in their medical care decisions. There is significant community support for this program and the program coordinator has built networks within the residential aged care sector, as well as with other groups, to increase community awareness of advance care planning.

ACT Health has adopted the Australian Charter of Healthcare Rights and is working with the Human Rights Commission and the Health Care Consumers Association of the ACT to implement the charter.

NSW Health

NSW Health has implemented two experience-based co-design programs.[74 124-125] Evaluations of these programs revealed that co-design was successful in:

- enabling frontline staff to better appreciate the impact of healthcare practices and environments on patients and carers
- engaging consumers in 'deliberative' or in-depth processes that were qualitatively different from conventional consultation and feedback

[j] www.health.act.gov.au

- achieving practical solutions that realise the wishes, advice and insights of consumers and frontline staff
- enabling staff to reflect on and improve their work practices
- involving consumers in developing or reviewing clinical pathways, leading to better experiences and patient flow.[74 124-125]

NSW Health has implemented a variety of other patient-centred initiatives. For example, the Essentials of Care Program aims to:

- encourage patient participation in decisions about their care
- focus on patients' needs and their experiences of care provided
- value the contributions from all involved in care
- support the ongoing review and development of practice
- use relevant research and evidence that is generated from practice and care settings.

The Clinical Leadership Program, an initiative of the Clinical Excellence Commission in NSW, helps healthcare professionals develop leadership strategies to introduce more patient-centred principles into their practice.

In response to the New South Wales inquiries, the NSW Health *Caring Together: Health Action Plan*[116] focuses on the patient as the centre of the healthcare system. This includes strategies to improve communication with patients, such as identifying categories of staff and types of uniforms worn, and developing patient fact sheets to support care in emergency departments. Nurse positions have been established and filled in the emergency department that focus on the needs of the patients in the waiting room, providing better communication on waiting times, initiating basic treatments and completing required admissions documentation.[116] The 'Take the Lead' project is designed to further facilitate the roles of the Nursing Unit Manager and Midwifery Unit Manager in improving the patient journey and carer experience. Some health services in NSW have also implemented Studer Group training and programs.

Queensland Health

Queensland Health acknowledges that measuring and reporting on the quality of care from a patient's perspective is essential to providing a high-quality healthcare system.

The Queensland Health strategy for improving the patient experience is informed by collecting consumer feedback through surveys, collecting patient stories and using an enterprise-wide computer-based information system to record and manage complaints and compliments. This information allows Queensland Health to assess performance along the patient journey at the service delivery level and to determine where it may be appropriate to concentrate improvement activities. It also allows the impact of these improvement activities to be measured over time. Management action plans developed in consultation with local Health Community Councils identify areas and activities for improvement, and state-wide trend analysis informs organisational strategic planning.

The Productive Ward and Transforming Care programs are key quality and safety improvement initiatives sponsored by the Patient Safety and Quality Improvement Service. Both focus on patient-centred care as a core pillar of their improvement strategy. The programs promote practice that improves the amount of time nurses spend delivering direct patient care (typically doubled). They provide strategies to improve the quality of patient-centred care and have established links with priority state and national safety initiatives. These include clinical handover, recognition and management of the deteriorating patient, malnutrition, patient identification, and falls and pressure ulcer prevention. Both programs have developed patient satisfaction and patient experience assessments to measure improvement in these indicators. The programs are spreading across all health service districts in Queensland.

Queensland Health actively engages consumers in service planning, design implementation and assessment, from executive committee membership to consultation in the assessment of survey outcomes. Health Consumers Queensland was established in 2008 to contribute to the continued development and reform of health systems and services in Queensland, by providing the Minister for Health with information and advice from a consumer perspective and by supporting and promoting consumer engagement and advocacy.

Victorian Department of Health

In 2006, the Victorian Department of Health developed the consumer participation in health policy, *Doing it With Us Not For Us*.[126] A strategic framework advocating a patient-centred approach to implementing this policy was drafted in 2009.[127] In addition, *The Victorian Mental Health Reform Strategy 2009–19* was developed in partnership with patients and carers.[128] The strategy and accompanying action plan and draft quality framework promote mechanisms to support patient-centred care in mental health services.

The Victorian Department of Health currently has several patient-centred projects and initiatives in place. For example:

- A series of ‘productive ward’ projects developed by the NHS in the UK are currently being piloted. The aims of the projects are to increase the time staff spend caring for patients, staff morale and retention and safety. The pilots also aim to reduce length of stay and patient complaints.
- In 2008, a stroke project aimed at improving the care pathway for people who have had a stroke and their carers was developed and implemented in consultation with patients and carers.
- A paediatric palliative care parent reference group is being established to ensure the needs and views of parents are included in the government’s policy for paediatric palliative services.

Several health services have conducted process-redesign projects as part of a broader Improving the Patient Experience Program. Under this program, emergency departments in Victoria had physical amenity upgrades and reviews of their internal environment. Three complementary activities to improve communication were initiated, including consistent patient-friendly signage improvements, consistent consumer-focused information materials, and communication workshops for frontline emergency department staff. Workshops were also held to improve clinicians’ understanding of the patient’s perspective of emergency department care and consumer communication needs.

The Ministerial Advisory Council on Safety and Quality, the Victorian Quality Council, has introduced an initiative to improve the patient journey. Objectives of this initiative are to promote the principles of patient-centred care across the Victorian health sector, improve and standardise information flow, and provide decision support tools to assist interfacility transfer of non-critical patients.

The Victorian Quality Council has also introduced the Consumer Leadership Program to develop patient-centred care at the system level. This program is based on research commissioned by the Rural and Regional Health and Aged Care Services Division[129] that cites the need for structured education and training for staff to develop and improve leadership, and effective consumer participation to improve the safety and quality of care. The most effective strategies were:

- consumer leadership programs that incorporate elements of formal learning, peer support and mentoring
- peer support networks (including local, regional and statewide meetings) for consumer leaders involved in safety and quality activities, in partnership with statewide and major metropolitan and rural health consumer organisations and networks
- guidelines for development and support of consumer leadership within the health system to ensure effective participation and leadership.[129]

Based on this research, the Victorian Quality Council has now contracted the Health Issues Centre to design a pilot program in health literacy and consumer leadership that meets the needs of health consumers and staff with varying experience levels in consumer participation, from beginner to experienced board member or health services staff. The pilot is expected to be completed by April 2011.

WA Health

WA Health has developed the Patient First Program.[130] This program places patients and their carers at the forefront of clinical processes by increasing patients' understanding of their condition, which facilitates better decision making through informed consent. The program aims to increase patients' health literacy and facilitate a quicker recovery time. Increasing patients' awareness of the risks inherent in their health care was seen to minimise the potential for adverse events and give them the ability to self-manage their health issues.

The program involved volunteer consumers ('Patient First Ambassadors') distributing a *Patient First* booklet directly to patients. The booklet provides information for patients on topics such as informed consent, issues to consider when making decisions about treatment, understanding the risks associated with treatment and procedures, patients' rights and responsibilities, medication safety, falls prevention, avoiding infection, improving emotional wellbeing, maximising recovery, health records, information and privacy.[131]

Using a Patient First Ambassador to distribute the booklets improved communication with patients, and the idea of one consumer talking to another consumer was a powerful tool. The ambassador was not affiliated with the hospital or health department, but identified with the health consumer and was therefore able to discuss the topics in the booklet in a non-threatening way.[130]

In 2009, the inaugural Australian PFPS Workshop was held in Perth. The workshop was a collaboration between the WHO Alliance for Patient Safety; the London, Chicago and Australian networks for PFPS; the Health Consumers Council WA; and the Office of Safety and Quality of the WA Department of Health. The workshop brought together a group of 40 health consumers, many of whom had experienced medical error or health system failure, healthcare providers and health policy makers from around Australia. The workshop resulted in the Perth Declaration for Patient Safety. The Perth Declaration builds on the London Declaration drafted at a similar workshop in London in 2005. The London Declaration is a pledge of partnership between consumers and providers. The Perth Declaration for Patient Safety continues and aligns with this Pledge of Partnership to form an Australian community of committed people who seek to ensure future health users are not harmed.

4.4 Case study 1: Australian Capital Territory — Calvary Health Care Simply Better Program

In 2006, staff at Calvary Health Care in the ACT — a mixed public and private service — reviewed their patient care experience results and identified a need for significant improvement. Analysis of surveys conducted through private vendor Press Ganey highlighted that patient satisfaction had fallen below the 10th percentile and that staff satisfaction was similarly poor. Assessment by Better Practice Australia identified a 'culture of blame' in the service.

In recognition of the link between patient satisfaction and staff satisfaction, Calvary launched the Simply Better Program using the Studer Group model to develop strategies to improve both the care experience of patients and the work environment experience of staff. Targets for performance improvement were established in both areas at the outset. Importantly, the Simply Better Program was linked to Calvary's strategic plan, organisational mission, and values and service accreditation processes.

Using the Studer Group principles for improvement, the program:

- obtained commitment from the executive team and departmental managers
- developed measurable goals and performance indicators (eg in preventable harm of patients and unplanned employee leave)
- built a service culture through training and coaching of staff and regularly providing staff with patient feedback
- developed leaders through providing managers with executive leadership training
- focused on staff satisfaction including staff rounding to engage staff in the improvement processes
- built accountability through review of measurable goals and meetings with line managers to identifier barriers and enablers
- aligned behaviours with goals and values
- provided open communication about strategies and program orientation for new staff
- instituted a process of formal recognition of staff excellence.

Specific strategies to improve patient care experience included rounding on patients (which is the systematic monitoring of patients that incorporates specific actions, done at specific intervals) and using patient whiteboards. Communication skills were improved through discharge calls and use of the AIDET tool (Acknowledge, Introduce, Duration, Explanation, Thank you) for all patient communication.

Calvary Health Care considers the Simply Better Program an organisational success, as witnessed through a range of improved outcomes. Improvements in patient care experience data occurred across the organisation, including the public mental health hospital (increased from 69th to 99th percentile), public day surgery (from 56th to 74th percentile) and private day surgery (from 47th to 88th percentile). In three years, employee satisfaction significantly improved from a 'culture of blame' to one of 'reaction/consolidation'. Following the implementation of the Simply Better program, Calvary achieved its best ever accreditation survey results and the program was awarded an 'outstanding achievement' during the accreditation. Other health services across the country have liaised with Calvary and are now adopting similar strategies.

In 2009, Calvary Health Care's Simply Better Program was awarded the Overall Winner of the ACT Quality in Healthcare Awards (Naree Stanton, Support Services Manager, Calvary Hospital, Canberra, ACT, pers comm, 2009).[132]

4.5 Case study 2: South Australia — Flinders Medical Centre Emergency Department Redesign project

Flinders Medical Centre established the Redesigning Care Program in 2003. The primary objective of this program was to address the increasing demand for emergency services in the hospital. The program used 'lean thinking' (ensuring that every step of a process adds value and serves the customer's needs) to redesign hospital processes across the entire spectrum of clinical care. Using these principles, Redesigning Care aimed to eliminate duplication and delays, and redesign patient flow to ensure that each step in the process added value and improved outcomes for patients and staff.[124] The program focused on the patient journey from admission to discharge, and one of its basic rules was to 'see things through the patient's eyes'. The work at Flinders Medical Centre has been undertaken in the emergency department, under the banner of experience-based design (EBD) based on Bate and Robert.[85]

In 2007, the Flinders Medical Centre Redesigning Care Program designed and implemented an approach to capture patient, carer and staff experience in the emergency department. The aim was to understand and improve patient, carer and staff experience when accessing the emergency department at Flinders Medical Centre; and to trial the EBD methodology as a sustainable approach to patient and carer participation within the Flinders Medical Centre management system.[124] In 2007–08, data were generated through observations of 22 patient and carer real-time experiences, 14 patient and carer reflective interviews and 29 staff interviews.

Some of the solutions for action related to recommendations for physical changes and resulted in a capital works redevelopment of the emergency department at Flinders Medical Centre to increase patients' comfort and improve their environment. These solutions included a physical redevelopment that reduced noise levels and improved visual management by staff. The main solutions outside of the redevelopment involved communication processes and team work. The program was considered successful in improving team work and improving communication between staff and patients.[74]

5 Making progress on patient-centred care in Australia

Current and emerging policy points to the need for health services to be patient centred, but healthcare services can often find it difficult to transform care delivery. Most services can readily put patient charters and informed consent policies in place, but many also find it hard to actively change the way care is delivered, and struggle to involve patients and learn from their experience.[2] There is little guidance on how to implement patient-centred care in health services in Australia.

This section highlights strategies and makes recommendations that practitioners, managers and policy makers can use to establish and maintain patient-centred care. Sections 5.1 and 5.2 consider the key issues of redefining quality in health care and in performance monitoring, and give policy-level recommendations. Practical strategies and service-level recommendations are discussed in Section 5.3, and checklists to help organisations assess their readiness to implement patient-centred care are presented in Appendix A.

5.1 Refocusing the way we look at quality in health care

Patient-centred care necessitates a change in the way policy makers and regulators think about the quality of health care. The traditional approach to health care focuses on clinical, therapeutic and diagnostic effectiveness, and cost-effectiveness as measures of health outcomes.[9] By contrast, patient-centred care takes a broader view; for example, the Next Stage Review defines quality as consisting of patient safety, clinical effectiveness and patients' experience.

In the US, the IOM report identified patient-centred care as one of six quality aims for improving care.[1] Berwick[133] states that incorporating patient-centred care into health care as we currently know it will involve some radical, unfamiliar, and disruptive shifts in control and power, out of the hands of those who give care and into the hands of those who receive it. The starting point for change will therefore be affirming patient-centred care as a dimension of quality in its own right, and not just through its effect on health status and outcomes.[133]

In Australia, it is recognised the traditional model of patient safety, drawn from high-risk industries, is not patient-centred enough to be used as a comprehensive approach to improving safety and quality.[134] Patient-centred care is now considered to be an integral dimension of quality health care in Australia: the ACSQHC's proposed National Safety and Quality Framework, described in Section 3.3, positions patient-centred care as one of three dimensions of safety and quality.[7] Various state and territory safety and quality frameworks acknowledge the importance of patient-centred concepts as a dimension of quality, although they do not generally define patient-centred care as a dimension of quality in its own right.

Recommendation 1

Policy makers and regulators should include patient-centred care as a dimension of quality in its own right in strategic and other policy documentation.

5.2 Performance monitoring — towards standardised measurement

All states and territories in Australia are increasing their activity in measuring healthcare quality, especially patient experiences. Most jurisdictions collect a variety of patient feedback, including national and local surveys, complaints data and web-based feedback.

In the US, the IPFCC[90] states that processes and outcomes associated with quality, safety and experience of care should be measured in order to embed patient-centred care in the health system. They also suggest that there is a need to link funding and accreditation to patient satisfaction measurements and to assure accountability by requiring accurate public reporting.[90]

According to the International Alliance of Patients' Organizations,[9] the traditional approach to defining quality has lacked the necessary concepts of patient-centred care, because there are no indicators or criteria to determine the level of patient-centredness in health care. Where indicators do exist, they are not standardised across service providers or jurisdictions, making the necessary comparisons and benchmarking needed to improve service delivery difficult.

In Australia, approaches to the measurement of quality in health care are increasingly being standardised across the country. There is also a need to standardise measures of patient care experience and indicators that can be used at the service level across the states and territories, and healthcare sectors.

Steps are being made in this direction. The new national-level Performance and Accountability Framework indicators will measure traditional operational and clinical performance indicators, as well as patient-reported quality based on the patient experience[94] (outlined in Section 3.2). These performance indicators will facilitate transparent monitoring across services and jurisdictions, setting a minimum standard of healthcare quality that each provider should adhere to. Adherence to the standards will be tied to funding.[94]

Currently, different organisation, jurisdictions, and healthcare sectors use different survey tools, preventing the compilation of national-level data. In standardising the collection of patient experience data, the data collection tools should address the principles of patient and family-centred care, including responsiveness, respect, information sharing and collaboration.

Recommendation 2

Patient survey tools should include a core set of items standardised at a national level to enable the collation and comparison of patient care experience data in key healthcare settings.

Recommendation 3

Patient surveys used to assess patient care experience need to include questions specifically addressing recognised patient-centred care domains and assess more than patient 'satisfaction'.

Recommendation 4

Implementation of healthcare funding models incorporating performance-based payments should include 'improving patient care experience' as an integral indicator of health service quality improvement.

Australia is also improving transparency. State and territory health departments are increasingly reporting patient survey and experience data publicly, with survey results available on departmental websites. In May 2010, the New South Wales Bureau of Health Information released a detailed report of patient care experience data for overnight and day-only patients. Data on specific area health services and individual hospital performance were made publicly available.

Some international initiatives may be useful models for Australia to follow to improve transparent data reporting. For example, in the UK, individual stories of patient and family perceptions of care complement survey data (see Section 2.1). In the US, the Department of Health and Human Services publicly reports all hospital data for patient care experience, processes of care (eg infection prevention) and outcomes of medical conditions and surgical conditions (eg mortality and surgical readmission rates) through the Hospital Compare website.

Recommendation 5

To improve transparency, Australian policy-makers and regulators should make data regarding patient care experience in health services publicly available via websites.

5.3 Organisational strategies

This section considers practical strategies that healthcare service providers can introduce to improve patient-centred care, as well as tools for assessing current care delivery. Key characteristics of organisations that have successfully transformed care delivery to patient-centred care include:

- having committed senior leadership
- using data to drive change — using regular collection and feedback of patient care experience
- engaging patients, family and carers as partners
- resourcing change responsively
- building staff capacity and a supportive work environment
- ensuring accountability at all levels for improving patient care experience
- fostering an organisational culture that strongly supports learning and improvement.[14 87]

Each organisation needs to reflect on its current care provision and feedback from patients before deciding on strategies to put in place to improve patient-centred care and how this should be done.

Data driving change: using regular collection and feedback of patient care experience

Internationally, few organisations have adequate systems for coordinating patient experience data collection and assessing its quality, or for learning from and acting on the results in a systematic way.[23] To gain a clear picture of patient care, it is important to use a range of sources to collect information about patient experience. Patient surveys that are conducted on a regular basis and reported throughout the organisation, from executive level to the ward, provide staff and management with feedback about care from the service users' perspective.

Patient surveys and complaints data are useful to reveal large-scale trends,[23] and help those responsible for service planning and governance. Although they are an important foundation, patient experience survey scores can present a limited picture. Detailed information about specific aspects of patients' experiences is likely to be more useful for monitoring performance of hospital departments and wards.[135]

These measures need to be complemented with patients' personal stories, which can have a direct impact on those responsible for care.[74] The stories allow carers to 'see the person in the patient': they bring patients' experiences, feelings and concerns to life in a way that connects with service providers' own experiences, feelings and concerns.[15 74] Other methods for collecting care experience information include conducting focus groups, providing patient journals, having patient advisers round on wards, using 'mystery shoppers' who report to governance committees, and asking patients to tell their story at executive meetings.

Recommendation 6

Healthcare service executives and managers should ensure that systems are in place for the regular collection and reporting of patient care experience data through quantitative patient surveys and qualitative, narrative-based sources.

Recommendation 7

Healthcare service executives and managers should ensure that organisational approaches to quality improvement include feedback about patient care experience — alongside clinical and operational data — when determining health service action plans.

Recommendation 8

Healthcare service executives and managers should contribute to the evidence base for patient-centred care by recording and publishing changes in key organisational and patient outcome metrics over time.

The importance of committed senior leadership

Schall et al[136] state that one of the best lessons learned at their facility in implementing patient-centred care is that 'leadership matters'. At the organisational level, the way in which leaders set policies and reinforce the importance of improvement greatly affects progress towards improvement. When leadership and staff priorities do not align, poor quality is likely.

Key characteristics of successful patient-centred care organisations are a strategic vision and mission that clearly articulate the organisation's focus on patients, and organisation leaders who continuously convey this message at all levels of the service.[14 87] Effective senior leaders often use a personal story to engage staff in the organisational mission.[87]

Changing organisational culture from a provider focus to a patient focus can be one of the most difficult barriers to establishing a patient-centred approach to care. Various change-management strategies can be used to shift culture over time, including Kotter's Eight Steps incorporating creating a vision for change, communicating the vision, removing barriers, creating short-term wins, and anchoring the change in organisational culture.[137] Programs by organisations such as the Studer Group (highlighted in Section 2.5) may help engage staff, particularly those in priority areas for improvement.

Designating a senior manager or executive with responsibility for implementing patient-centred care and designating champions who model patient-centred behaviours also sends a clear message to the organisation about the importance of this approach.

Recommendation 9

Healthcare service executives and managers should develop a shared patient-centred mission that senior leaders continually articulate to staff to promote the implementation of patient-centred care.

Engaging patients, families and carers as partners

Promoting patient engagement can focus on service-level improvement and individual care. For example, a service aiming to improve safety by partnering with individual patients may actively engage patients in handovers, medication reviews, and planning and managing their own care. Strategies for partnering with patients, family and carers at the service level include:

- partnering in service redesign and co-design projects
- involving patients, families and carers in educational programs for healthcare professionals and administrative leaders
- establishing a patient liaison office involving patient representatives
- establishing patient and family advisory councils or committees
- involving patients, families and carers in key organisational committees such as
  - strategic planning
  - safety and quality improvement
  - medical review
  - risk management

- facility planning and design
- staff interview panels and induction
- information technology
- ethics and research.

Appointing patients to advisory or governing structures is emerging as a successful consumer involvement activity that links the individual experience level of health care to the organisational and systems levels. For example, in the UK, experience-based co-design facilitated patients provision of feedback to the board of governors by telling their stories at board meetings.[138] In Canada, Patients For Patient Safety champions are appointed to consumer advisory committees of safety and quality structures,[139] and at the provincial level patients and families are being invited onto patient safety advisory committees at hospitals. In the UK, the *Local Government and Public Involvement in Health Act 2007*[70] empowers local networks to assess the healthcare needs and experiences of patients and citizens, and make recommendations to governors of health services.

Implementing organisational procedures and policies to support these patient-engagement strategies will ensure that the strategies are sustainable.[87] In this regard, the ACSQHC is developing a service-level standard for healthcare providers that outlines principles for partnering with patients, family and carers to improve quality of care within services. All the standards being developed as part of the set of National Safety and Quality Healthcare Service Standards need to be considered in the context of partnering with patients.

Healthcare services should be encouraged to evaluate the effect of these partnerships to build the evidence base for partnering with patients, families and carers to improve quality of care.

Recommendation 10

Healthcare service executives and managers should develop and implement policies and procedures for involving patients, families and carers in their own care and, at a service level, in policy and program development, quality improvement, patient safety initiatives and healthcare design.

Recommendation 11

Healthcare service executives and managers should ensure that the service meets the ACSQHC National Safety and Quality Healthcare Service Standard for 'Partnering with Consumers'.

Resourcing improvement of care delivery and environment

To successfully establish a patient-centred care approach, organisations need to address changes in response to the areas of need identified through patient feedback,[87] and consult with patient advisers and other relevant experts before deciding on strategies. Surprisingly, in successful patient-centred services, the improvements that patients suggested were not necessarily expensive,[87] and patient advisers in a number of US services are viewed as the force for making health care more affordable.

Examples of responsive changes made by hospitals to improve patient-centred care include:

- nurses hourly rounding on wards
- providing welcoming facilities for families
- reviewing hospital signage from a patient perspective
- providing a new style of hospital gown to afford dignity to patients
- engaging volunteers to act as concierges or patient navigators
- redesigning waiting areas
- introducing communication strategies to keep patients and families informed.[87]

These examples demonstrate the types of responsive changes that organisations can make and are not an exhaustive list. Other strategies are presented in sections 2 and 4, along with resources to help foster a patient-centred approach. Other resources, including checklists, are summarised in the appendices to this discussion paper.

Resourcing the improvement of the quality of the physical care environment strongly correlates with improved patient care experience and other health and business outcomes.[140] The Planetree approach in the US links architecture and physical space with healing, and outlines principles for healthcare service design (see Appendix B).When considering new healthcare facilities, or when renovating or reviewing existing facilities, organisations should consider patient-centred principles and seek patient adviser input; for example, through experience-based redesign and co-design strategies (see Section 2).

Recommendation 12

Healthcare service executives and managers should resource patient-centred changes to care delivery based on patient feedback and consumer input.

Building staff capacity and a supportive work environment

The strategies highlighted in Section 2 to support staff, such as staff and practice development, values training, communication skills training and staff satisfaction programs, will also help establish patient-centred care in Australia. This has already begun in New South Wales, where NSW Health is using a Studer Group program to improve communication between staff and patients.

According to Coulter,[141] the basic competencies required by individual health professionals to be patient-centred include:

- understanding the patient's perspective, expressing empathy and providing appropriate support
- guiding patients to appropriate sources of information on health and healthcare
- educating patients on how to protect their health and prevent occurrence or recurrence of disease

- eliciting and taking account of patients' preferences
- communicating information on risk and probability
- sharing treatment decisions
- providing support for self-care and self-management
- working in multidisciplinary teams
- managing time effectively.

Patient-centred organisations focus on increasing their staff skills to support patient-centred care delivery.[87] Strategies to achieve this include:

- training staff in communication skills
- adopting communication techniques such as AIDET (Acknowledge, Introduce, Duration, Explanation, Thank you)
- training all staff in patient-centred values
- training all staff in customer service techniques
- tailoring the workforce through selection of staff with a commitment to the mission of the organisation
- integrating discussion of patient-centred values into staff orientation sessions (for example, the chief executive could open the induction session with a discussion of values)
- actively involving patients and families in education programs for healthcare professionals, managers and executives
- holding education sessions for healthcare professionals where patients and families share their experience of care
- involving patients and families in educating junior healthcare professionals and in affiliated programs for undergraduates.

Focusing on the staff work environment is an important component of 'caring for the care givers'. Exemplary patient-centred health services achieve this by visibly celebrating the successes of staff in improving patient care experience (such as public acknowledgment by the chief executive, awards for achievements, or features in annual reports, hospital intranet and newsletters).[87] This approach recognises that the workforce is the healthcare service's most important asset and aligns with the move towards 'person-centred' organisations, where both patients and staff are valued.[14] Person-centred organisations also use feedback from staff surveys on the work environment to improve the work culture and processes. Giving significant attention to staff satisfaction also acknowledges the link between employee satisfaction and patient satisfaction.[142]

Recommendation 13

Healthcare service executives and managers should implement training strategies tailored to building the capacity of all staff to support patient-centred care.

Recommendation 14

Healthcare service executives and managers should focus on work environment, work culture and satisfaction of staff as an integral strategy for improving patient-centred care. Workforce surveys and review of staff recruitment and retention rates should be undertaken at regular intervals to monitor work environment.

Accountability at all levels for improving patient-centred care

Successful patient-centred care organisations establish clear lines of accountability for staff at all levels, making each person responsible for improving patient care experience.[87] Individual accountability can be reinforced through performance reviews. A range of strategies can be put in place to promote staff accountability, including:

- incorporating responsibility for improving patient care experience in job descriptions
- explaining at orientation for new staff that they are responsible for the experience of care that a patient has
- considering patient feedback during staff performance reviews, including sharing patient stories
- linking promotions or performance bonuses to improving quality indicators, including care experience
- incorporating patient care experience metrics into unit, departmental and organisational performance monitoring and reporting
- adopting a motto that embodies accountability (eg 'Every patient — Everyone's responsibility' [Pat Sodomka, Sebrio Vice President, Patient and Family Centered Care, MCG Health Inc and Director, Center for Patient and Family Centered Care, Medical College of Georgia, pers comm, 2009])
- linking quality metrics, including patient care experience, to performance reviews of organisational governance bodies and chief executives
- ensuring that the agenda for boards or governance committee meetings includes a strong emphasis on reviewing quality issues and quality performance data, including patient care experience.[87]

Although a number of options are outlined here, linking improvement of care experience to individual accountability through performance review is a starting point relevant to all staff.

Recommendation 15

Healthcare service executives and managers should integrate accountability for the care experience of patients into staff performance review processes.

An organisational culture that strongly supports learning and improvement

The leading organisations in patient-centred care have a culture of learning and strongly support change and improvement.[87] These organisations use many aspects of 'learning organisation' theory and have characteristics described in 'learning culture' frameworks. Learning organisations have systems, mechanisms and processes in place that are used to continually enhance the capabilities of those who work with it or for it, to achieve sustainable objectives.[143] They adapt to external environmental forces, constantly improve their own ability to change, promote individual and collective learning, and use these lessons to improve outcomes.

Healthcare organisations can commit to continuous quality improvement using approaches such as Deming's cycle of 'Plan, Do, Study, Act'. Focusing on needs analysis, performance measurement and improvement is crucial. Learning organisations also support a culture that values people, stimulates new ideas, develops teamwork and adopts staff recognition systems. Successful patient-centred organisations often take the next step of linking individual performance accountability with organisational performance.

Learning organisations are also open to learning from past failures.[144] A number of healthcare services have refocused care delivery for patient-centred care in response to tragic events in their facility, such as the death of a patient or severe harm of a patient during care. Such tragic events can cause an organisation to examine its own values and its processes for promoting safety and quality improvement. Listening to patients and involving patients and families in improving care delivery is a positive response to these tragedies.

Events at the Dana Farber Cancer Institute, Boston, US, are well known in the area of 'turning tragedy into a positive outcome.' In 1994, Betsy Lehman, a health journalist from the Boston Globe newspaper, died after she was accidentally given an overdose of cyclophosphamide, a chemotherapy drug she was receiving for breast cancer treatment. The institute began asking patients and families 'how can we do better?' The incoming chief operating officer, James B Conway, organised town hall meetings for patients and was inundated with people providing feedback about improving care delivery. Sixteen years on, the Dana Farber Cancer Institute is a leader in patient-centred care, with patient advisers involved throughout the organisation.

Learning from others' experiences can be invaluable and healthcare organisations can link with service providers who have experience of learning from patients and families.

Recommendation 16

Healthcare service executives and managers should foster a culture of learning within the organisation, equally learning from successes and failures, including tragic events, to promote patient-centred care.

6 Next steps

A patient-centred focus can improve healthcare quality and outcomes by increasing safety, cost-effectiveness, and patient, family and staff satisfaction.[145]

The ACSQHC developed this paper as a first step in a discussion of the concepts of, and evidence for, patient-centred care, and the applicability of international models to the Australian context.

The paper is aimed at practitioners, managers and policy makers and outlines the background, context, evidence and impetus for improving quality and safety through patient-centred care. It gives examples of both local and international success stories, where systems and practices have become more responsive to the needs and desires of patients. It also provides examples of models and tools, and recommendations that focus on what organisations can do to foster a patient-centred approach to health care.

To support and facilitate patient-centred care, the ACSQHC will work with consumers, patients and the general public to ensure that the principles, models and approaches it supports and endorses, including the work of other organisations, are appropriate, effective and modelled by the organisation's actions.

To this end, the ACSQHC is holding an open consultation on this discussion paper, and is seeking the views of a broad range of stakeholders including health services, practitioners, patients, consumers and policy makers on:

- the discussion paper itself, including the content, relevance and accuracy of the issues and examples raised in the paper; any gaps in the information or examples provided; and any areas that could be strengthened
- future directions, including opportunities for developing or supporting initiatives to improve the uptake of patient-centred care approaches within health services.

After the consultation process, the ACSQHC will prepare a consultation report that outlines the themes and key issues raised in the submissions and the overall response to the discussion paper. This report will be publicly available and information from the consultation process will contribute to further refinement of the discussion paper.

Both the discussion paper and the outcomes of the consultation process will be used to inform activities to improve patient-centred care in Australia, including the work of the ACSQHC and other organisations in this field.

Your views

The ACSQHC is seeking responses to this discussion paper. Questions of particular interest include:

The discussion paper

- Did you find the discussion paper useful?
- Were the principles of patient-centred care clear and understandable?
- Is there any terminology that needs further exploration or explanation?
- Are there any concepts that need further exploration or explanation?
- Would any of the approaches or strategies outlined in the discussion paper be particularly suitable or unsuitable for you or your organisation?
- Is the section on support provided by leading international organisations and their resources applicable to you or your organisation?
- Are there additional significant Australian or international initiatives or strategies that need to be highlighted in the discussion paper?
- Are there barriers to implementing patient-centred care in the Australian context that need to be explored in the discussion paper?
- How do the contents of the discussion paper, including the recommendations, apply to your service?

Future directions

- How could you or your organisation work better in partnership with patients?
- How do you promote greater uptake of patient-centred care models? Who needs to be involved in this?
- Is there more work that needs to be done in this field, specific to the Australian context?
- What type of tools or support would you need to make your organisation more patient centred?
- Are there activities, frameworks, strategies or protocols that could be provided that would help your organisation become more patient centred?
- Is there infrastructure or support that could be provided that would help your organisation become more patient centred?
- How could the ACSQHC or other organisations support or facilitate greater use of patient-centred care models?

Submissions do not have to address these questions and may respond to other issues raised in the discussion paper.

All submissions are welcome and will be accepted up to 17 December 2010. Submissions should be marked 'Patient-and Consumer-Centred Care' and forwarded to:

GPO Box 5480
Sydney NSW 2001

or emailed to:
mail@safetyandquality.gov.au

Appendix A Assessing organisational readiness to implement patient-centred care

Appendix A contains:

- Tool 1: a checklist for assessing organisational readiness for implementing patient-centred care, written by Dr Karen Luxford in the context of the Australian healthcare system.
- Tool 2: an assessment tool written by the Institute for Patient- and Family-Centered Care in 2008, entitled *Where do we stand? An Assessment Tool for Hospital Trustees, Administrators, Providers and Patients and Family Leaders*.

Tool 1: Patient-centred care organisational status checklist

Check your organisational readiness for improving patient care experience

1. Are you collecting patient care experience data? Yes No (go to Q8)
2. How are you collecting patient care experience data?
3. Why are you collecting patient care experience data?
4. How often are you collecting patient care experience data?
5. How are you using the data/information collected?
6. Is the data about patient care experience being reported?
7. To whom is it being reported?
8. Is staff satisfaction monitored?
9. Is there a ‘dashboard’ of performance metrics monitored by the organisation?
10. Does the ‘dashboard’/set of metrics include patient care experience indicators?
11. What is the mission/vision of the organisation?
12. What is the main message to staff from the leadership? CEO? Organisational governance?
13. Is the culture of the organisation supportive of change? Open to learning?
14. Are successes by staff visibly celebrated?
15. What is the current area of focus for staff development?
16. Have staff training activities included communication skills training or patient-centred values?
17. Are patients and families considered ‘partners’ in care?
18. Are any patient or family/carer representatives involved in any organisational committees?
19. If so, which areas do these committees cover?
20. Any future plans for engaging patients at a service level within the organisation?
21. Have there been any tragic events within the service from which lessons have been learnt? What did the organisation learn from these events?
22. Are families/carers considered ‘visitors’ to the service (ie restricted ‘visiting’ hours)?

Tool 2: Where do we stand?

An assessment tool for hospital trustees, administrators, providers, and patient and family leaders (IPFCC 2008)

Initial assessment

Organisational culture and philosophy of care

- Do the organisation's vision, mission, and philosophy of care statements reflect the principles of patient- and family-centered care and promote partnerships with the patients and families it serves?
- Has the organisation defined quality health care, and does this definition include how patients and families will experience care?
- Has the definition of quality and philosophy of care been communicated clearly throughout the healthcare organisation, to patients and families, and others in the community?
- Do the organisation's leaders model collaboration with patients and families?
- Are the organisation's policies, programs, and staff practices consistent with the view that families are allies for patient health, safety, and well-being?

Patient and family participation in organisational advisory roles

- Is there an organisational Patient and Family Advisory Council?
- If there is a Patient and Family Advisory Council, is patient safety a regular agenda item?
- Are patients and families members of committees and are they involved in initiatives for:
  - patient safety?
  - quality improvement?
  - facility design?
  - use of information technology?
  - pain management?
  - patient/family education?
  - discharge/transition planning?
  - palliative/end-of-life care?
  - staff orientation and education?
  - service excellence?
  - ethics?
  - diversity/cultural competency?
  - architecture and design
- Does the healthcare organisation's architecture and design:
  - create welcoming impressions throughout the facility for patients and families?
  - reflect the diversity of patients and families served?

- provide for the privacy and comfort of patients and families?
- support the presence and participation of families?
- facilitate patient and family access to information?
- support the collaboration of staff across disciplines and with patients and families?

Patterns of care

- Are family members always welcome to be with the patient, in accordance with patient preference, and not viewed as visitors?
- Are patients and families viewed as essential members of the healthcare team? For example, are they encouraged and supported to participate in care planning and decision-making?
- Do physician and staff practices reinforce that care will be individualised for patient and family goals, priorities, and values?
- Are patients and families, in accordance with patient preference, encouraged to be present and to participate in rounds and nurse change of shift?
- Is care coordinated with patients and families and across disciplines and departments?

Patient and family access to information

- Are there systems in place to ensure that patients and families have access to complete, unbiased, and useful information?
- Do patients and families, in accordance with patient preference, have timely access to medication lists, clinical information (e.g., lab, x-ray, and other test results), and discharge or transition summaries?
- Are informational and educational resources available in a variety of formats and media and in the languages and at the reading levels of the individuals served?
- Are patients and families encouraged to review their medical records and work with staff and physicians to correct inaccuracies?
- Are patients and families provided with practical information on how to best assure safety in health care?
- Are there a variety of support programs and resources for patients and families, including peer and family-to-family support?

Education and training programs

- Do orientation and education programs prepare staff, physicians, students, and trainees for patient- and family-centered practice and collaboration with patients, families, and other disciplines?
- Are patients and families involved as faculty in orientation and educational programs?

Research

- In research programs, do patients and families participate in:
  - shaping the agenda?
  - conducting the research?
  - analysing the data?
  - disseminating the results?

Human resources policies

- Does the organisation's human resources system support and encourage the practice of patient- and family-centered care?
- Are there policies in place to ensure that:
  - individuals with patient- and family-centered skills and attitudes

    are hired?
  - there are explicit expectations that all employees respect and
  - collaborate with patients, families, and staff across disciplines
  - and departments?
- Are there strategies in place to reduce the cultural and linguistic differences between staff and the patients and families they serve?

Appendix B Supportive resources

A variety of resources including assessment tools and checklists for health service providers are available online. Leading organisations and their resources are listed below in alphabetical order.

Leading organisations and their resources

Dana-Farber Cancer Institute (DFCI)

- Website:
  www.dana-farber.org
- *A Toolkit to Assist Organisations in Initiating Patient Safety Rounds:*
  www.dana-farber.org/pat/patient-safety/patient-safety-resources/patient-rounding-toolkit.html

Institute for Healthcare Improvement (IHI)

- Website:
  www.ihi.org
- *Partnering with Patients and Families to Design a Patient and Family-Centred Health Care System — Recommendations and Promising Practices* (2008):
  www.ihi.org/IHI/Topics/PatientCenteredCare/PatientCenteredCareGeneral/Literature/PartneringwithPatientsandFamilies.htm
- Resources, tools and improvement stories:
  www.ihi.org/IHI/Topics/PatientCenteredCare/PatientCenteredCareGeneral/PatientCenteredCareGeneralHome.htm

Institute for Patient- and Family-Centered Care (IPFCC)

- Website:
  www.ipfcc.org
- *Advancing the Practice of Patient - and Family-Centred Care — How to Get Started* (IPFCC 2008).
  www.ipfcc.org/pdf/getting_started.pdf
- *Partnering with Patients and Families to Design a Patient and Family-Centred Health Care System — Recommendations and Promising Practices* (Johnson, Abraham, Conway et al 2008 in conjunction with the Institute for Healthcare Improvement)
  www.ipfcc.org/pdf/PartneringwithPatientsandFamilies.pdf
- *Partnering with Patients and Families To Design a Patient- and Family-Centered Health Care System: A Roadmap for the Future — A Work in Progress* (Conway, Johnson, Edgman-Levitan, et al 2006 in conjunction with the Institute for Healthcare improvement)
  www.ipfcc.org/pdf/Roadmap.pdf
- Other IPFCC resources are available at:
  www.ipfcc.org/tools/downloads.html

Kenneth B Schwartz Center

- Website:
  www.theschwartzcenter.org
- Publications:
  www.theschwartzcenter.org/bibliography/index.html
- Schwartz Center Rounds:
  www.theschwartzcenter.org/programs/rounds.html
- Carepages website:
  www.carepages.com

The King's Fund

- Website:
  www.kingsfund.org.uk
- *Seeing the Person in the Patient: Point of Care Review Paper* (Goodrich and Cornwell 2008)
  www.kingsfund.org.uk/current_projects/the_point_of_care

Picker Institute

- Website:
  www.pickerinstitute.org/
- *The Patient-Centered Care Improvement Guide* (2008) (in conjunction with Planetree):
  www.patient-centeredcare.org/title.html
- *Self-assessment Tool*:
  www.patient-centeredcare.org/inside/selfassessment.html

Picker Institute Europe

- Website:
  www.pickereurope.org/
- A comprehensive review of the worldwide evidence of what works to engage patients and the public in healthcare.
  www.investinengagement.info

Planetree

- Website:
  www.planetree.org
- *The Patient-Centered Care Improvement Guide* (2008) (in conjunction with the Picker Institute):
  www.planetree.org/publications.html
- Other Planetree publications are available at:
  www.planetree.org/publications.html

Studer Group

- Website:
  www.studergroup.com

World Health Organization

- Patients for Patient Safety:
  www.who.int/patientsafety/patients_for_patient/statement/en/index.html

Patient-centred care strategies

Collecting patient feedback

United Kingdom

- UK Care Quality Commission's website:
  www.cqc.org.uk/usingcareservices/healthcare/patientsurveys.cfm
- UK survey coordination centre website:
  www.nhssurveys.org
- UK General Practice Patient Survey results:
  results.gp-patient.co.uk/report/main.aspx
- UK National Cancer Action Team Patient Experience:
  www.cancerinfo.nhs.uk/
- *Understanding What Matters — A Guide to Using Patient Feedback to Transform Services*, UK Department of Health (2009b)
  www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_099780

United States

- AHRQ CAHPS survey website:
  www.cahps.ahrq.gov/default.asp
- AHRQ National CAHPS Benchmarking Database:
  www.cahps.ahrq.gov/content/ncbd/ncbd_Intro.asp?p=105&s=5

Australia

- NSW Health patient survey website:
  www.health.nsw.gov.au/hospitals/patient_survey/index.asp
- NSW Bureau of Health Information website:
  www.bhi.nsw.gov.au
- Victorian Department of Health Patient Satisfaction Monitor survey website:
  www.health.vic.gov.au/patsat/
- ACT Health Healthcare Survey website:
  www.health.act.gov.au/c/health?a=da&did=10101609&pid=1254789026
- Queensland Health Patient Satisfaction survey website:
  www.health.qld.gov.au/quality/pat_sat_survey/patsat.asp
- SA Health Patient Evaluation of Health Services (PEHS) survey website:
  www.health.sa.gov.au/pros/Default.aspx?PageContentMode=1&tabid=44,

- WA Health *Toolkit for Collecting and Using Patient Stories for Service Improvement in WA Health*
  www.health.wa.gov.au/hrit/docs/A_toolkit_for_collecting_and_using_patient_stories.pdf

Real-time patient feedback

- UK website offering patient reviews for healthcare services:
  www.iwantgreatcare.com
- UK website offering patient reviews for healthcare services:
  www.patientopinion.org.uk

Patient choice of provider

- NHS Choices:
  www.nhs.uk/Pages/HomePage.aspx
- NHS survey results for patients' experience of choice
  www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsStatistics/DH_085329
- US Department of Health and Human Services Hospital Compare website:
  www.hospitalcompare.hhs.gov

Rights-based charters and codes

- *The NHS Constitution*:
  www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_113613
- American Hospital Association (AHA) Communicating with patients website:
  www.aha.org/aha/issues/Communicating-With-Patients/index.html,
- CMS *Important Message From Medicare Notice*: www.virginia.edu/uvaprint/HSC/pdf/CMS-R-193.pdf,
- *Australian Charter of Healthcare Rights*:
  www.safetyandquality.gov.au/internet/safety/publishing.nsf/Content/com-pubs_ACHR-pdf-01-con/$File/17537-charter.pdf
- Australian Commission on Safety and Quality in Healthcare publications and other information about the Australian Charter of Healthcare Rights:
  www.safetyandquality.gov.au/internet/safety/publishing.nsf/Content/PriorityProgram-01
- *London Declaration for Patient Safety*:
  www.who.int/patientsafety/patients_for_patient/London_Declaration_EN.pdf

Redesign and experience-based design

- NHS Institute for Innovation and Improvement Case Studies:
  www.institute.nhs.uk/quality_and_value/experienced_based_design/case_studies.html
- ACT Health Access Improvement Program:
  www.health.act.gov.au/c/health?a=&did=10109188
- NSW Health patient and carer experience:
  www.health.nsw.gov.au/performance/pcexperience.asp

- Victorian Department of Health Improving the patient experience program: www.health.vic.gov.au/managementinnovation/resources
- Queensland Health redesign projects: www.health.qld.gov.au/cpic/service_improve/projects.asp

Improving complaints processes

- UK Department of Health guide and advice sheets: www.dh.gov.uk/en/Publicationsandstatistics/Publications/index.htm
- *Listening, Responding, Improving: A Guide to Better Customer Care*, UK Department of Health (2009): www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_095408

Australian state and territory programs and initiatives

- ACT Health website: www.health.act.gov.au/c/health?a=&did=10101609
- Health Consumers' Council website: www.hconc.org.au/
- NSW Health Essentials of Care program: www.health.nsw.gov.au/nursing/projects/eoc.asp;
- NSW Health Clinical Leadership Programme: www.clinicalleadership.com/index.php?option=com_frontpage&Itemid=1,
- NSW Health 'Take the Lead' project: www.health.nsw.gov.au/nursing/projects/take_the_lead.asp
- Queensland Health Healthcare Experience Improvement Homepage for consumers: www.health.qld.gov.au/cpic/hlthcr_exp_improve/hlthcr_exp_improve.asp
- Victorian Quality Council website: www.health.vic.gov.au/qualitycouncil/activities/index.htm
- WA Office of Safety and Quality in Healthcare WA Patient First program: www.safetyandquality.health.wa.gov.au/involving_patient/patient_1st.cfm

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Editorials

Patient–clinician communication research for 21st century health care

INTRODUCTION

The pandemic has accelerated the shift to the unchartered waters of virtual healthcare delivery. Unprecedented numbers of consultations are now held remotely, introducing a range of novel challenges and uncertainties. One major area of concern relates to potential impacts on patient–clinician communication. Research on this has been limited so far, but early studies have indicated that there are critical differences between how patients and clinicians interact over telephone or video compared to face to face. These include patients raising fewer issues, clinicians seeking less information, and less psycho–social talk.[1]

The rapid development of a strong evidence base is urgently needed to understand if, when, and how communication is affected. With this objective in mind, we discuss major lines of inquiry in patient–clinician communication research, and argue why theoretical and methodological approaches from social sciences and linguistics will be particularly important in supporting future endeavours to identify and mitigate risks associated with virtual consultations.

DIFFERENT PARADIGMS IN PATIENT–CLINICIAN COMMUNICATION RESEARCH

Despite decades of research, patient–clinician communication remains imprecisely defined and operationalised in health research.[2] A large area of work has focused on clinician (and patient) skills and behaviours. Adopting umbrella constructs such as 'patient-centred communication' and 'shared decision making', these studies measure certain observable communication behaviours (such as information giving and socio-emotional talk) using observational coding and rating scales. Through such quantification, this strand of research has enabled the identification of associations between patterns of communication and critical outcomes (for example, satisfaction, malpractice claims, and adherence).[3,4]

Yet, while such research will continue to be important for understanding systematic differences in content and outcomes between consultation modes, they must be accompanied by a focus on the interactional, situated, and process-like functioning of communication.

Deviating from popular methods in mainstream health research, which tend to be informed by a positivist epistemology, communication research in dominant paradigms of (socio)linguistics and social sciences examine communication as a dynamic, interactional process that is simultaneously shaped by both interactants, the context in which it takes place, and wider sociocultural norms and rules.[5] As these fields recognise that people rely on context to interpret the meaning of talk, and can have very different interpretations of the same words, methods tend to focus on the processes through which patients and clinicians create and maintain shared understanding through multi-turn communication: using the interaction, rather than the individual, as the unit of analysis.

As we argue below, the operationalisation of communication as a situated, social process will be essential to identify and mitigate potential sources of miscommunication associated with telephone or video consults, and to understand how talk functions in virtual context.

RESEARCH PRIORITIES FOR PATIENT–CLINICIAN COMMUNICATION IN VIRTUAL CONSULTATIONS

From isolated behaviours to situated interactions

Behavioural approaches to communication do not give insight into problems at the interpersonal level (such as misunderstandings and missed information), which remain a problem in traditional care, and are expected to occur at an even greater rate in virtual consultation modes. Despite considerable investment in clinician competency and skills, studies consistently show that clinicians frequently overestimate the patient's understanding of their advice and often do not suspect miscommunication has occurred, resulting in missed opportunities to mitigate their communication accordingly.[6,7] These problems are likely greater in video or telephone consultations where the context is less shared (for example, the clinician and patient are in a different spatial location), the channels for communicating are reduced (for example, missing non-verbal cues), and signals of communication problems (such as hesitations, vagueness, agitation, and silence) may be harder to detect and interpret (for example, due to transmission delays and absence of non-verbal cues). Indeed, a recent study found that latency in video consultations can result in misinterpretations of silences and gaze behaviour, resulting in misplaced practices for resolving conversational issues.[8]

While popular coding methods, such as the Roter interaction analysis system,[9] are critical for understanding systematic differences in observable behaviour between consultation modes, they give little indication of the success of those behaviours in terms of whether a shared understanding was achieved. This is significant as the quality of mutual comprehension will in many cases be a mediator between communication patterns and distal outcomes such as treatment adherence or disease self-management.[3] Similarly, rating scales can help judge clinician behaviours against a predefined standard but do not provide insight into whether these behaviours were adaptive in the consultation's specific context, or how it was interpreted by the patient. While patients and clinicians agree on the components that comprise 'good communication' (such as information seeking and socio-emotional communication), they often disagree on whether these skills are present in a consultation.[10]

To capture relational discrepancies between patients and clinicians, we require methods that focus on the interactive and procedural nature of communication: how patients and clinicians simultaneously contribute to shaping interaction sequences (for example, through responding and turn-taking), how they negotiate their potentially differing individual (or cultural) interpretations of what is said, and how this is interdependent with the conversation's (in)direct context. By adopting the interaction (rather than behaviour) as the unit of analysis, dialogical methods (such as interpersonal perception methods and conversation analysis) can offer insight into how misunderstandings emerge and can be resolved, bringing into focus reciprocal behaviours, such as mutual checking, verifying, and repairing, to secure and maintain a mutual understanding of issues discussed. If misunderstandings are indeed more likely to occur in video or telephone consultations, then discovering and repairing misunderstandings through dynamic and interactive approaches becomes all the more important.

The increased adoption of virtual care also represents a major opportunity for advancing this field, which is distinctive in requiring naturally occurring data, with minimal interference from researchers. The operational challenge of obtaining such data has historically limited progress, but with unprecedented rates of consultations now held virtually there is an imminent opportunity to advance this body of work.

Towards contextual and multilevel approaches

A further consideration, especially critical for virtual healthcare delivery, is the need to adopt methods that allow for a broader understanding of how communication interrelates with context at multiple levels. This includes direct factors, such as patient and clinician characteristics (for example, language proficiency and clinician expertise), their interaction (for example, race/gender concordance and quality of their relationship), and their surrounding environment (such as characteristics of the consultation mode). It is evident that technical factors in virtual consultations, such as latency, sound or camera quality, lower consultation length, or transmission delays are fundamental. Other non-technical factors, such as language proficiency, degree of privacy, extent of shared background information, or the presence of health advocates (such as relatives and caregivers), may also play a greater or different role in online platforms compared to face-to-face consultations, and can introduce new challenges to equity in the quality of care. The sudden widespread adoption of virtual care also represents a major shift in indirect macro-level structures (for example, organisation of health delivery, media, and power relations), which have remained somewhat understudied in communication research.[11] Neglecting these contextual dimensions, or treating them in isolation, risks leaving novel hazards associated with remote care undetected.

Studies that combine ethnography and analysis of talk to gain a contextual understanding of patient–clinician interactions remain relatively underrepresented, but are growing in number, and are a promising development towards detecting, preventing, and resolving conversational problems associated with virtual modes. For example, a recent study identified a number of conversational strategies to address interactional challenges that occur in video consultations, such as acts of repeating and paraphrasing to prevent potential loss of information due to audio or video glitches.[12] Similarly, an ethnomethodological study on video consultations by Pappas and colleagues highlighted the centrality of the opening phase in video consultations as a moment of negotiating roles and structuring talk.[13] Future work requires wider investment in such interactional and context-sensitive approaches to communication across a range of circumstances to ensure safe and high-quality virtual care.

CONCLUSION

Meeting the challenges posed by healthcare's digital transformation demands rapid investment in multidisciplinary research to ensure novel risks are identified, prevented, and mitigated. Studies of patient–clinician communication that adopt an interactional and multifaceted approach will have ever greater importance. The shift to virtual consultations also represents a huge opportunity for advancing the field of patient–clinician communication. Never before have consultations via audio and video platforms been implemented at such a large scale, offering a low-cost and feasible means to capturing and analysing naturally-occurring patient–clinician interactions across different contexts, settings, and groups. If studied appropriately, this can fundamentally recalibrate and enrich our understanding of patient–clinician communication, its functioning across different contexts, and how it can be improved.

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"Best Practice" for Patient-Centered Communication: A Narrative Review

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Abstract

Background Communicating with patients has long been identified as an important physician competency. More recently, there is a growing consensus regarding the components that define physician-patient communication. There continues to be emphasis on both the need to teach and to assess the communication skills of physicians.

Objective This narrative review aims to summarize the work that has been conducted in physician-patient communication that supports the efficacy of good communications skills. This work may also help to define the physician-patient communication skills that need to be taught and assessed.

Results A review of the literature shows it contains impressive evidence supporting positive associations between physician communication behaviors and positive patient outcomes, such as patient recall, patient understanding, and patient adherence to therapy. There is a consensus about what constitutes "best practice" for physician communication in medical encounters: (1) fostering the relationship, (2) gathering information, (3) providing information, (4) making decisions, (5) responding to emotions, and (6) enabling disease- and treatment-related behavior.

Conclusions Evidence supports the importance of communication skills as a dimension of physician competence. Effort to enhance teaching of communication skills to medical trainees likely will require significant changes in instruction at undergraduate and graduate levels, as well as changes in assessing the developing communication skills of physicians. An added critical dimension is faculty understanding of the importance of communication skills, and their commitment to helping trainees develop those skills.

Introduction

For decades, medical educators and patient advocates have stressed the importance of communicating with patients and their families during medical encounters, and communication has emerged as an important physician competency.[1–5] There is a consensus about the essential elements of communication skills relevant to medical encounters and the need to teach those skills to medical trainees.[6–11] Further, the assessment of communication skills has become an explicit component of medical education curricula and of formative and summative examinations for physicians in the United States and Canada.[10,12–15]

At the same time, patients report that many of their informational and emotional needs remain unmet during encounters with their physicians.[16–20] Currently, training and role modeling of communication and interpersonal skills in medical education is relatively brief, is placed early in the curriculum, and often is not reinforced in the latter stages of training.[21–25] The decline in empathy and communication as trainees progress through programs has been well documented across multiple studies.[26,27] If the medical education community is dedicated to renewing its commitment to teaching excellence in the communication skills of physicians, which some have called for,[22] there is a need for better understanding of the evidence that supports the efficacy of good communication skills, and use of that evidence to define what should be taught to medical students and residents. In this era of milestone development, it seems timely to offer scientific support for the efficacy of good communication skills.

Terminology

Early work to define communication skills relevant to medical practice used the terms *physician-patient*

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communication or *excellence in communication*. The term *patient-centered communication* has emerged in more recent writing on the subject. This usage followed the pronouncement from the Institute of Medicine in 2001 that medical care should become more patient-centered, that is, more responsive to patient needs and perspectives, with patient values guiding decision making.[28] Although definitions of patient-centered communication vary,[29,30] there are common core components. Epstein and Street[31] offered an operational definition of patient-centered communication: (1) eliciting and understanding patient perspectives (concerns, ideas, expectations, needs, feelings, and functioning), (2) understanding the patient within his or her unique psychosocial and cultural contexts, and (3) reaching a shared understanding of patient problems and the treatments that are concordant with patient values.

More recent empirical studies and reviews couch their findings in patient-centered terminology. One way of conceptualizing the patient-centered method for encounters is to consider both parties' agendas: (1) the physician's focus on explaining the illness in terms of the taxonomy of disease, and (2) the patient's focus, which encompasses perspectives on his or her illness, the need for information and understanding, and the desire for partnership in management. The physician's task is to address both agendas, resolving any conflict between the 2 by further dialogue and negotiation.[32]

What Is the Evidence in Support of Communication Skills?

Patient Satisfaction Research that seeks to link physician communication skills during encounters to outcomes after those encounters has been conducted since the late 1960s, with many of these studies focusing on patient satisfaction after encounters with physicians.[33–38] Although patients may be satisfied (or dissatisfied) because of factors unrelated to communication (eg, wait times and other attributes of the encounter), the available evidence indicates that patient satisfaction is strongly associated with the communication behaviors that occur during the physician-patient interaction. Several studies that focused on patient satisfaction were able to outline specific behaviors associated with satisfaction, grouping them into 2 categories: task-oriented behaviors (eg, drawing out patients with active listening responses and providing detailed information), and affective behaviors (ie, socioemotional exchanges, such as responding empathically, showing caring, and addressing the patient's main concerns).[39,40] The Stewart and Roter[41] and Roter et al[42] studies involving interactional analysis showed that task behaviors were the more important contributors to patient satisfaction, but, to be effective, those behaviors needed to be couched in patient-centered terms, including not being overly directive while being attentive to the patient's receipt of information. Did the patient understand? Was the information congruent with the patient's beliefs and circumstances? Affective behaviors also were associated with satisfaction, but the relationship was less strong. Nonverbal communication behaviors, such as eye contact and listening attentively, are also linked to increased patient satisfaction.[43]

Recall, Understanding, and Adherence Recent research on communication has sought to link clinician behaviors and skills to other important, but less immediate, clinical outcomes, such as patient recall, understanding treatment recommendations, and adherence to those recommendations.[41,42,44–52] In these studies, the association of outcomes with clinicians' communication skills was less consistent and less strong than in studies that focused on patient satisfaction. The most consistent findings suggest that communication needs to be consistent with basic principles of information transfer. Successful communication should

- Be uncomplicated
- Be specific
- Use some repetition
- Minimize jargon
- Check patient understanding

In addition, communication should simultaneously employ a patient-centered approach and interpersonal interaction to promote patient satisfaction. Overly directive communication and teaching appeared to have negative consequences.[41,42]

Health Outcomes Several recent studies and reviews have found positive associations between communication skills and improved health outcomes for patients, such as physiologic measures (eg, blood pressure, blood glucose levels), health status (eg, headache frequency, depression), and measures of functional status, including less patient distress with the illness experiences.[25,31,45,53–67] Most recently, Weiner and colleagues,[68] in an observational study, showed improved health care outcomes in patients in primary care clinics who had their cues about psychosocial factors attended to. This data set must be viewed as intriguing and suggestive but incomplete. Many of these studies are small, and there is a lack of consistency in findings across studies. It is challenging to attribute the positive outcomes studied to the earlier clinician communication behaviors the patient experienced, and there are multiple influences and potential confounders, many of which reside outside the clinician's sphere of influence.

Communication and Negative Outcomes Studies have shown that patients of physicians with high malpractice claims have twice as many complaints related to communications than do physicians with low malpractice claims.[20] Similarly, Tamblyn et al[69] reported a link between poor patient-physician communication, as measured by the Medical Council of Canada's clinical skills licensure examination, and subsequent rates of malpractice claims. Levinson and colleagues[70] studied audiotapes of physicians with low and high malpractice claims and found that primary care physicians with no malpractice claims used more facilitation, encouraged the patient to talk, checked his or her understanding of explanations and instructions, and solicited his or her opinions. As a group, studies about malpractice claims show a link between poor communication and later complaints.[71,72] This body of research, however, does not offer guidance on the specific communication skills that are most important in reducing the risk of malpractice.

Mazor and colleagues[73] studied breakdowns in cancer care in 416 patients. Of the 93 patients who experienced problems with their treatment, 44 (47%) reported communication breakdowns, including fundamental problems in information sharing, emotional support, and care coordination. From the patients' perspectives, some of these were reported to be as harmful as traditional adverse events or errors.

Another negative outcome of poor physician communication skills may be the missed opportunities to improve self-management by patients with chronic disease. This area is less well studied. At the same time, given the burden of chronic disease from smoking, eating disorders, and alcohol consumption, combined with the evidence regarding positive outcomes attributable to physician communications about these behaviors, ineffective behaviors and missed opportunities have become an area of concern.[17,74–77] Studies in this area will need to link specific clinician behaviors to positive and negative outcomes, as well as to disease-specific, self-management outcomes.

Patient Views of Physician-Patient Communication Patients are the recipients of health care services, making it relevant to identify what they want from their interactions with providers. In addition, if we consider the suffering and anxiety associated with illness and the difficulty seeking some types of care, patients' statements about what they wish for and are unhappy about in their interactions with physicians are worthy of consideration.

Two surveys[75,76] concluded that patients strongly desire a patient-centered communication approach. In these studies, patients expressed the wish that their physicians would

- Explore the patient's ideas about the problem (thoughts, worries, feelings, expectations) and take the patient's input seriously
- Try to understand the whole person and family influences and how the problem affects the patient's life
- Tell the patient what is wrong in plain language
- Seek common ground and partnership: agree on the nature of the problem, on the priorities, and on the goals of treatment; make management decisions and clarify the respective roles of the physician and patient
- Strive for an enhanced physician-patient relationship: be approachable and friendly, share decision making, show genuine care, and be respectful

In 2005, Mazor and colleagues[78] asked lay participants to serve as analog patients and rate the relative importance of a list of desired clinician behaviors. This study provided strong support for the same encounter behaviors that have emerged from empirical studies of patients: soliciting patient input on the agenda and expectations for the visit, allowing the patient to speak without interruption, presenting information clearly without jargon, providing specific advice and recommendations, asking for patient input about treatment plans, checking patient understanding, and acknowledging patient emotions. The patient-centered components of the Good Medical Practice—USA document[79] echoed these same desired medical behaviors. It also emphasized the need for physicians to

- Discuss the costs of different tests, medications, and treatment options, and take into account what the patient's health care insurance will cover
- Understand and be responsive to the patient's living circumstances and support structure
- Offer involvement and support for other caregivers of the patient's choosing[79]

The Evidence in Total Taken together, there is an impressive body of evidence that supports a positive association of physician communication behaviors with multiple outcomes of interest (TABLE 1). Good communication skills clearly lead to more satisfied patients. Satisfaction is a desired outcome in its own right. The data suggest it is also a necessary (yet perhaps not a sufficient) condition for other patient outcomes, such as recall, patient understanding, and adherence to therapy. Although the evidence base is weaker, there are data to support the effect of good communication skills on "more distant" outcomes, including physiologic parameters and health status.

Consensus Statements and "6-Function Model" Based in part on the evidence that has emerged to date, there is a remarkable consensus about what constitutes competency in physician-patient communication in general and in patient-centered communication, in particular.[7,9–11,31,80] This consensus is organized around 6 core functions

TABLE 1 **Evidence Supporting Use of Communication Skills in Medical Encounters**

Communication Outcome	Source, y	Study Findings
Patient satisfaction	Korsch et al,[33] 1968 Bertakis,[36] 1977 Stiles et al,[34] 1979 Buller and Buller,[35] 1987 Roter et al,[42] 1987 Stewart and Roter,[41] 1989 Rowland-Morin,[40] 1990 Wanzer et al,[37] 2004 Brédart et al,[38] 2005 Roter et al,[43] 2006 Tallman et al,[39] 2007	▪ Studies as a group show strong, consistent association of physician behaviors with patient satisfaction ▪ Both "task behaviors" (facilitating patient talk, giving detailed instructions) and "affective behaviors" (socioemotional exchanges, being empathic, showing caring) shown to be positively associated, with varying primacy of each ▪ Nonverbal behaviors also positively associated (eg, eye contact) ▪ Negative behaviors (being overly directive, not addressing patients' main concern) can attenuate effect
Recall, understanding, adherence	Francis et al,[47] 1969 Ley et al,[44] 1976 Garrity,[51] 1981 Bartlett et al,[48] 1984 Tuckett et al,[46] 1985 Roter et al,[42] 1987 Stewart and Roter,[41] 1989 Kjellgren et al,[50] 1995 Stewart,[45] 1995 Silk et al,[49] 2008 Zolnierek and DiMatteo,[52] 2009	▪ Studies show weaker, but positive, association of physician behaviors with recall of encounter events, giving of medical advice, and adherence ▪ Information given needs to be clear, simple, jargon-free ▪ Checking patients' understanding of physician explanations and instructions is fundamental but less frequently performed by physicians ▪ Attending to patient satisfaction is a necessary accompaniment
Health outcomes	Orth et al,[55] 1987 Kaplan et al,[54] 1989 Fallowfield et al,[53] 1990 Stewart,[45] 1995 Kinmonth et al,[56] 1998 Stewart et al,[57] 2000 Epstein and Street,[31] 2007 Levinson et al,[25] 2010 Weiner et al,[68] 2013	▪ Some studies show positive association with physiologic measures (blood pressure, blood glucose), health status (headache frequency, depression), functional status (levels of distress with illness); others unable to find such effects ▪ Data set is suggestive but incomplete—effect sizes are small, studies are inconsistent
Negative outcomes	Beckman et al,[72] 1994 Hickson et al,[20] 1994 Levinson et al,[70] 1997 Ambady et al,[71] 2002 Whitlock et al,[77] 2002 Tamblyn et al,[69] 2007 Mazor et al,[73] 2012	▪ Physicians with higher malpractice complaints have twice as many complaints about communication ▪ Physicians with poor communications scores on Canadian licensing examination have higher subsequent malpractice claims ▪ Physicians with few malpractice claims use more facilitation, encourage patients to talk, and check understanding ▪ Patients report harmful breakdowns in communication (insufficient information and lack of emotional support)

(or goals) for medical encounters: (1) fostering the relationship, (2) gathering information, (3) providing information, (4) making decisions, (5) responding to emotions, and (6) enabling disease- and treatment-related behavior (self-management; TABLE 2).

A "Best Practice" Approach The framework above has, in turn, stimulated the outline of domains and subdomains of specific clinician behaviors, organized under the 6 key functions.[61,81–88] Similar lists can be derived from consideration of instruments used to assess communication skills.[89]

It is possible to create an amalgam of these skills, which can be considered as a best practice for communication in medical encounters (TABLE 3). The concept of *best practice* acknowledges what de Haes and Bensing[80(p293)] have concluded: "Although in much of the literature on medical communication a compelling case is made about the relevance of adequate communication for high quality care, the evidence base of this argument is underdeveloped." A true evidence base for communication will articulate "the hypothesized relationship between specified communication functions and elements and concrete endpoints (including the assumed mechanism and pathways)."[80(p293)] Smith and colleagues[88] agree that better-described, behaviorally defined skills are a critical next step in advancing the field.

Teaching Communication Skills

The literature demonstrates that physicians can learn patient-centered communication skills.[23–25,63,90–92] A review by Levinson, Lesser, and Epstein[25] cited several studies that showed a strong association between physician training in, and later use of, patient-centered communication skills in medical encounters. High-intensity interventions (ie, multi-method, longer instructional time) had a stronger effect

TABLE 2 **Goals for Communication in Medical Encounters**

The 6-Function Model of Epstein and Street,[31] 2007	The 6-Function Model of de Haes and Bensing,[80] 2009
Fostering healing relationships	Fostering the relationship
Exchanging information	Gathering information
…	Information provision
Making decisions	Decision making
Enabling patient self-management	Enabling disease- and treatment-related behavior
Responding to emotions	Responding to emotions
Managing uncertainty	…

than did lower-intensity teaching. Health care institutions (hospitals, physician group practices) have developed training programs to teach patient-centered communication skills. Such programs have led to improved physician self-confidence in communication skills, as well as improvement in patient satisfaction.[83,93,94]

Reports of patients as well as our own observations of performance on the United States Medical Licensing Examination Step 2 Clinical Skills Examination suggest that there remains a large gap between what is perceived as best practice by experts and what is actually used in clinical settings. This suggests that communication skills training needs to be increased, particularly in the later years of medical school and in residency. We also need to better understand why more physicians fail to embrace or prioritize patient-centered communications. Certainly, practice setting and problem urgency affect the scope of communication skills appropriate for any given encounter. There is a widespread perception by physicians that being patient-centered will take more time. At least 1 study[95] has shown a weak association between visit length and use of patient-centered techniques. Levinson et al[25] and educators assert that communication is like any other skill: it requires practice for efficient use. Another study supports this notion, showing that consultation time actually decreased once physicians developed better facility asking about and responding to patient questions.[96] Thus, effective approaches for teaching communication skills require a much better understanding (at the individual and system levels) of the reasons why there has not been more robust adoption of patient-centered communication skills by physicians.

Discussion

We assert that a compelling case can be made for a set of communication skills and behaviors; this emerges from the empirical evidence and the statements of experts and patients (TABLE 3). This amalgam may be considered a best practice for communication in medical encounters, and at present, is our best guide. This assertion is supported by data outlining the prevalence of nonadherence to medications in US health care, by the growing use of patient satisfaction studies as a quality of care measure, and finally, by the attention of self-management in chronic disease.[97,98] These issues touch all clinical disciplines and can be addressed by state-of-the-art communication between physician and patient. The specific skills listed in the right-hand column of TABLE 3 form a *basic* skill set around which our collective curricular and assessment efforts should be directed.

Once learned, basic skills can be flexibly modified based on patient, disease, health care setting, and other differences that are inevitable in patient care. In some circumstances, the emotional reaction to illness is minimal and requires very little response; patients may have little to say about their current problem if it is minor and/or of recent onset, other than to express a desire to have it resolved. Not all patients desire a role in decision making, and not every clinical situation calls for a shared decision-making approach. All specialties have been challenged to develop curricula and assessment methods for interpersonal and communication skills as part of the Accreditation Council for Graduate Medical Education (ACGME) 6 competencies.[99,100] Each specialty likely has key issues that require particular emphasis and/or some modification of a basic communication skills set. A focus on these discipline-specific needs for physician-patient communication is likely to emerge from the ACGME Milestone Project, currently being implemented in 7 specialties that entered the Next Accreditation System on July 1, 2013 (emergency medicine, internal medicine, neurological surgery, orthopedic surgery, pediatrics, diagnostic radiology, and urology).[101]

A best practice approach allows us to move forward, while acknowledging—as have de Haes and Bensing[80] and Smith et al[88]—that the evidence base underlying the efficacy of patient-centered communication is underdeveloped. Issues, such as inadequate conceptual grounding, small sample sizes, varying definitions and assessment methods,

TABLE 3 **Best Practice for Communication in Medical Encounters**[a]

Functions of the Medical Interview	Roles and Responsibilities of the Physician	Skills
Fostering the relationship	▪ Build rapport and connection ▪ Appear open and honest ▪ Discuss mutual roles and responsibilities ▪ Respect patient statements, privacy, autonomy ▪ Engage in partnership building ▪ Express caring and commitment ▪ Acknowledge and express sorrow for mistakes	▪ Greet patient appropriately ▪ Maintain eye contact ▪ Listen actively ▪ Use appropriate language ▪ Encourage patient participation ▪ Show interest in the patient as a person
Gathering information	▪ Attempt to understand the patient's needs for the encounter ▪ Elicit full description of major reason for visit from biologic and psychosocial perspectives	▪ Ask open-ended questions ▪ Allow patient to complete responses ▪ Listen actively ▪ Elicit patient's full set of concerns ▪ Elicit patient's perspective on the problem/illness ▪ Explore full effect of the illness ▪ Clarify and summarize information ▪ Inquire about additional concerns
Providing information	▪ Seek to understand patient's informational needs ▪ Share information ▪ Overcome barriers to patient understanding (language, health literacy, hearing, numeracy) ▪ Facilitate understanding ▪ Provide information resources and help patient evaluate and use them	▪ Explain nature of problem and approach to diagnosis, treatment ▪ Give uncomplicated explanations and instructions ▪ Avoid jargon and complexity ▪ Encourage questions and check understanding ▪ Emphasize key messages
Decision making	▪ Prepare patient for deliberation and enable decision making ▪ Outline collaborative action plan	▪ Encourage patient to participate in decision making ▪ Outline choices ▪ Explore patient's preferences and understanding ▪ Reach agreement ▪ Identify and enlist resources and support ▪ Discuss follow-up and plan for unexpected outcomes
Enabling disease- and treatment-related behavior	▪ Assess patient's interest in and capacity for self-management ▪ Provide advice (information needs, coping skills, strategies for success) ▪ Agree on next steps ▪ Assist patient to optimize autonomy and self-management of his or her problem ▪ Arrange for needed support ▪ Advocate for, and assist patient with, health system	▪ Assess patient's readiness to change health behaviors ▪ Elicit patient's goals, ideas, and decisions
Responding to emotions	▪ Facilitate patient expression of emotional consequences of illness	▪ Acknowledge and explore emotions ▪ Express empathy, sympathy, and reassurance ▪ Provide help in dealing with emotions ▪ Assess psychological distress

[a] Modified using Makoul,[7,8] Levinson et al,[25] Epstein and Street,[31] McCormack et al,[61] and Smith et al.[88]

and inconsistent results, have been noted by various investigators.[30,61,80,102] Many of the studies assert causation of certain outcomes by selected behaviors, when the research designs only allow conclusions about *associations* of behaviors with outcomes. Future research should address these weaknesses and address less-studied areas, such as shared decision making and patient self-management.[30] In addition, there are areas important to modern practice for which we have very little evidence: the ever-shortening time available for medical encounters and the use of electronic health records. Further, there is a need for a much better understanding of how the needs and concerns of patients with specific diseases and conditions interact with the communication behaviors outlined in TABLE 3. What behaviors by clinicians result in the best patient understanding of an explanation of a complex disease, such as rheumatoid arthritis? What skills result in the best acceptance by patients offered a choice of therapies for cancer? What communication works best to optimize self-management in chronic diseases, such as asthma or depression?

Conclusion

The field of physician-patient communication has come a long way since the pioneering work of Korsch et al,[33] Lipkin et al,[11] Stiles et al,[34] Kaplan et al,[54] Roter et al,[42] and

Francis et al.[47] The scope of communication skills important for successful clinical encounters has been broadened and better defined.[103] It has been demonstrated that these behaviors can be taught, and strategies for both instruction and assessment are developing. There is also a growing body of medical educators and researchers contributing to the field as well as national (American Academy on Communication in Healthcare)[96] and international (European Association on Communication in Health Care)[104] academies, each with their own periodicals (*Medical Encounter*,[105] *Patient Education and Counseling*[103]).

An effort to make significant enhancements to medical trainees' communication skills likely will require significant changes in instruction at both the undergraduate and graduate levels of training. It will also require changes in the approaches for assessing communication skills during residency. The challenges of assessing communication skills should not be underestimated.[95] At the very least, assessment should use well-established instruments for measurement of trainees' communication skills in patient encounters.[89,103] A prerequisite for ultimate success will be faculty acceptance of the importance of communication skills and a commitment to development. In some settings, implementation of learned skills will require system changes to allow sufficient time in encounters for patient-centered approaches to be used. The time to begin is now. As Levinson and Pizzo[22(p1803)] have stated, "If the medical profession wishes to maintain or perhaps regain trust and respect from the public, it must meet patients' needs with a renewed commitment to excellence in the communication skills of physicians. It is time to make this commitment."

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Counseling Patients in Primary Care: Evidence-Based Strategies

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Family physicians spend substantial time counseling patients with psychiatric conditions, unhealthy behaviors, and medical adherence issues. Maintaining efficiency while providing counseling is a major challenge. There are several effective, structured counseling strategies developed for use in primary care settings. The transtheoretical (stages of change) model assesses patients' motivation for change so that the physician can select the optimal counseling approach. Structured sequential strategies such as the five A's (ask, advise, assess, assist, arrange) and FRAMES (feedback, responsibility of patient, advice to change, menu of options, empathy, self-efficacy enhancement) are effective for patients who are responsive to education about health risk behavior. For patients ambivalent about change, motivational interviewing is more likely to be successful. Capitalizing on a teachable moment may enhance the effectiveness of health behavior change counseling. The BATHE (background, affect, troubles, handling, and empathy) strategy is useful for patients with psychiatric conditions and psychosocial issues. Patients should be referred for subspecialty mental health or substance abuse treatment if they do not respond to these brief interventions. (*Am Fam Physician*. 2018;98(12):719-728. Copyright © 2018 American Academy of Family Physicians.)

Illustration by Todd Buck

Counseling patients on lifestyle modification and psychosocial problems is a fundamental competency for family physicians.[1-4] Approximately 40% of primary care office visits are for chronic illness[5] in which psychosocial factors play a major role in etiology and disease progression.[6] Counseling patients about health risk behaviors and health education is a core component of 18% of all primary care office visits.[5] Although counseling regarding weight management, diet, smoking, and alcohol use is an important part of clinical practice, a survey found that only between 31% and 56% of primary care physicians rated themselves as having significant expertise in counseling about these issues.[4]

See related *FPM* articles at https://www.aafp.org/fpm/2011/0500/p21.html and https://www.aafp.org/fpm/2016/0900/p32.html.

CME This clinical content conforms to AAFP criteria for continuing medical education. See CME Quiz on page 711.

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In the past decade, primary care counseling strategies have been refined,[7-10] and some have been empirically evaluated.[10-12] This article describes practical counseling strategies typically requiring no more than five to 10 minutes that can be integrated into a typical office visit[1,3,9,13,14] *(Table 1[3,15])*. Research and clinical guidelines suggest that smoking cessation can be effectively addressed in three

WHAT IS NEW ON THIS TOPIC

Counseling Strategies

Research based on the transtheoretical (stages of change) model suggests that it is possible to change multiple health risk behaviors concurrently.

Application of the FRAMES counseling protocol to French primary care patients found reductions in cannabis use among patients up to 18 years of age at the six-month follow-up, whereas use increased among adolescents receiving routine care.

SORT: KEY RECOMMENDATIONS FOR PRACTICE

Clinical recommendation	Evidence rating	References	Comments
The five A's (ask, advise, assess, assist, arrange) technique has been associated with reduced smoking and alcohol use as well as modest weight loss.	B	10, 25, 26	Multisite studies; recommended by the U.S. Preventive Services Task Force
The FRAMES (feedback about personal risk, responsibility of patient, advice to change, menu of options, empathy, self-efficacy enhancement) technique has been associated with reductions in alcohol-related risk behavior and reduced cannabis use.	B	31, 32	Leads to harm reduction
The use of motivational interviewing in primary care is associated with decreases in weight, blood pressure, and alcohol use.	A	16, 33, 35	Meta-analyses and systematic reviews specific to the primary care setting
The BATHE (background, affect, troubles, handling, empathy) technique is associated with increased patient satisfaction.	B	11, 45	Two recent studies have found this pattern
The transtheoretical (stages of change) model increases coaction of health behavior change for weight management.	B	3, 49	Three studies combined for analysis

A = consistent, good-quality patient-oriented evidence; **B** = inconsistent or limited-quality patient-oriented evidence; **C** = consensus, disease-oriented evidence, usual practice, expert opinion, or case series. For information about the SORT evidence rating system, go to https://www.aafp.org/afpsort.

TABLE 1

Summary of Primary Care Counseling Strategies

Counseling strategy	Problem type	Major features
Transtheoretical (stages of change) model	Specific health behavior and adherence	Assumes that patients have varying levels of motivation for change; assesses patients' pros and cons for changing behavior
Five A's (ask, advise, assess, assist, arrange)	Substance use; lifestyle modification; lack of adherence to medication, medical testing, or a procedure	Assumes that patients lack complete knowledge of the impact of health risk behavior, nonadherence, etc.; patients will respond to direct advice
FRAMES (feedback about personal risk, responsibility of patient, advice to change, menu of options, empathy, self-efficacy enhancement)	Substance use; lifestyle modification; lack of adherence to medication, medical testing, or a procedure	Provides new information; encourages patients to select personalized treatment or lifestyle modifications from a menu to increase likelihood of a behavior change
Motivational interviewing	Substance use; health behavior and adherence	Recognizes and directly acknowledges patients' ambivalence about change; systematic approach to increasing patients' motivation; relates health behavior to patients' core values
BATHE (background, affect, troubles, handling, empathy)	Psychosocial problems and their social, emotional, and cognitive dimensions	Specific statements and questions that quickly develop rapport with patients; focuses on a specific dimension of a problem; encourages improved coping

Information from references 3 and 15.

minutes, problem alcohol use in five minutes, and dietary fat consumption and lipid levels in eight minutes.[13] These strategies may also be adapted to multiple patient contacts. For example, the U.S. Preventive Services Task Force guidelines[16] and a recent Cochrane review[17] for addressing alcohol misuse suggest that several 10- to 15-minute counseling appointments are most effective for reducing alcohol consumption.

Transtheoretical (Stages of Change) Model

The stages of change model *(Table 2*[3,15,18-20]*)*, originally developed from studying successful smoking cessation

TABLE 2

Transtheoretical (Stages of Change) Model for Addressing Health Risk Behavior

Stage	Description	Counseling statements	Goals
Precontemplation	Patient has no plans to change health-related behavior; may not be interested or may oppose change	"Is it okay if I talk to you about your smoking?" "What are your thoughts about your weight?" "Has your drinking been a problem to you or anyone else in your life?" "What would be the first sign that would tell you that it might be time to cut down on your use of marijuana?" "How long do you think you will keep smoking two packs of cigarettes per day?"	Increases patient awareness of the health-related issue; emphasizes patient choice to continue the behavior
Contemplation	Patient considering behavior change in the next six months; likely ambivalent about behavior change; has not taken specific action	"What do you see as the pros and cons of continuing to smoke?" "What do you think the hardest thing would be about reducing your drinking?" "What do you like about smoking marijuana?" "What would be the biggest challenge you would face if you decided to change your diet?"	Helps the patient articulate ambivalence about health behavior change; increases pros and decreases cons of behavior change
Preparation	Definitely planning on making behavior change in the next 30 days; notifies others in social network Making environmental changes (e.g., removing ashtrays and lighters from home; considering possible challenges to behavior change	"How can I help you be successful in quitting smoking?" "What would be the best day in the next month to stop smoking?" "Once you quit smoking, what do you see as the major challenges?" "What could happen that might make you start drinking again?"	Elicits specific commitment to change; encourages collaboration with the patient in developing a concrete plan Helps patient recognize and prepare for challenges with smoking, alcohol, and cannabis abstinence
Action	Engaging in new behavior for a continuous period of six months; has successfully negotiated unanticipated challenges to behavior change	"You've done really well in cutting down your drinking." "What have been the major challenges in staying away from cigarettes?" "Have there been any situations in which you were tempted to overeat? How did you handle the situation?"	Reinforces patient success; highlights patient self-efficacy; encourages patient to consider unanticipated challenges to abstinence
Maintenance	New behavior has been in place for more than six months; the patient has likely had several lapses	"Some lapses are normal when you change a long-standing habit. After you smoked those 10 cigarettes, how did you regain control?" "It sounds like there were some new situations that could trigger heavy drinking that you had not anticipated. That is entirely understandable. The important thing is that after that night of heavy drinking, you went back to having only one drink per day."	Reinforces patient success; normalizes lapses as part of the change process; highlights patient success in preventing lapses from becoming relapses; emphasizes that the new behavior is the result of the patient's efforts and is under his or her control

Information from references 3, 15, and 18 through 20.

techniques,[18,19] recognizes that many patients are currently unmotivated or ambivalent about habit change. The stages of change model provides a framework for assessing the patient's degree of commitment to change, and can guide physicians in choosing from counseling strategies such as the five A's, FRAMES, or motivational interviewing. By asking questions assessing the patient's motivation, and determining his or her specific stage in the change process (i.e., precontemplation, contemplation, preparation, action, maintenance), physicians can move the patient toward initiating action or support ongoing health behavior change.[15,18,19]

Patients in the precontemplative stage pose a particular challenge. If a patient appears unconcerned about health risks, the physician may be tempted to emphasize the consequences of a behavior such as continued smoking or excess alcohol use. However, aggressive education may increase resistance and has the unintended effect of reducing patient openness to physician input.[3,15,16,18,19,21]

In precontemplation, the cons of changing outweigh any perceived benefits.[18] Therefore, counseling at this stage should emphasize the benefits of change. Achieving "decisional balance" involves eliciting the patient's stated pros and cons for changing his or her behaviors. To move the patient from contemplation to action, the physician should address obstacles to change. Studies of multiple health behaviors conclude that moving through these stages involves a crossover, with an increase in the pros and a decrease in the cons of changing behaviors.[18]

In the action and maintenance stages, patients contend with possible lapse and relapse. When a lapse occurs, patients are more likely to relapse to their former habit when they attribute the lapse to internal causes such as personality, genetics, or lack of willpower. Patients attributing their lapse to external factors such as peer pressure or work-related stress are less likely to relapse.[20] Physicians should frame lapses and relapses as learning experiences and help patients recognize formerly unanticipated challenges as well as develop a plan for preventing future episodes.

Although the stages accurately predict health-related behaviors, the effects of specific stage-matched interventions are unclear.[22,23] For example, the stages of change are useful in approaching smoking cessation counseling, with precontemplative patients responding best to a brief motivational intervention accompanied by written information.[23] Patients in the preparation or action stage benefit most from a combination of focused advice, written guidance, and prescription medication when indicated.[23]

The Five A's

The five A's (ask, advise, assess, assist, arrange; *Table 3*[3,10,15,24,25]), is a stepwise protocol for primary care physicians to efficiently assess and counsel patients about smoking cessation,[3,10,15,26] alcohol intake,[3,19] and weight loss.[24,25] Whenever possible, advising and assessing should link the patient's presenting problem (e.g., gastrointestinal distress, knee pain with a body mass index above 30 kg per m^2) to objective, factual standards (e.g., safe vs. unsafe levels of alcohol use, recommended daily caloric intake). Patients are likely to respond more favorably to "I" statements ("I recommend…") rather than "You" statements ("You should…").[3]

In one study, using some of the five A's in a brief emergency department intervention required a median of seven minutes that included screening, describing the relationship between alcohol and the patient's presenting problem, enhancing patient motivation, and goal setting. This physician-delivered intervention, when assessed during a

TABLE 3

Five A's for Addressing Health Risk Behavior

Technique	Physician intervention
Ask	"How many alcoholic drinks have you had in the past week?" "How long does a pack of cigarettes last for you?" "When was the last time that you exercised for half an hour straight?" (May also use structured surveys such as the CAGE questionnaire or Fagerstrom Scale)
Advise	Describe, in factual terms, the patient-relevant health risks of continuing the current behavior. Provide written patient education information as appropriate. "As your doctor, I recommend that you stop smoking (reduce alcohol use or begin exercising for 30 minutes at least five times per week)." When appropriate, link the presenting problems to the recommendation (e.g., "Cutting back on alcohol use has been found to reduce blood pressure," "Patients using marijuana as much as you describe often do have chronic cough.")
Assess	"What do you know about how drinking/smoking/lack of exercise affects health?" "What do you know about the level of alcohol use that is considered safe for men?"
Assist	"Do you feel you are ready to quit smoking in the next month?" "Strategies that have helped many patients stop smoking include medication and educational support groups. Would you like to hear more about these?"
Arrange	"I would like to see you again in about two weeks. At that time, we can see how your exercise program is going and if there is any help you need with it. We can also discuss diet and whether speaking with a nutritionist might help."

Information from references 3, 10, 15, 24, and 25.

six-month follow-up visit, was associated with an average of 3.3 fewer binge episodes in the previous 28 days compared with 1.5 fewer episodes for patients receiving standard care.[27]

Some evidence suggests that depending on the clinical context and problem, the impact of the specific A's may vary.[24,25] For example, when the five A's impact on dietary change was assessed at the three-month follow-up, weight loss of 3.3 lb (1.5 kg) occurred only when the intervention included the "arrange" step, whereas small, statistically significant self-reported changes in fat and fiber intake occurred with the first four A's alone.[25]

FRAMES

FRAMES (feedback about personal risk, responsibility of patient, advice to change, menu of options, empathy, self-efficacy enhancement) is a precursor to motivational interviewing, and was originally developed to address alcohol misuse. However, it has been applied to other health issues such as reducing stroke risk[28] *(Table 4[3,15,27,28])*. To facilitate collaboration, the physician obtains the patient's permission before providing information. By granting permission, patients maintain control, implicitly demonstrate interest, and are less likely to experience a threat to their independence.[3,15,21,29,30] How open the patient is to change will usually be evident by his or her answer to the responsibility statement that emphasizes the patient's autonomous choice to address health risk behavior. Specific, individualized feedback is presented in a factual, nonjudgmental manner. For some patients, seeing numerical data, such as blood pressure readings or body weight, in the context of normal values, may be adequate for eliciting motivation to change.[3,15,27]

The FRAMES strategy incorporates elements of motivational interviewing[21] and patient-centered care's emphasis

TABLE 4

FRAMES for Addressing Health Risk Behavior

Technique	Physician intervention
Feedback about personal risk	Provide information in a factual manner; for example, use a CAGE questionnaire or laboratory test results to explain alcohol use. Make the connection, "Do you see any connection between drinking and your blood pressure, your history of falling and breaking bones, your stomach pain and indigestion?" If yes, briefly elaborate again using facts. "There is a pretty clear association between unintentional injuries and the amount of alcohol people drink." If no, provide information and then ask, "What do you think of that?"
Responsibility of patient	"Making a change in your alcohol use is a choice that only you can make."
Advice to change	Advice should be conveyed neutrally, but based on objective indicators such as the National Institute on Alcohol Abuse and Alcoholism standards for moderate drinking (one drink per day for women and two for men[27]). "Although stopping marijuana altogether would probably be the best thing that you can do, cutting down would benefit you."
Menu of options	"Different strategies work better for different patients based on their lifestyle. Here are some strategies that have been successful for stopping smoking." Options may include medication and/or educational groups for smoking cessation or following written guidelines or attending self-help groups for dietary changes.
Empathy	"You sound like you have a lot of stressors in your life. It is hard to make a major change when you're feeling all these demands." "It sounds like you are at a point where you want to make a change and are really motivated to quit smoking and at the same time are a bit nervous about going through nicotine withdrawal."
Self-efficacy enhancement	"Two years ago, you were able to quit smoking for six months. You succeeded despite the worst part for many patients—nicotine withdrawal. That tells me that you have a lot of strength and follow-through. I genuinely believe you can be successful this time."

Information from references 3, 15, 27, and 28.

on understanding patients' values and preferences.[29,30] The menu of options step reflects the growing importance of shared decision making in clinical practice—particularly when there are several therapeutic options.[3,15,30] Research has shown that providing choices increases patient adherence and satisfaction.[29]

Although abstinence remains the standard treatment for excessive alcohol use, reduced consumption has health benefits and diminishes risk of harm.[31] In a randomized controlled trial, FRAMES was associated with 40% to 50% lower alcohol intake with a corresponding reduction in alcohol-related arrests and automobile crashes.[31] A recent application of FRAMES to French primary care patients found reductions in cannabis use among patients up to 18 years of age at the six-month follow-up, whereas use increased among adolescents receiving routine care.[32]

Motivational Interviewing

Motivational interviewing has been successfully adapted to a range of behaviors including medication adherence, dietary change, and safe sex practices.[33,34] Motivational interviewing was developed from the observation that substance abuse counselors who communicated empathy, understanding, and objective information were more successful in reducing patients' substance use than those relying on confrontation.[21] In the past decade, this strategy has developed further and its evidence base, particularly in primary care, has grown.[33] A meta-analysis of outcomes across multiple medical conditions found that compared with usual care, motivational interviewing strategies resulted in an average of 10% to 15% added benefit in patient outcomes such as alcohol consumption, blood pressure readings, body weight, and human immunodeficiency virus viral load.[33,35]

Similar to the stages of change model, motivational interviewing assumes that patients are ambivalent about change[21,34] and are more likely to change if they consider and explain their own reasons for new behavior.[21] In the context of a supportive relationship, physicians ask Socratic-style questions that help patients articulate their own reasons and goals for addressing health risk behaviors[21,34,36] *(Table 5[3,15,21,30])*.

The initial stage of motivational interviewing focuses on understanding the patient's concerns and developing rapport *(Table 6[21,34,36-38])*. Effectiveness is optimized when physicians maintain a ratio of two reflective statements per question.[38] When providing information, the physician should demonstrate a neutral, objective tone: "The amount that you reported drinking is at the 95th percentile for your age and sex, which means that you drink more than 95% of men your age in the United States."[39] In focusing the conversation, the physician emphasizes the patient's own reasons for and against change: "After learning about your blood pressure, you are concerned about the overall effect that drinking is having on your health and that drinking beer is the way that you relax after work. Do I have that right?" The use of "and" rather than "but" is deliberate; "but" is likely to increase defensiveness.[38,39]

TABLE 5

General Motivational Interviewing Skills: Establishing Rapport and Eliciting Patient Values

Technique	Examples of physician statements and questions	Goals
Open-ended question	"What brings you in today?" "How can I help you? "How do you view your drinking?"	Elicit patient's agenda; determine key values
Affirmations	"You have been smoking since you were 15 years old; I applaud your determination to quit at this point in your life."	Validate patient's interest in changing; highlight strengths and past successes
Reflections	"You have become more uneasy about the amount you drink at night and don't like the queasy feeling in the morning."	Indicates to the patient that he or she is being heard; helpful in redirecting if the encounter goes off track
Summarization	"You have tried to make regular exercise a part of your life in the past, and the demands of work and family have often gotten in the way. Now, you are worried about your weight and blood pressure and are more determined to start and stick with a regular exercise routine."	Ties together discussion and sets stage for the next step

Information from references 3, 15, 21, and 30.

TABLE 6

Motivational Interviewing: Guidelines for Presenting Information and Developing Discrepancy

Technique	Examples of physician statements and questions	Goals
Ask permission before presenting information; present information in a neutral manner	"Would it be okay if I shared some information about drinking alcohol during pregnancy?" "Would you mind if I gave you some information about diet and cholesterol?"	Increase patient's sense of control and investment in addressing the concern
Follow-up question after presenting information; question posed in noncritical tone in a spirit of curiosity	"What do you think of what I have just shared with you?"	Assess whether the patient heard and understood information; a basis for developing discrepancy
Pair health risk behavior/symptoms with important aspects of patient's life when the patient does not make a connection between information and current health concerns or between core values and health risk behavior	"How does your smoking fit with your desire to be part of the lives of your grandchildren?" "How does your alcohol use fit with your desire to be more productive at work?"	Creates cognitive dissonance; to resolve this, the patient must change behavior to coincide with values

Information from references 21, 34, and 36 through 38.

Patient values are a key component of developing an effective treatment plan *(Table 7[34,38])*. Asking the patient about his or her ideas for habit change and continuing to ask permission before presenting options further the patient's investment in change. This approach is also more likely to result in patient adherence. Although motivational interviewing techniques can be learned, maintaining the overall spirit of this strategy is more demanding.[27] In primary care settings, benefits have been demonstrated with one to three 15- to 20-minute sessions.[35] Descriptive reviews of motivational interviewing training for primary care physicians report a median of nine hours of training time[40]; however, basic skills have been acquired with two hours of training.[41]

BATHE

At least one-half of all mental health encounters occur in primary care,[1,3,42] and patients with psychiatric conditions are twice as likely to be treated by primary care physicians than by mental health professionals.[42] The BATHE (background, affect, troubles, handling, empathy) strategy is a structured set of questions and statements created specifically for the primary care setting and its inherent time constraints to address a range of psychosocial problems[3,9,43] *(Table 8[9,43,44])*. Compared with usual care, the BATHE technique has been associated with improved patient satisfaction in several medical settings.[9,45]

The background question deliberately encourages the patient to focus on a specified present topic or event since the most recent office visit. It is not usually productive or practical to pursue a detailed history, and physicians should generally avoid the psychotherapist's, "Tell me more about that?" query.[3,19]

Affect is the next step. Nonverbal cues such as tone of voice, posture, and facial expression can be clues to emotional states that the physician can encourage the patient to consider, because some patients may have difficulty articulating their feelings.[9,15]

The troubles question is particularly useful with patients who are overwhelmed or are responding to a major life stressor, and it provides focus for the remaining queries. The physician should not assume that he or she knows what is most troubling to the patient.[9,15] For example, if a patient's spouse has just announced that she is filing for divorce, the key concern may not be loss of the relationship but financial worry about how to survive on a single income.

Handling is primarily an issue of coping. When there are specific actions that would render a situation less distressing, a problem focus is most helpful (e.g., learning about available treatment options for newly diagnosed cancer). However, for uncontrollable situations such as the death of a loved one, emotion-focused coping may be associated with less distress.[46] In this approach, the patient is encouraged to consider activities that can help him or her get through an emotionally difficult period. An expression of empathy closes the interview.

Approach to the Patient

These counseling strategies should be implemented in a stepped care model.[47] For patients who respond to direct physician advice, the five A's and FRAMES are likely to be

TABLE 7

Motivational Interviewing: Guidelines for Establishing a Treatment Plan

Technique	Examples of physician statement and questions	Goals
Reflectively restate patient's desire to change	"You have thought about quitting smoking before. The combination of having these frequent coughs and the money spent on cigarettes is telling you it is now time to quit."	Establishes that the patient is currently ready to change; affirms that you and the patient are working toward a shared goal
Ask the patient for his or her ideas about change strategies	"What approaches do you know about for quitting smoking?" "Do you think any of the strategies you mentioned would work well for you?"	Increases patient investment in change; emphasizes patient self-efficacy
Ask if patient is interested in hearing about effective interventions or treatments	"Different approaches work best for different people. You know yourself better than I do." "I can describe several approaches to quitting smoking that I have seen work with my patients. Are you interested in hearing about them?"	Highlights patient choice and emphasizes that they should consider approaches that are consistent with their lifestyle; further emphasizes patient control over change
Provide a menu of evidence-based options; if the patient seems to be seeking direction, provide a recommendation based on knowledge of the patient	"I have some printed dietary guidelines for men your age. I can also refer you to a nutritionist who can work with you to develop an eating plan. There are also several self-help groups. What do you think would work best for you?"	Provides information; continues to emphasize patient choice

Information from references 34 and 38.

TABLE 8

BATHE for Addressing Health Risk Behavior

Technique	New patient or established patient with new presenting problem	Follow-up visit
Background	"What is going on in your life?"	"What has been happening since our last visit?"
Affect	"How do you feel about … (the situation patient described)? "Many people in that situation might feel … ?" Name some emotional states, then ask, "Do any of these words fit with how you are feeling?"	"How are you feeling today (or during the past week) about the situation?" "Is the emotional reaction the same or different as previously? Is the intensity different?" "It sounds like you feel more hopeful today."
Troubles	"What bothers you most about the situation?"	"When we met last week, you said that _________ was the most upsetting part of this situation; How do you see that today?" "What has been the most troubling issue that has arisen since we last met?"
Handling	"How are you managing or coping with the situation?"	"How could you approach the situation?"
Empathy	"That sounds very frustrating."	"That sounds very difficult."

Information from references 9, 43, and 44.

effective. When patient motivation is a factor, the stages of change can guide counseling. Motivational interviewing techniques for increasing patient awareness of risk and resolving ambivalence are indicated for patients who are not interested in addressing health risk behavior. For patients with psychiatric conditions and broader psychosocial issues, the BATHE technique is a useful starting point.[3,15]

A common dilemma is how to proceed when patients exhibit comorbid problems such as smoking and excessive alcohol use. Historically, physicians were advised to target one behavior at a time.[48] Recent research suggests that this limitation may not be necessary.[49] In a recent study clinicians used transtheoretical model–based counseling to address three weight-related behaviors simultaneously (e.g., reducing fat intake by less than 30% of overall calories consumed, increasing exercise to 30 minutes at least five times per week, and reducing daily food intake by 500 calories). Patients progressing to the action or maintenance stage with one behavior were twice as likely to move to the action or maintenance stage with a second behavior during a 24-month period compared with a usual care group.[49]

Physicians capitalizing on the teachable moment, estimated to occur in 10% of primary care office visits, may have greater counseling success.[50] These moments are typically acute medical events or pregnancy that increase patients' motivation for change.[50] For example, a patient presenting with acute bronchitis may be more open to discussing smoking cessation, particularly when the physician connects the acute symptoms to cigarettes.

Patients with psychiatric conditions or psychosocial stressors will benefit from the BATHE technique, combined with psychotropic medication when appropriate.[51] These patients should be closely monitored to assess their response to this initial intervention. Patients demonstrating a minimal response to brief psychosocial counseling should be referred for subspecialty mental health care.

This article updates a previous article on this topic by the author.[3]

Data Sources: Search terms included counseling and psychotherapy paired with primary health care. Some of the counseling strategies described (five A's, FRAMES, BATHE) were entered alone. Motivational interviewing and transtheoretical (stages of change) model were paired with primary medical care. The author's previous articles (references 3 and 15) were reviewed for recent citations. In addition to Medline, searches were run in the Agency for Health Care Research and Quality database, Cochrane Database of Systematic Reviews, U.S. Preventive Services Task Force, and the Substance Abuse and Mental Health Services Administration. Finally, to search for applications of newer counseling approaches to the primary care setting, acceptance and commitment therapy, solution focused therapy, and positive psychology were paired with primary medical care in Medline. Search dates: October 2017 through January 2018, and April, May, and September 2018.

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A machine learning-assisted system to predict thyrotoxicosis using patients' heart rate monitoring data: a retrospective cohort study

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Previous studies have shown a correlation between resting heart rate (HR) measured by wearable devices and serum free thyroxine concentration in patients with thyroid dysfunction. We have developed a machine learning (ML)-assisted system that uses HR data collected from wearable devices to predict the occurrence of thyrotoxicosis in patients. HR monitoring data were collected using a wearable device for a period of 4 months in 175 patients with thyroid dysfunction. During this period, 3 or 4 thyroid function tests (TFTs) were performed on each patient at intervals of at least one month. The HR data collected during the 10 days prior to each TFT were paired with the corresponding TFT results, resulting in a total of 662 pairs of data. Our ML-assisted system predicted thyrotoxicosis of a patient at a given time point based on HR data and their HR-TFT data pair at another time point. Our ML-assisted system divided the 662 cases into either thyrotoxicosis and non-thyrotoxicosis and the performance was calculated based on the TFT results. The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of our system for predicting thyrotoxicosis were 86.14%, 85.92%, 52.41%, and 97.18%, respectively. When subclinical thyrotoxicosis was excluded from the analysis, the sensitivity, specificity, PPV, and NPV of our system for predicting thyrotoxicosis were 86.14%, 98.28%, 94.57%, and 95.32%, respectively. Our ML-assisted system used the change in mean, relative standard deviation, skewness, and kurtosis of HR while sleeping, and the Jensen–Shannon divergence of sleep HR and TFT distribution as major parameters for predicting thyrotoxicosis. Our ML-assisted system has demonstrated reasonably accurate predictions of thyrotoxicosis in patients with thyroid dysfunction, and the accuracy could be further improved by gathering more data. This predictive system has the potential to monitor the thyroid function status of patients with thyroid dysfunction by collecting heart rate data, and to determine the optimal timing for blood tests and treatment intervention.

Abbreviations

ATD	Anti-thyroid drug
HR	Heart rate
LOOCV	Leave one-out cross validation
ML	Machine learning
NPV	Negative predictive value
PPV	Positive predictive value
SNUBH	Seoul National University Bundang Hospital
T3	Triiodothyronine
T4	Thyroxine

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TFT	Thyroid function test
TSH	Thyroid stimulating hormone

Thyrotoxicosis is a clinical syndrome resulting from the high concentration of free thyroxine (T4) and/or free triiodothyronine (T3). The prevalence of thyrotoxicosis is approximately 2%, and its most common cause (60–90%) is Graves' disease, an autoimmune disease that stimulates the thyroid gland to produce and release thyroid hormone[1–3]. Thyrotoxicosis cannot be diagnosed with clinical symptoms and signs alone because there are various non-specific symptoms and signs such as fatigue, anxiety, palpitations, sweating, sleep disturbance, and weight loss[4]. In real clinical practice, blood tests for serum thyroid hormone levels (thyroid function test, TFT) are required for diagnosing thyrotoxicosis and the treatment goal is to maintain normal TFT results in the patients[5]. Since radioimmunoassay is generally required for TFT, unlike diabetic patients who can measure blood glucose by themselves, patients with thyroid dysfunction do not have an objective indicator to check their disease status.

Thyroid hormone has a positive chronotropic and inotropic effect. It stimulates the rate and force of systolic contraction and the rate of diastolic relaxation[6]. Therefore, thyrotoxicosis increases systolic blood pressure and heart rate (HR) and widens pulse pressure[7]. Although palpitation and increased HR are among the most frequent symptoms and signs of thyrotoxicosis[6], and the easiest to quantify, HR can easily be affected by other factors such as the patient's emotional state, body position, and physical activities. To overcome this limitation of HR parameters, our previous study demonstrated that resting HR monitored by a wearable device that can measure the user's HR and physical activity using photoplethysmography and a 3-axis accelerometer is associated with serum free T4 level in patients with thyrotoxicosis[8]. This means that HR parameters from 'high definition' data obtained by wearable devices can be an objective indicator for monitoring the thyroid function status of the patients.

In this study, we expanded our previous study and developed a machine learning (ML)-assisted system to predict thyrotoxicosis using patients' HR data monitored by wearable devices. Through this study, it will be possible to develop digital therapeutics that can be prescribed in actual clinical practice for monitoring thyroid function status of patients with thyroid dysfunction.

Results

Model validation

The performance of our ML-assisted system for detecting thyrotoxicosis was summarized in Tables 2 and 3. Our ML-assisted system divided 662 cases into thyrotoxicosis or non-thyrotoxicosis. The sensitivity, specificity, PPV, and NPV of our system to predict thyrotoxicosis were 86.14%, 85.92%, 52.41%, and 97.18%, respectively (Table 1). In the 79 cases which showed false positive results, 74 cases were subclinical thyrotoxicosis. When subclinical thyrotoxicosis was excluded in the analysis, the sensitivity, specificity, PPV, and NPV of our system to predict thyrotoxicosis were 86.14%, 98.28%, 94.57%, and 95.32%, respectively (Table 2).

The distribution of free T4 and TSH levels of data pairs according to the decision of our ML-assisted system was demonstrated in the Fig. 1. Among 561 cases showing serum free T4 level $\leq$ 1.78 ng/dL, 79 cases were divided into thyrotoxicosis and 482 cases were divided into non-thyrotoxicosis. The cases divided into thyrotoxicosis showed higher free T4 and lower TSH levels compared with the cases divided into non-thyrotoxicosis (free T4,

	Predicted thyroid function state	
	Thyrotoxicosis	**Non-thyrotoxicosis**
True thyroid function state		
Thyrotoxicosis	87	14
Non-thyrotoxicosis	79	482

Table 1. Prediction of thyrotoxicosis by the ML-assisted system. The ML-assisted system divided 662 data into thyrotoxicosis or non-thyrotoxicosis (predicted thyroid function state). True thyroid function state is based on the thyroid function test and thyrotoxicosis was defined as serum free T4 level > 1.78 ng/dL. *ML* machine learning.

	Predicted thyroid function state	
	Thyrotoxicosis	**Non-thyrotoxicosis**
True thyroid function state		
Thyrotoxicosis	87	14
Non-thyrotoxicosis	5	285

Table 2. Prediction of thyrotoxicosis by the ML-assisted system excluding subclinical thyrotoxicosis. Among 662 data, 263 data of subclinical thyrtoxicosis defined as serum free T4 level of 0.89–1.78 ng/dL and serum TSH level < 0.3 mIU/L were excluded in the analysis. The ML-assisted system divided 399 data into thyrotoxicosis or non-thyrotoxicosis (predicted thyroid function state). True thyroid function state is based on the thyroid function test and thyrotoxicosis was defined as serum free T4 level > 1.78 ng/dL. ML, machine learning.

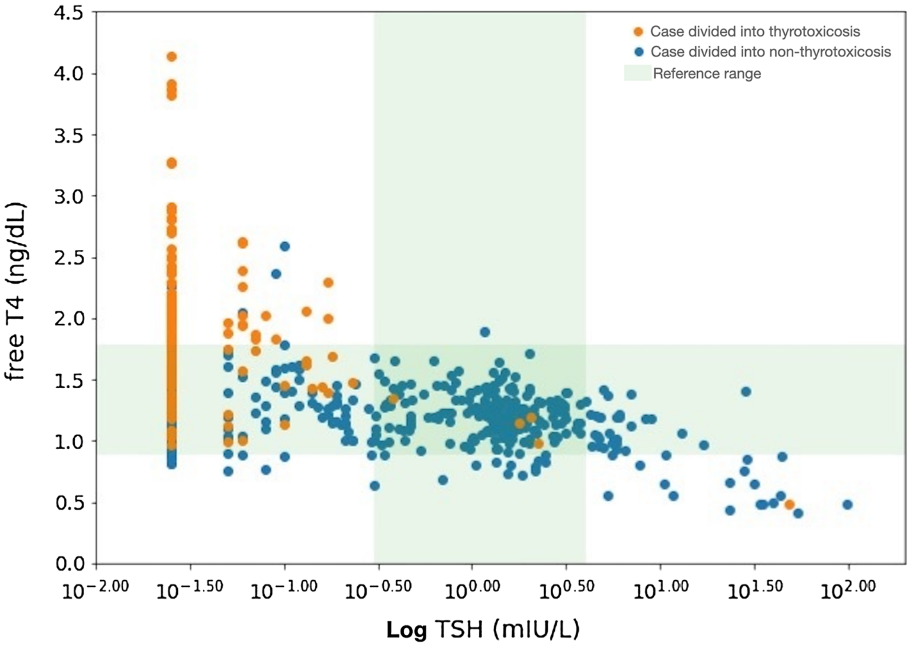

Figure 1. Distribution of free T4 and TSH levels of data pairs according to the decision of the ML-assisted system. *T4* thyroxine, *TSH* thyroid stimulating hormone, *ML* machine learning.

1.48 ± 0.27 vs. 1.21 ± 0.24 ng/dL, $p < 0.001$; TSH, 0.025 (0.025) vs 0.755 (1.730) mIU/L, $p < 0.001$). The proportion of cases with TSH < 0.1 mIU/L was higher in the cases divided into thyrotoxicosis compared with cases divided into non-thyrotoxicosis (82% vs. 33%, $p < 0.001$).

Collection of HR data

In order to determine the optimal duration for collecting HR data to predict thyroid function, the diagnostic performance was measured by varying the number of days for collecting HR data. It should be noted that the use of HR data collected during N days means utilizing the HR data for N consecutive days prior to the date of referred TFT and the target date, regardless of missing values. Figure 2A displays the diagnostic performance when the duration of collecting HR data varies for both training and test data, while Fig. 2B shows the diagnostic performance when the duration of collecting HR data is fixed at 10 days for the training data and varies for the test data. Both sensitivity and specificity are highest when the HR data is collected for 10 days.

Feature importance

Figure 3 displays the feature importance of the ML-based classifiers constructed to predict the thyroid function of 662 test data. Based on the criteria of gain and split, the serum free T4 level of the referred TFT (average

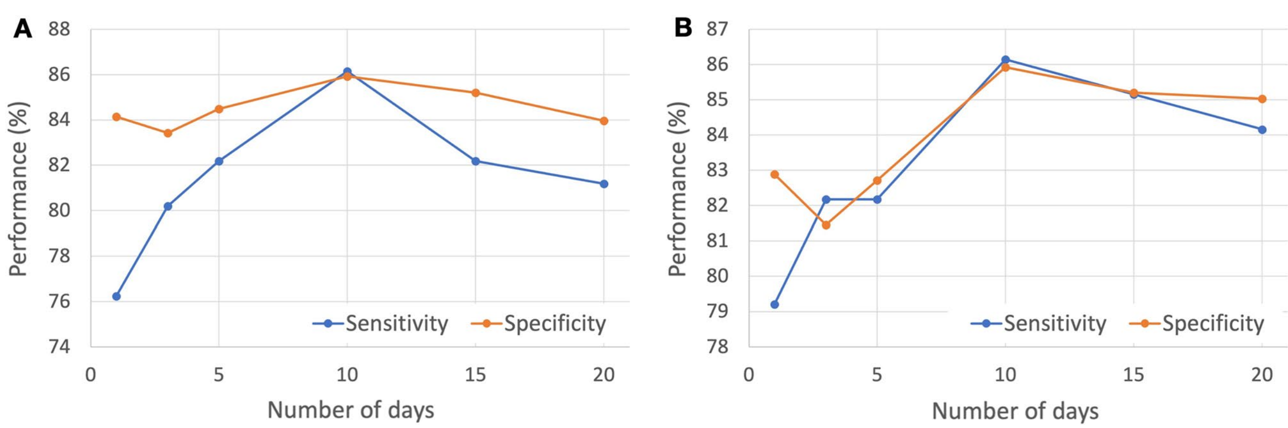

Figure 2. Sensitivity and Specificity for varying the number of days of collecting HR (**A**) for both training and test data and (**B**) for the test data when the duration of collecting HR data is fixed at 10 days for the training data. HR, heart rate.

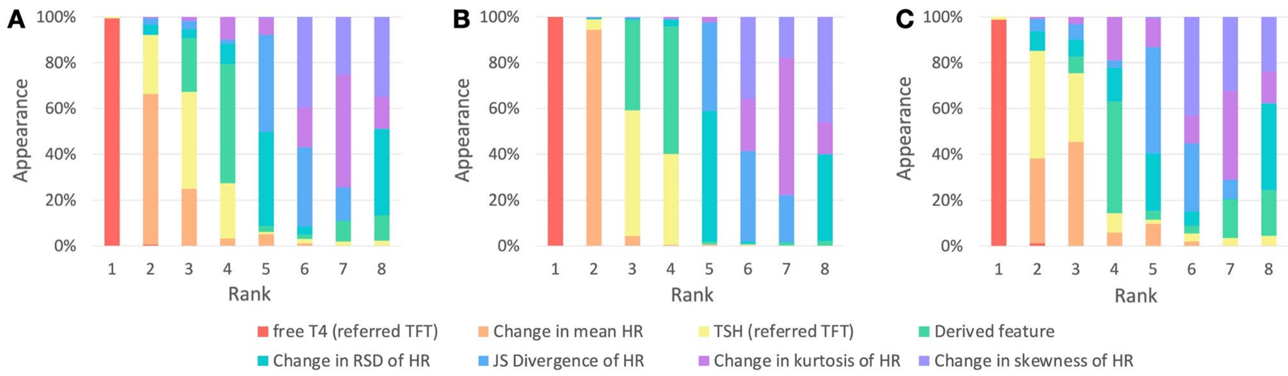

Figure 3. Rank for each feature in the experiment. Feature importance considering (**A**) gain and split, (**B**) gain, and (**C**) split. Gain, relative contribution of a feature in the model; Split, relative count of times a feature occurs in the model.

rank: 1.01), the change in mean HR (average rank: 2.51), and the serum TSH level of the referred TFT (average rank: 3.24) are identified as the most significant features. The average ranks of the other five features are similar.

Discussion

In this study, a ML-assisted system was developed to predict thyrotoxicosis using HR data collected from a wearable device, which showed promising results for clinical applications. Previously, an association between serum free T4 levels and resting HR from wearable devices in patients with thyrotoxicosis was reported, where an increase of 11 beats per minute in 5-day mean resting HR was correlated with an increase in serum free T4 level of about 0.5 ng/dL and a three-fold increase in the risk of thyrotoxicosis[8]. Although a rule-based classifier using the statistical association was developed, the accuracy was not high enough for predicting thyroid function state in individuals. Therefore, ML methodology was applied, and various features of HR monitoring data were used to predict thyrotoxicosis. Although the ML-assisted system predicted thyrotoxicosis with reasonable accuracy (85.15% of sensitivity and 85.03% of specificity), it showed low PPV, indicating a high probability of false positive cases. However, most false positive cases were subclinical thyrotoxicosis, which suggests that the ML-assisted system can predict not only the current thyrotoxicosis but also the potential risk of thyrotoxicosis based on the changes of biosignals considering that subclinical thyrotoxicosis can progress to overt thyrotoxicosis[9–15]. Moreover, higher free T4 and lower TSH levels in false positive cases compared with true negative cases support that the ML-assisted system may screen more severe cases of subclinical thyrotoxicosis, which is likely to cause comorbidities[12,16–21] and progress to overt thyrotoxicosis[13–15].

In this study, the ML-assisted system performed best when using HR data from the 10 days preceding the date of referred TFT and target date. This result is clinically valid, as using HR data from a shorter period, such as 5 days or 1 day, may incorporate variations caused by exercise, alcohol consumption, emotional states, or other acute illnesses, potentially lowering the ML model's performance. Similarly, using HR data from a longer period than 10 days may include data distant from the date of referred TFT and target date, which could also reduce the model's performance. The essential features in our machine learning model for predicting thyroid function were the free T4 level of referred TFT and the change of mean HR. This can be reasonably predicted considering the well-known chronotropic effect of thyroid hormones on the myocardium[6] and our previous research, which showed a correlation between resting HR collected using wearable devices and serum free T4 levels[8,22]. Therefore, our machine learning model can be considered as reflecting thyroid hormone physiology and previous clinical research results effectively.

This study has some important clinical implications. Our ML-assisted system can be integrated into a mobile app that helps patients self-manage thyroid dysfunction. In this case, the mobile app integrated with wearable devices can inform users of the risk of thyrotoxicosis and guide them to take medication diligently or encourage remote consultations or hospital visits depending on the risk level, thereby facilitating successful disease management. Additionally, our ML-assisted system can be used in remote monitoring solutions in hospitals. Although the risk of hyperthyroidism based on heart rate data from wearable devices cannot replace blood tests, our system can monitor the disease status based on the predicted risk level in patients who cannot visit the hospital or have infrequent hospital visits. The system we developed can also be used in the treatment process, such as adjusting medication. For example, it can monitor the response of anti-thyroid drug (ATD) in patients with Graves' disease who are starting or undergoing treatment, verify the effectiveness of medication, and determine the optimal timing for hospital visits and blood tests. It can also be employed to detect disease relapse in patients with Graves' disease who are in remission considering that approximately 50% of patients are reported to relapse within 2 years after discontinuing ATD despite following the recommended guidelines[23]. Additionally, our ML-assisted system could have implications for monitoring drug-induced subclinical thyrotoxicosis in patients with differentiated thyroid cancer (DTC) who are undergoing TSH-suppressive therapy. TSH-suppressive therapy involves maintaining a subclinical thyrotoxic state through thyroid hormone over-replacement and is recommended to prevent the growth of DTC[24]. However, excessive suppression of TSH elevates the risk of fractures and cardiovascular disease[25–27]. Therefore, a simple monitoring tool for TSH-suppressive therapy, such as a wearable device that can be used in daily life, could aid in maintaining the appropriate level of TSH suppression.

This study has some limitations which should be considered when interpreting the results. Firstly, the number of cases used in our study is relatively small compared to the number typically used for constructing ML models. This is because our study is not based on pre-existing medical databases, such as medical image readings, but on collecting biosignal data from wearable devices during unstable periods of thyroid hormone levels in patients with thyroid dysfunction. Therefore, our attempt to predict thyroid dysfunction using biosignals collected from wearable devices is unique to our research group and includes the most significant number of cases in related studies. Since we are continuously collecting data, we expect to present more advanced ML models in the future. Furthermore, since the current data is collected from a single institution, it is not immune to overfitting issues. To overcome this limitation, additional research for external validation is warranted in the future. Secondly, the data in this study predominantly collected from an age group familiar with the use of wearable devices and the associated smartphone apps. Additionally, the symptoms of thyrotoxicosis are often less apparent in the elderly population, which may limit the prediction accuracy and usability of our ML-assisted system for elderly thyrotoxicosis patients. However, in elderly patients, less evident thyrotoxic symptoms are primarily associated with mood-related symptoms, and even in elderly patients diagnosed with apathetic hyperthyroidism, the effects of thyroid hormones on the heart are not different from those in the general population with typical hyperthyroidism, as increased heart rate is observed[28–30]. Furthermore, according to a study conducted in Austria that measured resting heart rate in patients undergoing coronary angiography, it was observed that the positive correlation between resting heart rate and fT4 levels remains consistent even in relatively older patients with an average age of 62.8 years, indicating the presence of this relationship in an older age group[31]. Therefore, it is expected that the performance of our ML-assisted system will be maintained in elderly patients as well.

In conclusion, a ML-assisted system has made reasonably accurate predictions of thyrotoxicosis in patients with thyroid dysfunction. The accuracy of the system could be further improved by obtaining more data. This predictive system can be used for monitoring patients' thyroid function status without requiring blood tests, either as a smartphone-based self-monitoring tool or as a remote monitoring solution in hospitals.

Subjects and methods

Study design and participants

This is a retrospective cohort study that combines data from three different studies: a proof study that demonstrated the clinical feasibility of monitoring resting HR using a wearable device in patients with thyrotoxicosis[8], an extension study that collected additional data on patients with thyroid dysfunction, and an app-feasibility study that evaluated the clinical feasibility of a disease-managing mobile application for thyroid dysfunction. Participants in the proof study were recruited from the outpatient clinic of the endocrinology department at Seoul National University Bundang Hospital (SNUBH) between November 2016 and June 2017. Patients between the ages of 15 and 60 who had been newly diagnosed with or had a recurrence of thyrotoxicosis were eligible to participate. The extension study, which aimed to collect more data, recruited patients between the ages of 18 and 60 who had been newly diagnosed with thyroid dysfunction, including thyrotoxicosis or hypothyroidism, or were being treated for thyroid dysfunction. This study began recruiting participants at SNUBH in January 2021, and recruitment is ongoing. The feasibility study to evaluate the disease-managing mobile application recruited patients aged 18 years and older who had been newly diagnosed with Graves' disease at SNUBH from March 2022, and recruitment is also ongoing. The data from all three studies were combined to develop a machine learning (ML)-assisted system to predict thyrotoxicosis. All participants included in these studies needed to own a smartphone and to be able to use a wearable device and its mobile application. TFT were performed more than 3 times at more than 4-week intervals in the participants and their activity, sleep, and HR data were continuously monitored with a wearable device during the study period. In these studies, participants utilized a mobile application for self-management of thyroid dysfunction, Glandy™ (THYROSCOPE INC., Ulsan, Republic of Korea), to manage the data collected from wearable devices. Prior to using the app, participants agreed to the terms and conditions and the privacy policy and provided their clinical information, including data collected from their wearable devices and results from thyroid function tests. Personal identifying information was not provided, and the analysis of data was conducted using only the information that participants agreed to provide. The studies included in this retrospective study were approved by the SNUBH Institutional Review Board (IRB number, B-1609-363-004, B-2012-654-303, and B-2201-735-304) and registered on Clinicaltrials.gov (trial registration number, NCT03009357, NCT04806269, and NCT05828732). All methods were carried out in accordance with relevant guidelines and regulations and informed consent was obtained from all subjects.

Wearable devices and applications

In the proof study[8], we used the Fitbit Charge HR™ or Fitbit charge 2™ (Fitbit, San Francisco, CA) and the Fitbit application for iOS™ (Apple, Cupertino, CA) or Android™ (Google, Mountain View, CA). The firmware versions of these devices were 18.128 for Fitbit charge HR™ and 22.53.4 for Fitbit charge 2™ at the end of the study, and the latest version was maintained continuously over the study period. In the extension study and the app-feasibility study, we used Fitbit inspire™ or Fitbit inspire 2™ (Fitbit) and the Fitbit application for iOS™ (Apple) or Android™ (Google). The firmware versions of these devices were updated to the latest version during the study period (1.88.11 for Fitbit inspire™ and 1.124.28 for Fitbit inspire 2™). All models share a common sensor and data processing algorithm for both activity tracking and HR measurement. Activity and HR data are collected by the 3-axis accelerometer and photoplethysmography sensor, respectively. Fitbit also provide sleep data, including total time asleep and total number and time of awakening using their own algorithm to detect sleep from activity and HR data. To collect interday summarized data of activity, HR, and sleep and intraday detailed data of activity

Age (year)	38.9 ± 9.2
Sex (male/female)	28/147
Thyroid function state of data	
Overt thyrotoxicosis (n = 101)	
Free T4 (ng/dL)	2.51 ± 1.30
TSH (mIU/L)	0.049 (0.000)
Subclinical thyrotoxicosis (n = 263)	
Free T4 (ng/dL)	1.37 ± 0.24
TSH (mIU/L)	0.055 (0.025)
Euthyroid (n = 229)	
Free T4 (ng/dL)	1.22 ± 0.16
TSH (mIU/L)	1.488 (0.900)
Subclinical hypothyroidism (n = 33)	
Free T4 (ng/dL)	1.11 ± 0.15
TSH (mIU/L)	6.995 (1.730)
Overt hypothyroidism (n = 36)	
Free T4 (ng/dL)	0.70 ± 0.14
TSH (mIU/L)	16.236 (28.905)

Table 3. Participants and data characteristics. Data are expressed as mean ± SD or median (interquartile range). *T4* thyroxine, *TSH* thyrotropin.

and HR, we used a mobile application for self-management of thyroid dysfunction, Glandy™ (Thyroscope inc., Ulsan, Republic of Korea) for iOS™ (Apple) or Android™ (Google). This mobile application uses the application programming interface provided by Fitbit and makes it possible to collect the data mentioned above after Fitbit user's authorization.

Biochemical measurements

Serum concentrations of free T4 and thyroid-stimulating hormone (TSH) were measured using immunoassays (free T4: DiaSorin S.p.A.; TSH: CIS Bio International). The free T4 assay had an analytic sensitivity of 0.05 ng/dL, and TSH had an analytical sensitivity of 0.04 mIU/L and functional sensitivity of 0.07 mIU/L. The reference ranges for free T4 and TSH were 0.89–1.78 ng/dL and 0.3–4.0 mIU/L, respectively. Thyroid function status was defined based on the results of the TFT. Overt thyrotoxicosis was defined as higher free T4 level than reference ranges; subclinical thyrotoxicosis as normal free T4 and lower TSH levels; euthyroid as normal free T4 and TSH; subclinical hypothyroidism as normal free T4 and higher TSH levels; overt hypothyroidism as lower free T4 level.

Data preparation

We utilized a dataset consisting of "HR-TFT pairs", paired TFT results and HR data measured using a wearable device over a period of 10 days. Specifically, we paired the result of the TFT, which is carried out in the daytime, with the sleeping heart rates measured during the 10 days prior to the test date. For each day, a series of sleeping heart rates for 1 sleep session belongs to the date when the sleep ends. We extracted these HR-TFT pairs from a total of 175 patients and a total of 662 HR-TFT pairs were collected. Among these, the number of euthyroid, subclinical thyrotoxicosis, and overt thyrotoxicosis based on TFT results were 229, 263, and 101 pairs, respectively (Table 3). Because we included the subjects with thyrotoxicosis and hypothyroidism in the extension study, 36 data pairs of overt hypothyroidism and 33 data pairs of subclinical hypothyroidism were included in the dataset (Table 3). The HR data which we used was detailed HR data during sleep based on each user's sleep time data provided by Fitbit.

Development of the ML-assisted system

The proposed system is designed to classify whether a user has thyrotoxicosis by using the relationship between the change of serum free T4 level and the change of HR data, where the relationship is represented by the data combining each two HR-TFT pairs (one for the date of the referred TFT and another for the target date) of a user. The input of the system consists of the results of the referred TFT (i.e., serum free T4 level and serum TSH level) and the change of HR data between the two different time points, while the binary output of the system indicates whether the user has thyrotoxicosis. The data with free T4 level > 1.78 ng/dL at the target date is labeled as positive. As shown in Fig. 4, in the input, the change of HR data is expressed with five features (changes in mean HR, changes in the relative standard deviation of HR, changes in the skewness of HR, changes in the kurtosis of HR, Jensen-Shannon divergence) to utilize the difference between two distributions of HR data. Additionally, we newly derived one feature from existing ones (e.g., product/quotient of two different features) to enhance the prediction. Among candidates, we selected the quotient of changes in mean HR divided by TSH level of referred TFT.

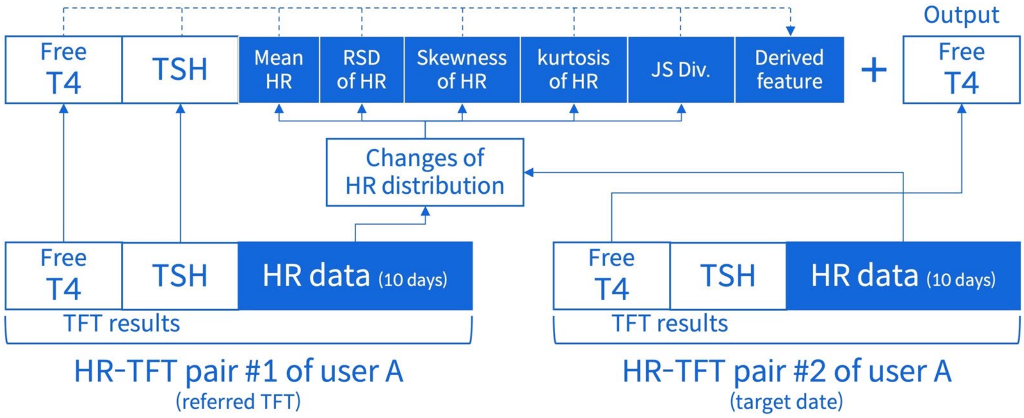

Figure 4. Design of a ML-assisted system to predict the occurrence of thyrotoxicosis using HR data collected from wearable devices. *ML* machine learning, *HR* heart rate, *T4* thyroxine, *TSH* thyroid stimulating hormone, *RSD* relative standard deviation, *JS Div* Jensen–Shannon divergence, *TFT* thyroid function test.

Light gradient boosting machine, tree based gradient boosting algorithm, is used to establish a classifier, and each input feature is transformed by using quantile transformer. The training data are augmented by creating HR-TFT pairs consisting of free T4 levels and TSH levels calculated by linear interpolation of those of two adjacent TFTs, and actual HR data of corresponding dates. During our training process, we combined data from two time points of a single patient to create one case for training. As a result, with the initial 662 HR-TFT pairs from 175 individuals, we had a total of 2182 cases for training. When including the additional data generated through interpolation, we were able to use a total of 2,711 HR-TFT pairs, resulting in a pool of 31,138 cases available for training.

Statistical analysis

Values with a normal distribution are expressed as mean ± SD, and values with a non-normal distribution are expressed as median (interquartile range). We used Student t test or the Mann Whitney U test for continuous variables. The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of the ML-assisted system for detecting thyrotoxicosis were computed based on the definition of overt thyrotoxicosis. Diagnostic values were analyzed using leave one-out cross validation (LOOCV), but the LOOCV was modified so that the training data did not include the test data. For example, if we have two participants A and B, and these participants have 3 HR-TFT pairs at three different time points (T1, T2, and T3). Our ML-assisted system predicts thyrotoxicosis in participant A at T2 using HR data at T2 and the HR-TFT pair at T1 (Ta1 → Ta2) or T3 (Ta3 → Ta2). When we test Ta1 → Ta2, learning data are as follows; Ta1 → Ta3, Ta3 → Ta1, Tb1 → Tb2, Tb2 → Tb1, Tb1 → Tb3, Tb3 → Tb1, Tb2 → Ta3, and Tb3 → Tb2). We did not include Ta3 → Ta2 in the training data to prevent the inclusion of the test data in the training data. Our ML-assisted system predicts thyrotoxicosis at T2 in participant A with the average value of probability derived from Ta1 → Ta2 and Ta3 → Ta2. A two-tailed $p < 0.05$ was considered statistically significant. All statistical analyses and data preparation were performed using IBM SPSS Statistics (version 28.0; IBM Corporation, Armonk, NY, USA).

Data availability

The raw data including medical records are not available upon request due to ethical and legal restrictions imposed by the SNUBH IRB. The original data are derived from the institutions' electronic health records and contain patients' protected health information. Deidentified data are available from the SNUBH for researchers who meet the criteria for access to confidential data and have a data usage agreement with the hospital. Data requests for this study should be made by contacting the corresponding author, Jae H.M.

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Author contributions

J.H.M. led the conception and design of the study and provided study supervision. J.H.M., T.J.O., S.H.K., C.H.A., J.H.M., and M.J.K. collected the data. K.S., J.K., J.P., and J.H.M. analyzed and interpreted the data. K.S. and J.H.M. wrote the manuscript. All authors had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Administrative, technical, and material support were provided by J.H.M. All authors reviewed the manuscript and provided edits and revisions. All authors take responsibility for the integrity of the work as a whole, from inception to the finished article, and all authors approved the final version submitted. J.H.M. was responsible for the decision to submit for publication.

Competing interests

THYROSCOPE INC. has a patent for the ML-assisted system to predict thyroid dysfunction. J.P. and Jae H.M. are the stock owners and board members of THYROSCOPE INC. K.S., J.K., T.J.O., S.H.K., C.H.A., Joon H.M., and M.J.K. declares no competing interests.

Additional information

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Association between change in heart rate over years and life span in the Paris Prospective 1, the Whitehall 1, and Framingham studies

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Heart rate, a measure of the frequency of the cardiac cycle, reflects the health of the cardiovascular system, metabolic rate, and activity of the autonomic nervous system. Whether changes in resting heart rate are related to lifespan has not yet been explored to our best knowledge. In this study, we examined the association between resting heart rate and lifespan using linear regression in the Paris Prospective Study I, the Whitehall I Study, and the Framingham Heart Study. We used Cox proportional hazards regression to relate changes in heart rate over years to mortality risk. We observed a statistically significant association between increases in resting heart rate over a 5-year period and risk of mortality in the Paris Prospective Study I (HR mortality per 10 bpm increase over time: 1.20; 95% CI: 1.13 to 1.27) and over an 8-year period in the Framingham Heart Study (HR: 1.13; 95% CI: 1.07 to 1.19 for men and HR: 1.09; 95% CI: 1.04 to 1.15 for women), after adjusting for classical risk factors and resting heart rate. Our study shows that men and women who increase their resting heart rate over time increase their risk of mortality.

Across mammalian species, larger body size is associated with lower heart rate and a longer life span[1–3]. Heart rate, a measure of the frequency of the cardiac cycle, reflects health of the cardiovascular system, metabolic rate, and activity of the autonomic nervous system. Evolution has shaped optimal body size and heart rate in various species across millennia such that there is a neat inverse association between the two. The astonishing increase in human life span in the past two centuries raises the question of whether the human species is an outlier in the inverse heart rate-life span association[4]. Improvements in health and social care underlie improvements in life span but there are concerns that the omnipresent obesogenic environment may undermine further progress. In the original Framingham Heart Study cohort of 5070 subjects, a greater death rate was observed in individuals with faster heart rate, who were free of cardiovascular disease at the entry of the study and followed for 30 years[5]. However, to our best knowledge, whether change in resting heart rate is related to lifespan has not been yet explored. This would be an important step to increase our knowledge on a potential causal relationship between HR and lifespan, which may have clinical and public health implications.

To understand the association between heart rate and lifespan, we had 2 aims: (1) to confirm the inverse association between resting heart rate and lifespan within human populations; and (2) to look for any association between increase in heart rate (over a 5-year period in the Paris Prospective Study I and an 8-year period in the Framingham Study) and risk of mortality in modern humans.

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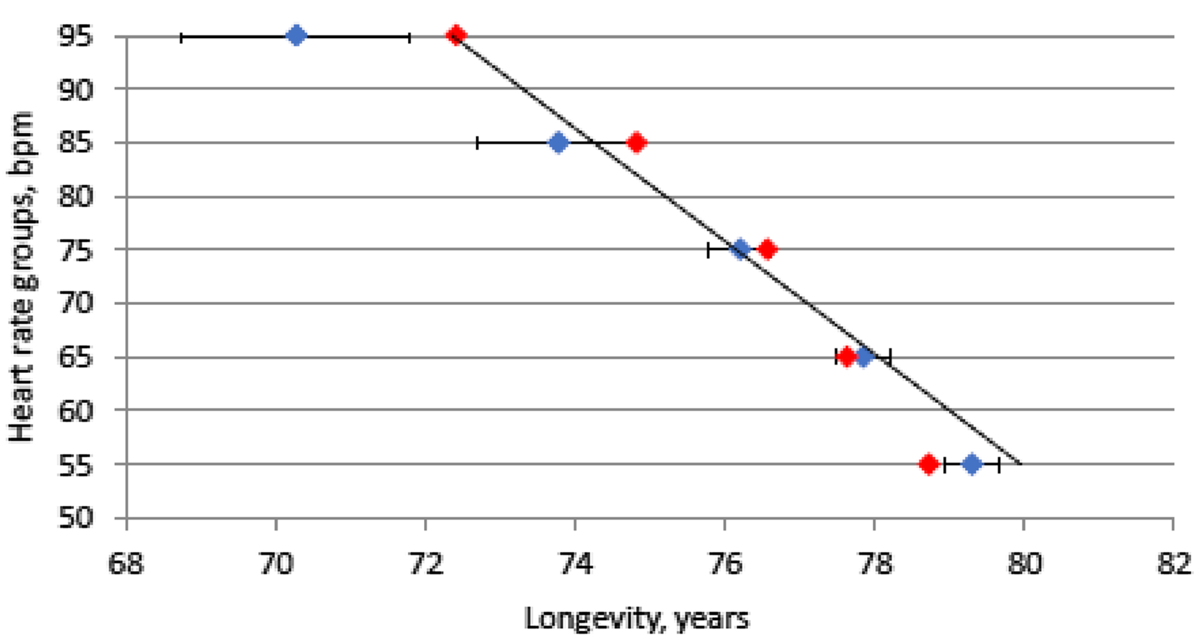
Results

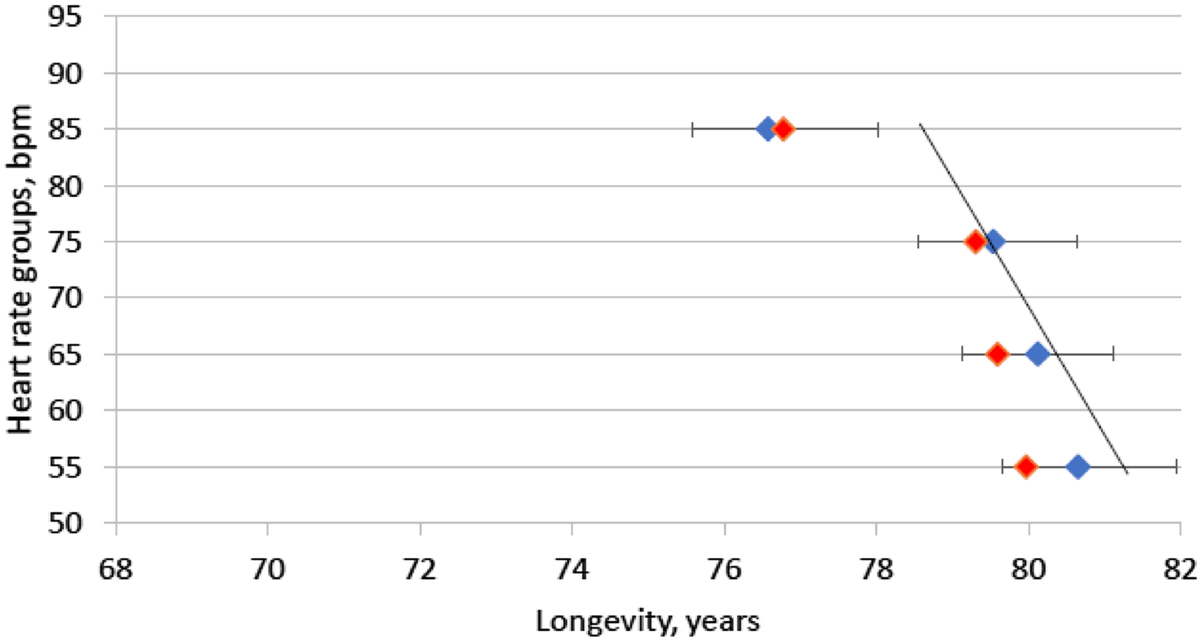
We investigated whether differences in resting heart rate within human populations are associated with life span using data from three population-based cohorts based in three different countries: the Paris Prospective Study I (France, Fig. 1), the Whitehall I study (UK, Fig. 2), and the Framingham Heart Study (USA, Fig. 3). In the Paris Prospective Study I, among 7976 healthy French men aged 42 to 53 years, recruited in the study from 1967 to 1972, 2387 deaths were recorded by 31[st] December 1993. As expected, there was an inverse linear relation between resting heart rate and life span, which remained significant after adjustment for potential confounders. Subjects



Resting heart rate at inclusion	N	Life span of population, Mean (std dev)
≤60	2137	79.3 (8.4)
]60-70]	2762	77.8 (9.6)
]70-80]	2371	76.2 (10.5)
]80-90]	434	73.8 (11.6)
>90	272	70.3 (12.8)
P for trend		<0.0001

- A total of 2258 deaths in 7976 men over a 25-year follow-up. Baseline follow up was between 1967 and 1972
- The men are divided into 5 groups according to their resting heart rate at inclusion.
- For deceased people, life span is defined by the actual life span of subjects; for people alive at the end of study, life span is defined by the life span of the French population at the corresponding generation and age of subjects at the end of, or when they left, the study.
- Blue dots are mean of life expectancy by groups; Red dots are least square means (general linear model) of life span by groups; dotted line is linear regression of heart rate values by life span.
- Factors of adjustment are smoking (mean consumption g/days over the 5 last years), age, diabetes, body mass index, current sport, systolic blood pressure, total cholesterol.

Figure 1. Relation between resting heart rate (at study baseline) and life span in 7967 men from the Paris Prospective study I.



Resting heart rate at inclusion	N	Life span of population, Mean (std dev)
≤60	236	80.7 (10.7)
]60-70]	437	80.1 (10.2)
]70-80]	329	79.5 (10.6)
>80	224	76.6 (11.7)
P for trend		<0.0001

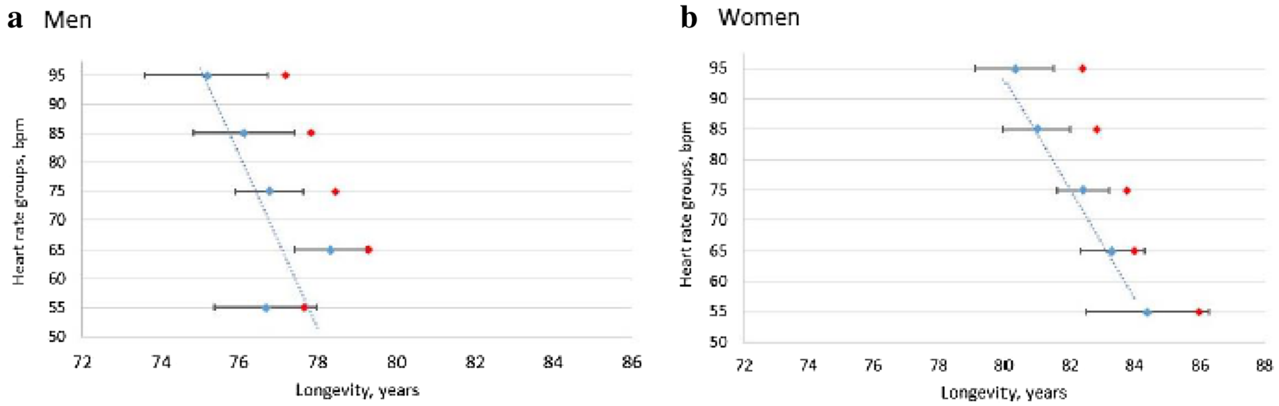
- A total of 977 deaths in 1226 persons over a 40 years follow-up. Baseline follow up was at 1967.
- Subjects were divided into 4 groups according to their heart rate at inclusion.
- Blue dots are mean of life span by groups; red dots are adjusted mean (general linear model) of life span by groups. Dotted line is linear regression of heart rate values by life span.
- For deceased people, life span is defined by the real-life span of subjects; for alive people at the end of study, life span is defined by the life span of the UK population at the corresponding generation and age of subjects at the end of, or when they left, the study.
- Factors of adjustment are smoking, age, diabetes, body mass index, current sport, systolic blood pressure, total cholesterol.

Figure 2. Relation between resting heart rate (at study baseline) and life span in the 1226 men from the Whitehall I study.

with a resting heart rate over 90 bpm had a life span of 70.27 ± 12.78 whereas subjects with a resting heart rate of less than 60 bpm had a life span of 79.30 ± 8.43 (HR mortality for 10 bpm difference: 1.25; 95% CI: 1.21 to 1.30; $p < 0.0001$) (Fig. 1), corresponding to an average difference in life span of 9 years.

In the Whitehall I study, among the 1226 healthy British men aged 52 ± 7 years, recruited between 1967 and 1969, 977 had died at last follow up in 2008. Similar to the Paris Prospective Study I, an inverse resting heart rate-life span relation was also observed in both the unadjusted and adjusted models (Fig. 2).

The analysis in the Framingham Heart Study extended the results to women. The participants were recruited from 1971 to 1975. Among 4001 US women aged 47 ± 16 years, a total of 2409 deaths occurred over a mean follow-up time of 32.8 years. Among 3299 men aged 45 ± 15 years, a total of 2172 deaths occurred over a mean follow-up of 30.9 years. An inverse association was observed between resting heart rate and life span in both



Men			Women		
Resting heart rate at inclusion	N	Life span of population, Mean (std dev)	Resting heart rate at inclusion	N	Life span of population, Mean (std dev)
≤60	311	76.7 (12.1)	≤60	150	84.4 (11.4)
>60-70	610	78.3 (11.4)	>60-70	536	83.3 (11.1)
>70-80	725	76.8 (11.8)	>70-80	813	82.4 (11.5)
>80-90	312	76.1 (11.1)	>80-90	523	81.0 (11.9)
>90	214	75.2 (11.7)	>90	387	80.3 (12.3)
p for trend		0.0076	p for trend		<0.0001

- A total of 2172 men and 2409 women had died in 3299 men and 4001 women over a mean follow-up of 30.9 years for men, and 32.8 years for women. Baseline follow up was between 1971 and 1975.
- Subjects were divided into 5 groups according to their resting heart rate at inclusion.
- Blue dots are mean of life span by groups; red dots are adjusted mean (general linear model) of life span by groups.
- For deceased people, life span is defined by the real-life span of subjects; for alive people at the end of study, life span is defined by the life span of the US population at the corresponding generation and age of subjects at the end of, or when they left, the study.
- Dotted line is linear regression of heart rate values by life span. Life span is only given for deceased people.
- Factors of adjustment are smoking, age, diabetes, body mass index, physical activity index, systolic blood pressure, and total cholesterol.

Figure 3. Relation between resting heart rate (at study baseline) and life span in 3299 men (**a**) and 4001 women (**b**) from the Framingham Original and Offspring cohorts.

women and men. Women experienced longer longevity than men (5 years on average) for each category of resting heart rate (Fig. 3).

Furthermore, we examined the association between change in resting heart rate over a 5-year and 8-year period and risk of mortality using data from the 5589 men in the Paris Prospective Study I as well as 3299 men and 4001 women in the Framingham Heart Study (Table 1). We observed a statistically significant association between increase in resting heart rate over a 5-year period and risk of mortality in French men from the Paris Prospective Study I (HR mortality per 10 bpm increase over time: 1.20; 95% CI: 1.13 to 1.27) and over an 8-year period in American men and women from the Framingham Heart Study (HR: 1.13; 95% CI: 1.07 to 1.19 for men and HR: 1.09; 95% CI: 1.04 to 1.15 for women), after adjusting for classical risk factors and resting heart rate. This hypothesis was not tested out in the British cohort.

Cohort	Sex	Hazards ratio per 10 bpm increase	95% Confidence interval	*p* value
PPSI	Male	1.20	(1.13, 1.27)	< 0.0001
FHS	Female	1.09	(1.04, 1.15)	0.0003
	Male	1.13	(1.07, 1.19)	< 0.0001

Table 1. Multivariable-adjusted hazard ratios for change in heart rate over 5–8 years and risk of mortality in the Paris Prospective Study I and the Framingham Heart Study. PPS I: A total of 1256 deaths among 5589 over a 25 years follow-up. Baseline follow up was at 1967. FHS: A total of 1582 men and 1840 women had died over a mean follow-up of 30.9 years for men, and 32.8 years for women in 2461 men and 3068 women. Baseline follow up was between 1971 and 1975. Heart rate change was determined using two measurements, the heart rate baseline and after 5 years in PPSI, and after 8 years in FHS. Subjects whose resting heart rate increased by 10 bpm over a 5-year period for PPS I and 8-year period for FHS, increased their risk of mortality ($p < 0.001$). Models are adjusted for age, smoking (mean consumption g/days 5 last years for PPS I; current smoking (yes/no) for FHS), diabetes, body mass index, physical activity, systolic blood pressure, and total blood cholesterol, and baseline heart rate.

Ethics declaration
All methods were carried out in accordance with relevant guidelines and regulations, informed consent was obtained from all subjects and/or their legal guardians.

Statistical analysis
Life span in all three cohorts was defined by age at death among participants who died over the follow-up period. For participants in the Paris Prospective and Whitehall studies who were alive at the end of the follow-up, life span was estimated based on national age and sex in the corresponding country.

Aim 1 and 2
As the sample size and number of deaths were larger in the Paris Prospective Study I and Framingham Study, heart rate was categorized into 5 groups: ≤ 60, 60–70, 70–80, 80–90, > 90 bpm. In the Whitehall I study smaller numbers led us to use 4 categories: ≤ 60, 60–70, 70–80, > 80 bpm. Participants with known or suspected cardiovascular disease or with any of the following conditions were excluded from the study: resting systolic blood pressure > 180 mm Hg, resting 12-lead standard electrocardiogram abnormality (Minnesota code). A linear regression was used to examine the association between resting heart rate and life span in each cohort study, first unadjusted and then adjusted for age, baseline body mass index, smoking, physical activity, diabetes, systolic blood pressure, and total cholesterol level.

Aim 2
In the Paris Prospective Study I and Framingham Heart Study, multivariable Cox proportional hazards regression models were used to assess the change in heart rate (over a 5-year period in the Paris Prospective Study I and 8-years period in the Framingham Study) and risk of mortality. These analyses were adjusted for the covariates used in aim 2 along with resting heart rate at baseline. In the Paris Prospective Study, I, change in heart rate was defined as resting heart rate at the last follow-up visit minus resting heart rate at baseline. In the Framingham Heart Study, heart rate change (increase or decrease) was defined as resting heart rate at examination 16 (1979–1982) minus resting heart rate at examination 12 (1971–1974) for the Original cohort, and examination 2 (1979–1983) minus examination 1 for the Offspring cohort (1971–1975). A positive difference in heart rate indicated an increase in heart rate at the subsequent examination, while a negative difference indicated a decrease in heart rate.

SAS, Version 9.2 and 9.4 (Statistical Analysis System, Cary, N.C.) was used for analysis. P-values were 2-sided.

Discussion
In this study, using three long-term historical and large cohorts from France, UK and USA, we reported 2 main findings. We confirmed the inverse relation between resting heart rate and life span, firstly, within human populations. Secondly, we determined that men and women who increase their resting heart rate over a 5-year period in the Paris Prospective Study I and 8-years period in the Framingham Study, respectively, increase their risk of mortality.

The concept of total energy expenditure throughout the life span may be a part of the explanation of the association between heart rate and life span. Larger organisms lose heat as a function of their body surface area but generate heat as a function of their body mass. To conserve body temperature, smaller mammals with a higher surface area to mass ratio require an increased metabolic rate, which in turn determines heart rate. Thus, heart rate may be a "surrogate marker" for total energy expenditure, albeit much easier to measure accurately.

The same linear relation is observed between body size and longevity. Milnor proposed an explanation for the relation between heart rate, body size and aortic input impedance: the pulse wave length and arterial length are matched in a way that minimizes cardiac work[6]. The lower maximal rate in larger animals probably reflects the longer time required for propagation and reflection of pulse wave as heart rate size increases[6]. In our three studies, adjustments were performed for body mass index and did not alter the relation between heart rate and life span. In support of the key rate of energy expenditure as a determinant of mammalian life span, Rubner first noted in 1883 that total energy expenditure in a life span was approximately constant for 5 different species of domestic animal[7]. In 1994, Azbel determined little variation in the total number of heart beats per lifetime in non-human mammals and found that energy consumption per body atom, or per heartbeat, was relatively constant, at approximately 10 O_2 molecules per atom per lifetime or approximately 10^{-8} O_2 molecules per atom per heartbeat[2]. It is noteworthy, however, that fish and insects appear to have significantly fewer heart beats per life time than mammals (by approximately one order of magnitude), perhaps reflecting different environments and/or bioenergetic profiles.

Some experimental evidence has addressed the question of whether slowing heart rates in mammals might increase life span. Coburn et al.[8] found that slowing heart rate in mice by approximately 50% (using the pharmacologic agent digoxin) increased life span by approximately 23%, however, this study was confounded by decreased body weight in the digoxin-treated mice. Beere et al.[9] reported that lowering heart rate in monkeys (by surgical interference with the cardiac pacemaker cells) led to slowed development of coronary atherosclerosis. In humans, inducing a slow heart rate has only been studied in the context of serious illness. In this setting, using beta adrenergic receptor blocker have been associated with decreased mortality risk in post-myocardial infarction patients with heart failure and a high resting heart rate[10,11]. Furthermore, Fox et al.[12] reported that although reduction in heart rate with ivabradine, a pure bradycardic I_f channel blocker agent, has been shown to decrease the incidence of coronary artery disease outcomes in a subgroup of patients who have heart rates of 70 bpm or higher, it has not been shown to improve cardiac outcomes in all patients with stable coronary artery diseases and left-ventricular systolic dysfunction. However, the effects of a slowing heart rate on life span have not been studied in a healthy human population, and causality has not been addressed.

In addition to the biological plausibility argument summarized above, assessing the causality in epidemiology requires 3 steps. The first step is observational, and in the present study, we show independent association between higher heart rate and shorter life span. The second step is also observational and we newly demonstrate that individuals who increase their heart rate over a 5-year period in the Paris Prospective Study I and 8-years period in the Framingham Study, respectively, decrease their life span. Importantly, these results were obtained in populations from different countries and different risk factors distribution supporting the generalizability of the findings. Also, the differences in lifespan according to baseline or change in resting heart rate were meaningful in magnitude, which supports the clinical and public health relevance of our findings. The third step is interventional: when an intervention decreases heart rate, we observe an increase in life span in the long run. We acknowledge that this step will be much more difficult to demonstrate. Still, on the basis of the present findings, and given the simplicity of heart rate assessment, heart rate might be a target to monitor humans health and life span. Given the obesogenic environment of our societies, creating favourable environment to promote healthy behaviours that impact heart rate (i.e. physical activity) is more than ever a priority.

It is important to mention that other trials have tackled this question with different findings, such as Whitehall II, that showed no association between changing heart rate and mortality. More investigations on this topic should prove useful to understanding the conclusions that should be drawn from the collective scene.

Online methods
Data
Paris prospective study I

This study is based on 7976 participants, aged 42 to 53 years, recruited to the study from 1967 to 1972 with a 93.4% response rate[13,14]. Briefly, oral informed consent was given by each participant, and the research protocol was approved by the institutional board, Commission Nationale Informatique et Liberté. Clinical examination consisted of electrocardiograms, physical examination, and collection of blood samples for laboratory tests; participants also responded to a questionnaire administered by trained interviewers. Resting heart rate was determined by measurement of the radial pulse during a one-minute recording, after a five-minute rest in supine position. The covariates assessed were age, smoking (mean consumption g/days over the 5 last years), diabetes (self-report of history of diabetes, whether treated), body mass index, current sport, systolic blood pressure, total cholesterol. Participants were invited back for a clinical examination each year for the following 4 years using the same procedure; a total of 5589 provided a heart rate at the final follow-up (five years from the baseline follow-up).

Mortality data were collected from medical records in hospital departments or general practitioners and death certificates. The end of the follow-up period was 1st January, 1994. The vital status could not be determined for 355 subjects (4.6%). The baseline characteristics of those 355 subjects lost to follow up and of those who did not complete the final follow-up examinations (five years from the baseline follow-up) were not different from the remaining 5589 men studied in the present analysis. Vital status was followed on all subjects until the last recorded survey in 1994 (25 years of average follow up), through the administrative department in charge of the study population until retirement age and then thereafter via the French national death registry.

Whitehall I study

Data were collected from 19,019 male London-based government employees aged 40 to 69 at screening between 1967 and 1970, with a 77% response rate. Participants provided informed written consent and research ethics approval was obtained from the National Health Service London—Harrow Research Ethics Committee. Screening involved the completion of a study questionnaire and participation in a medical examination, both of which have been described in detail elsewhere[15]. Heart rate was measured in a random subset of 1263 men (mean age 52, SD 7 years) in whom radiologic heart volume was calculated[16]. The covariates assessed in this age, smoking, diabetes, body mass index, current sport, systolic blood pressure, total cholesterol.

All cohort participants were traced using the National Health Service Central Registry for mortality for 40 years until 31st October 2008 by which time 977 men had died.

Framingham heart study
The final sample size for the current investigation consisted of 7300 participants (3299 men and 4001 women) from the Original (Examination 12, 1971–1974) and Offspring (Examination 1, 1971–1975) cohorts in the Framingham Heart Study after exclusion of participants < 20 years of age (n = 245), with prevalent cardiovascular disease (n = 698), or a systolic blood pressure > 180 mm Hg (n = 198). Heart rate was measured by echocardiogram and data on covariates (age, sex, systolic blood pressure, current smoking, diabetes status, body mass index, physical activity index, and total cholesterol level) were drawn from the same wave. Written informed consent was provided by all participants and the study protocol was approved by the Boston University Medical Center Institutional Review Board.

Death were confirmed from death certificates and supporting information was obtained from hospital records[17]. Participants were followed until 31st December 2017, an average follow-up of 30.9 years for men and 32.8 for women.

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Author contributions

BG and XJ had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: BG, XJ Acquisition, analysis, or interpretation of data: BG, EV, VX, MCP, DSC, MS, EM, RJS, JPE, VSR, XJ Drafting of the manuscript: BG Critical revision of the manuscript for important intellectual content: BG, EV, VX, MCP, DSC, MS, EM, RJS, JPE, VSR, XJ Study supervision: BG, XJ Additional Information Competing financial interests: The authors declare no competing financial interests.

Competing interests

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Additional information

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A Prediction Algorithm Using a Combined Heart Rate and Movement Index to Estimate the Energy Expenditure of Anaerobic Exercises

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A B S T R A C T

In recent times, as healthcare is perceived as important, exact information on the proper amount of exercise is necessary. Therefore, if an algorithm can predict real-time energy expenditure and is used in portable equipment, it can help athletes, the general public, and patients create individualized exercise plans. Physical activities must be evaluated to accurately predict energy expenditure. This study propose a new energy expenditure prediction algorithm (EEPA) of combining the heart rate and movement index and applying them simultaneously to several anaerobic exercises to address the disadvantages of preceding energy expenditure prediction studies. A total of 53 subjects (43 males and 10 females) were recruited for this study. The participants used a wireless patch-type sensor (AIRBEAT System) and a wireless gas analyzer (K4b2: Cosmed, Srl, Italy). AIRBEAT System consists of a sensor board, rubber board, and communication module. The sensor is patched onto the participant's chest to obtain physical activity data, including heart rate, movement index, humidity, and temperature. The system was only applied to measurement of heart rate and movement index, and application of energy expenditure prediction algorithm has been limited so far. The relation test for energy expenditure prediction algorithm proposed in this study yields an error rate within ±5% compared with the gas analyzer (K4b2: Cosmed, Srl, Italy) and proves to be more accurate algorithm to estimate physical activity EE. The algorithm developed for energy expenditure prediction is applied not only to anaerobic exercise but also to evey exercise. It is expected that the algorithm developed for estimating energy expenditure will present new application areas for portable heart-rate measurement equipment, such as the AIRBEAT system, and wireless healthcare monitoring devices.

K E Y W O R D S : Energy expenditure, exercise tests; heart rate, movement sensors, energy estimations algorithms, AIRBEAT system

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1. 서 론

전 세계적으로 경제 및 기술 발전에 의한 삶의 질이 높아지면서 개인 건강에 대한 관심이 고조되고 있다. 그러나, 경제적인 풍요와 헬스 케어 시스템의 향상에도 불구하고 대사 장애를 비롯한 고혈압, 당뇨, 비만, 심장병과 같은 고질병으로부터 고통 받고 있는 비율이 급속히 증가하였다[1,2].

이러한 고질병들의 가장 중요한 원인은 신체활동과 운동 부족으로 인한 에너지 불균형에서 비롯된다. 에너지 불균형은 개인의 에너지 소비, 칼로리 섭취와 밀접한 관련이 있다. 심박수와 운동 지수는 신체활동을 측정하기 위해 사용된다. 따라서, 심박수와 운동지수 그리고 에너지 소비에 대한 정확한 평가와 이들 상호간의 연관성을 분석하는 것이 필요하다[3,4].

최근 건강에 대한 기대가 높아지면서 고질병 예방을 위한 에너지 소모량 측정 Wireless healthcare monitoring Device, Ubiquitous healthcare system이 개발 되고[5,6], 에너지 소모를 예측하는 알고리즘이 연구되고 있다[7-13].

가속도 동작감지기를 이용한 선행연구는 에너지 소모를 예측하는 Freedson(1998), Swartz(2000), Courter(2006)[14-16]의 알고리즘에 사용되어 각 가속도를 이용하여 Walking 과 Running 조건에 대한 에너지 소비를 예측하지만, Fitness club에서 수행되는 다양한 무산소운동의 경우 각 가속도 동작감지기로는 에너지 소비를 측정하기 어렵다.

심박수를 이용한 에너지소모 예측 선행 연구는 일상생활 동안의 심박수(90-150b/m)안에 일어나는 운동에 대해서만 선형 관계 (Ceesay et al., 1989; Renneie, Hennings, Mitchell, & Wareham, 2001; Spurr et al., 1988)[17,18]로 예측이 가능하지만, 일상생활 (누워 있음, 앉음, 서있음 등)에서 감정, 자세, 환경적인 조건 (Hebestreit & Bar-Or, 1998)에 의해 심박수가 변화하여 운동과는 상관없이 증가하는 경우 에너지 소비 측정에 사용할 수 없는 단점이 있다[19]. 또한, 모든 사람에게 동일하게 적용하여 신체활동 에너지를 계산하기 때문에 개인별 신체 능력에 따른 정확한 에너지 소비를 측정할 수 없었다.

심박수와 각 가속도 동작감지기를 이용한 에너지 소비 예측 방법의 단점을 보완하기 위하여 심박수와 가속도 동작감지기를 결합한 시스템과 알고리즘이 연구되고 있다[20-22]. 그러나 일부 결합된 시스템은 많은 연구에도 불구하고 일부 특정 운동의 신체활동에서는 정량적으로 가용 데이터를 측정 하지 못하는 단점을 갖고 있어 야외 및 유산소 운동에만 적용되고 있다[23,24].

AIRBEAT 시스템[4]은 에너지 소비량을 추정하기 위하여 결합된 심박수와 운동지수를 이용하였으나 심박수와 운동지수 측정에만 적용 되었으며, 지금까지 에너지소비를 예측할 수 있는 알고리즘 개발이 제한적이었다. 본 연구는 무산소 운동에 대한 에너지 소비 예측 알고리즘 개발을 통하여 AIRBEAT 시스템의 다양한 분야에 적용 가능한 새로운 응용분야를 제시하고자한다.

본 논문의 구성은 다음과 같다. 제 2장에서는 시스템구조에 대해서 살펴본다. 제 3장에서는 실험 및 결과를 논의한다. 제4장에서는 결론을 기술한다.

2. 에너지소비 평가를 위한 시스템구조

2.1 AIRBEAT 시스템

ARIBEAT 시스템은 심박수와 운동지수의 무선 모니터의 Patch-type sensor module이 내재되어 있다[4,6]. <그림 1>과 같이 이 시스템 모듈은 센서보드, 러버보드, 그리고 통신모듈로 구성되어 있다.

센서보드는 Li-ion 전지, USB interface connector, Zigbee RF 모듈, ECG acquisition, 마이크로컨트롤, 전압 조정기, 전압변환기, 3축 가속도 동작감지기를 포함한다. 이 센서는 심박수, 운동지수, 습도, 온도 등 신체활동 데이터를 측정하기 위하여 피험자들의 가슴에 부착하였다. 마이크로컨트롤러는 KB 플래시 메모리를 장착한 Texas Instruments MSP 430 chip, 2048 SRAM 메모리, 8MHz 클록 주파수를 사용하였다. 입력 신호는 signal processing algorithm을 사용하여 심박수, 열 압박, 운동지수를 적절하게 나타냈다.

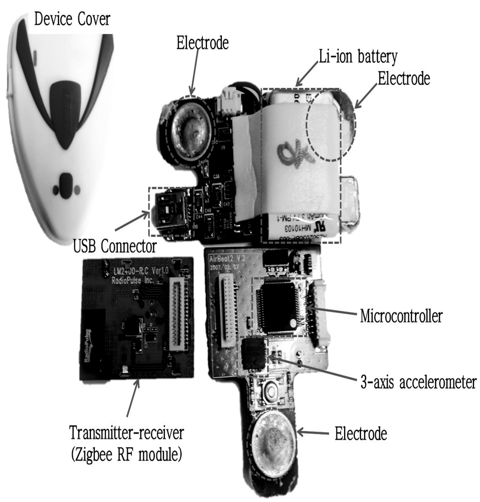


그림 1. 에어비트 디바이스와 3축 가속도 동작감지기, 심전도의 패치 타입 센서 보드
Figure 1. ARIBEAT device and Patch-type sensor board with 3-axis accelerometer, ECG

<표 1>과 같이 동시에 400m 이내 거리에 있는 8명까지 실시간으로 멀티 통신이 가능 하다. 측정된 정보는 각각의 센서 모듈에 저장되며 Zigbee 통신 모듈을 통해 중앙컴퓨터로 전송된다. 상업적인 Zigbee 통신 모듈은 데이터 변환에 사용되었다[4,6].

표 1. AIRBEAT시스템의 설명
Table 1. The specification of AIRBEAT system

Items	Performance
Channel	8ch
Resolution	12bit
Sampling Rate	200/s
Frequency B.W.	1Hz~50Hz
Power	LI-ION
Max HR	250/m
HR Detection Error	Below 10%
Power	±3.3V, 3.3V
Comm. Module	ZigBee
Comm. Distance	400m
MCU	MSP430(TI,USA)
Electrode	Jumper setting available
Size	6cm*9cm, 20g

2.2 AIRBEAT 시스템 구조도

인체에 부착되어 있는 H/W device는 신체에서 발생되는 심박수(Heart rate)와 인체의 움직임(Movement Index)에 의해 발생되는 signal을 측정한다. <그림 2>와 같이 측정된 Signal은 Analog로 되어 있어 Airbeat 장치 내에 있는 converter를 이용하여 디지털 signal로 변화 후 데이터를 패킷화한다. 패킷이 완성되면 H/W는 Embedded와 Zigbee통신하여 데이터를 송신한다. Embedded S/W에서 수신된 데이터를 MI algorithm과 HR algorithm을 통해 embedded 화면에 display하도록 설계하였다. 이 시스템은 심박수와 가속도감지기를 동시에 사용하여 감정변화에 의한 오류와 센서의 부착 범위에 따른 사용제한 문제를 개선하였다. AIRBEAT 시스템 알고리즘은 심박수와 운동지수만 측정이 가능하도록 설계되어 상업화가 아닌 연구목적으로 사용되고 있다. 일부 선행 연구에서 이 시스템을 사용하여 에너지 소비 예측 알고리즘을 적용하였으

나 실험 참여 인원과 달리기, 러닝머신 등 운동의 제한으로 인하여 대중적 기반 연구에 사용이 제한적이었다[3,4,6].

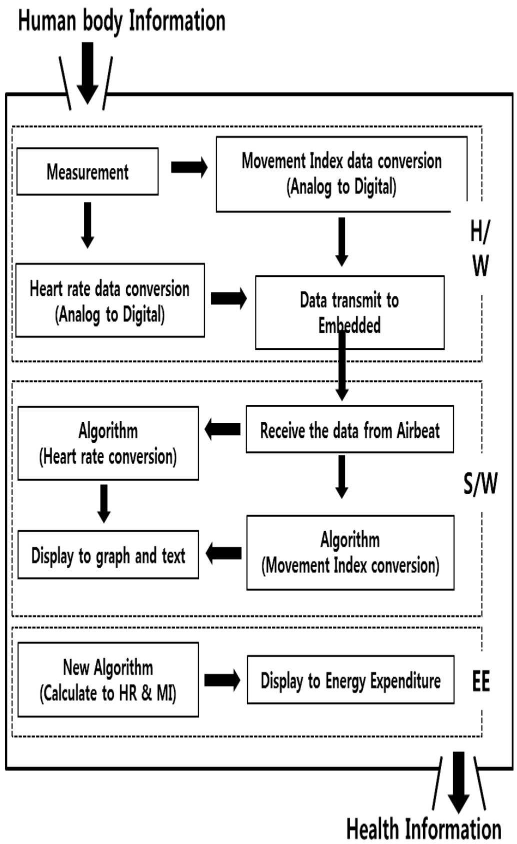


그림 2. 에어비트 시스템 구조도
Figure 2. Structure of AIRBEAT system

3. 실험 및 결과

3.1 실험디자인

실험은 남자의 경우 나이 는 26세에서 52세, 신장은 160 cm에서 180 cm, 몸무게 65 kg에서 80 kg과 여자의 경우 나이 는 24세에서 32세, 신장은 150 cm에서 165 cm, 몸무게 45 kg에서 56kg의 신체적 특성을 갖는 참가자들이 참여 하였다. 모든 참가자들은 실험의 일관성을 위해 식사 후 2시간이 지난 뒤 개인별 정상 심박수를 측정 하고 실험을 실시하였다. 실험 프로그램은 ACSM' Guidelines for Exercise Testing and Prescription (2000)[25]을 참고 하였다. 실험 프로토콜은 <표 2>와 같은 실험프로그램으로 진행되었으며, 조선대학교 IT융합 신기술 연구센터(IFRC-2013-01) 윤리위원회에 의해 승인되었다.

표 2. 운동 프로그램
Table 2. Training Program for Tests

Progression	Composition	Training Program	Duration	ACSM Guidelines
Warm up	Stretching	Seconds /per stretch	20 Seconds	10~30 seconds and 3~4 times repetition
Aerobic training	Treadmill	1~5min 3km/h 6~15min 4km/h 16~20min 5km/h	20 min	Walking 50~10m/min (3~6km/HR)
Weight training	Butterfly	Chest	12RM for 50 minutes by setting 7 kinds of exercises into 1 set	8~12RM 8~10 kinds of exercises into 1 set for below one hour
	chest press			
	Shoulder press	Shoulders		
	Long pull	Arms		
	Leg press	Legs		
	Leg extension			
	Incline press	Chest		
Endurance training	Cycling	60% VO_2R	20 min	40/50~80% HRR or VO_2R 20~60min

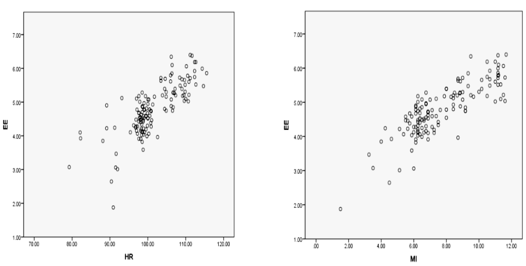
RM=Repetition Maximum
HRR=Heart Rate Reserve
VO_2R=Maximal Oxygen Consumption Reserve

실험 참가자들은 각 운동에 대한 에너지 소비량 측정을 위해 Wireless Patch-type sensor를 가슴에 부착하였으며 실제 에너지 소비량을 측정하기 위해 산소 섭취량 oxygen consumption(VO2), 이산화

탄소 발생률 carbon dioxide(VCO2)를 직접 측정 가능한 휴대용 무선가스분석기 (K4b2: Cosmed, Srl, Italy)을 사용하였다[26]. 모든 데이터 분석을 위하여 SPSS(v20.0) 통계 프로그램[27]을 사용하였다. 유의 수준(significance level)은 유의 확률(P- Value)이 0.05일때 유의하며[28], 평균 오차율은 신뢰 수준(CI: Confidence Level)95%, 표준 오차 ±5에 허용하였다[29].

3.2 결 과

10가지 실험 운동에 대한 평균 에너지 소비량과 심박수(HR), 평균 에너지소비량과 운동 지수(MI)의 선형 연관성 척도를 분석 하였다. 〈그림 3〉은 심박수와 에너지소비량, 운동지수와 에너지 소비량이 0<p≤+1로 절대 값 1에 가까워 높은 것임을 보여주며 이 둘의 연관성은 양의 선형 관계를 나타냈다.



(A)Linear correlation of EE-HR (B)Linear correlation of EE-MI

그림 3. 에너지소비와 심박수, 에너지소비와 운동지수의 선형연관성 척도
Figure 3. The measurement of linear correlation of EE-HR(A) and EE-MI(B)

에너지소비량이 운동지수와 심박수에 대한 회귀계수를 분석하였다. 〈표 3〉과 같이 회귀 계수를 이용한 각 운동 별 유의성 분석결과 에너지 소비량에 대한 운동지수와 심박수는 유의하는 것으로 나타났다.

표 3. 회귀분석 통계결과
Table 3. Statistical Results of Regression Analysis

Activity	HR	MI	HR	P-Values
Treadmill	0.027	0.225	0.308	0.006
Stretching	0.035	0.133	0.020	0.001
Butterfly	0.028	0.214	0.344	0.003
Chest press	0.004	0.980	2.122	0.006
Shoulder press	0.029	0.863	-0.706	0.042
Long pull	0.056	0.774	-4.152	0.000
Leg press	0.029	0.430	0.117	0.010
Leg extension	0.030	0.592	-0.817	0.000
Incline press	0.025	0.493	0.218	0.000
Cycling	0.037	0.240	2.317	0.000

HR = Heart Rate
EE = Energy Expenditure
MI = Movement Index

심박수 및 운동 지수에 대한 에너지 소비 알고리즘은 관계식(1)로 표현 하였다.

$$EE = (A * HR) + (B * MI) + C \quad (1)$$

여기에서 A, B는 운동지수 및 심박수가 한 단위 변함에 따라 에너지소비에 미치는 영향력 크기를 의미한다. 〈그림 4〉와 같이 에너지 소비량을 예측하기 위한 알고리즘으로 pseudo code을 사용하였다. EE는 에너지 소비량을 저장하는 변수이며 begin은 변수를 사용하기 위한 선언 및 초기화이다. I는 1분 동안 Heart rate 값과 Movement Index 값을 누적하기 위한 반복문 변수이며, 1부터 60까지 증가하도록 설정했다. 운동 종목에 관련된 변수는 4개의 종목만 설정하였다. 각 종목을 선택하면 C값이 결정될 수 있으며, 1분 동안 누적된 Heart rate 값과 회귀계수 A값, Movement Index 값과 회귀계수 B값이 계산되고 운동 종목에 따른 C값이 결정되어 EE 변수에 대입함으로써 무산소 운동이

나 모든 운동에 적용될 수 있는 통용적인 에너지 소비량 알고리즘을 구하였다.

```
EE <- 0;

begin
I <- 1;
HR <- HR + I;
MI <- MI + I;
I = I+1;
if I is not 61 goto begin

find item of exercise{
if exercise = treadmill
C <- 0.308;
if exercise = Stretching
C <- 0.020;
if exercise = Butterfly
C <- 0.344;
if exercise = Chest press
C <- 2.122;
else
C <- 0.706;
}

EE <- (A * HR) + (B * MI) + C;
```

그림 4. 에너지소비 예측 알고리즘 Pseudo code
Figure 4. Pseudo code of energy expenditure prediction algorithm

실제 에너지소비량과 본 논문에서 제시한 에너지소비량 예측 알고리즘간의 정확성을 측정하기 위해 검증과정을 수행 하였다. <표 4>와 같이 10가지 운동의 가스 분석기를 이용한 에너지소비량 측정값과 새로운 알고리즘을 이용한 에너지소비량 예측 값은 표준오차가 모든 운동에서 ±5% 이내로 나타났다. 본 논문에서 제시하는 에너지소비량 예측 알고리즘은 실제 실험에서의 에너지 소비량과의 평균 오차율이 낮아 에너지소비량 예측 알고리즘의 정확성이 매우 높은 것을 알 수 있다.

표 4. 10가지 운동의 측정된 에너지소비량과 예측 에너지소비량과의 오차율
Table 4. Error ratio between measured EE and estimated EE during testing.

Activity	Average of Measured EE (Kcal/min)	Average of Estimated EE (Kcal/min)	Error ratio (%)
Treadmill	10.625	10.610	0.146
Stretching	3.984	3.830	3.872
Butterfly	1.446	1.514	4.712
Chest press	1.186	1.169	1.365
Shoulderpress	1.909	1.829	4.207
Long pull	3.055	3.086	0.991
Leg press	2.020	2.037	0.853
Leg extension	1.446	1.425	1.496
Incline press	1.898	1.895	0.112
Cycling	17.592	17.839	1.405

4. 결 론

최근 건강관리에 대한 관심이 높아짐에 따라 에너지소비에 대한 정확한 분석이 요구 되고 있다. 따라서 HR과 MI측정이 가능한 휴대용 기기들에 실시간 에너지 소비량 측정 및 이를 계산할 수 있는 알고리즘을 이용할 경우 맞춤형 운동 계획을 세우는데 도움을 줄 수 있다. 본 논문에서는 AIRBEAT 시스템을 사용하여 참가자들의 심박수와 운동지수를 통해 10가지 운동 종류에 따른 에너지 소비량을 측정 하였고, 가스 분석기와 AIRBEAT 시스템을 이용한 운동 별 실행 에너지 소비량 측정값과 본 논문에서 제안한 에너지 소비량 예측 값과의 오차율 ±5% 이내의 높은 정확도를 가진 새로운 알고리즘을 개발하였다. 본 논문에서 제안한 알고리즘은 심박수와 운동지수를 동시에 사용하여 에너지 소비량 예측 결과의 오차율을 최소화 할 수 있다. 따라서 제안된 알고리즘은 AIRBEAT 시스템과 같은 실시간

통신이 가능한 다양한 생체신호 측정 기기들에 적용할 경우 유산소 및 무산소 운동에 상관없이 정확한 에너지를 예측 할 수 있을 것으로 기대된다.

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무산소운동의 에너지소비량을 추정하기 위해 결합된 심박수와 운동지수를 이용한 예측 알고리즘

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요 약

최근, 헬스 케어는 중요한 것으로 인식되고 있기 때문에 적절한 운동량에 대한 정확한 정보가 필요하다. 따라서, 만약 알고리즘이 실시간 에너지 소비를 예측할 수 있고 휴대용 장치에 사용될 수 있다면 운동 선수와 일반인, 그리고 환자들이 개인적인 운동 계획을 수립하는데 도움을 줄 수 있다. 신체 활동은 에너지 소비를 정확하게 예측하기 위해 평가 되어야 한다. 심박수와 운동 지수는 일반적으로 에너지소비를 측정하기 위해 사용된다. 본 연구는 심박수와 운동 지수를 결합하여 선행 에너지 소비 예측 연구들의 단점들을 해결하기 위해 여러 무산소 운동에 적용하는 새로운 에너지 소비 예측 알고리즘을 제안한다. 총 53명(남자 43, 여자 10명)의 피험자들이 이 연구에 모집되었다. 참가자들은 wireless patch-type sensor (AIRBEAT System) 와 wireless gas analyzer (K4b2: Cosmed, Srl, Italy)를 사용하였다. AIRBEAT System은 센서보드, 러버보드, 그리고 통신모듈로 구성되었다. 센서는 심박수, 운동지수, 습도, 온도 등 신체활동 데이터를 측정하기 위하여 피험자들의 가슴에 부착하였다. 이 시스템은 심박수와 운동지수 측정에만 적용되었으며, 지금까지 에너지소비를 예측할 수 있는 알고리즘 개발이 제한적이었다. 본 연구에서 제안한 에너지 소비 예측 알고리즘은 무선 가스분석기wireless gas analyzer (K4b2: Cosmed, Srl, Italy)와 비교해서 ±5% 이내의 오차율을 나타내어 정확성이 매우 높은 것을 알 수 있다. 에너지 소비 예측을 위해 개발된 알고리즘은 무산소 운동에 뿐만 아니라 모든 운동에 통합적으로 적용되어 AIRBEAT System 과 무선 헬스 캐어 모니터 장치와 같은 휴대용 심박수 측정 장치를 위한 새로운 응용분야를 제시할 것으로 기대된다.

감사의 글

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